

Protected Reconstruction from Codazzi Defects: Equianharmonic Quantization and a Minimal Standard-Model Carrier

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Abstract

Protected reconstruction is formulated for a codimension-three timelike Codazzi defect in a four-dimensional Lorentzian branch. The oriented rank-three normal geometry yields a primitive projective link and, after scalar traces are removed, a graded source $V_1[1] \oplus V_2[2]$. Its principal $SU(2)$ content identifies compact A_2 , while Borel-Weil visibility and faithful central marking reconstruct the minimal $3 + 2$ carrier. Determinant and Clifford closure give $S(U(3) \times U(2))$, its global $\mathbb{Z}_6$ form, and $k_Y = 5/3$. The equianharmonic Coxeter lattice supplies the three-point family torsor and the first primitive norm-seven commensurate class. In an Alena-Codazzi collar, structural reconstruction is separated from realization by projector, Riesz-response, Rees-Picard, Hodge, and neutral-relay residuals. The selected projector has an intrinsic polynomial certificate and reconstructs the packet $(1, 4, 5; 7)$; the finite first two responses determine the primitive Rees geometry, while a global $2:3:5$ principal lift links family determinant holonomy with the projected weak-leg rate. After the marked operator gates are passed, one universal dimensionless scalar u is fixed once and the remaining CKM, neutral, and PMNS quantities are evaluated without retuning. On the frozen reference branch, the downstream readings include $\delta_{CKM} = 65.36^\circ$ and $\Delta m_{21}^2 / \Delta m_{31}^2 = 0.029655$. These values are retained as falsifiers of the completed branch rather than structural inputs. Absolute normalization and the global Riesz-Callias attachment remain separate.

Keywords: Codazzi defects; Lorentzian geometry; protected reconstruction; equianharmonic quantization; finite internal geometry; Eisenstein lattice; Schur-Feshbach reduction; flavor mixing; Standard Model; Alena Tensor

1 Introduction

The geometrization of interactions is commonly obtained by enlarging the geometric structure. Gauge variables are represented by higher-dimensional or bundle-metric data in Kaluza-Klein mechanisms [1]. Forced dynamics may be represented as geodesic motion on an extended space through the Eisenhart-Duval lift [2]. Randers and Finsler descriptions encode charged trajectories by an effective geometry [3], while finite-resolution curvature defects provide a related local comparison class [4]. A narrower inverse problem is considered here: which finite internal data are forced by a local defect when only its resolved boundary observables, their source grading, and their closure relations are retained? In each of the constructions above the connection belongs to the assumed geometric data. Here it does not. The retained datum is a local defect together with the grading of its boundary source; the carrier, its structure algebra, and the transport acting on it are selected by the same minimality, so that the connection appears as the projected transport of the isolated sector rather than as an added bundle. Enlargement of the ambient geometry is thereby replaced by closure of the retained boundary description.

The local datum is a codimension-three timelike defect in a four-dimensional Lorentzian branch. Its oriented rank-three normal bundle has a two-sphere link, read in the optical Codazzi setting as a projective spinor line $\mathbb{CP}_T^1$. After the scalar trace has been separated, a normal source of transverse order at most two has two principal non-scalar components: an order-one current moment and an order-two trace-free Codazzi moment. The aim is to determine the finite carrier generated by these graded boundary data and to separate this structural result from its geometric realization and subsequent low-energy completion.

The organizing principle is self-reconstruction. A physical configuration determines an observable description, and the retained description is required to reconstruct the same physical configuration:

$$\Phi \longmapsto (\mathcal{A}_\Phi, \mathfrak{S}(\mathcal{A}_\Phi), \delta_\Phi, \text{Sec}_{\text{iso}}(\mathcal{A}_\Phi), \nabla_\Phi) \longmapsto \hat{\Phi}. \quad (1)$$

Here $\mathcal{A}_\Phi$ is the generated observable algebra, $\mathfrak{S}(\mathcal{A}_\Phi)$ is its state space, δ_Φ is the retained dynamics, $\text{Sec}_{\text{iso}}(\mathcal{A}_\Phi)$ denotes the isolated stable sectors, and ∇_Φ denotes their natural transport data. Equality of Φ and $\hat{\Phi}$ is understood on the physical quotient, after gauge, diffeomorphism, and unitary equivalences have been removed.

Under the intersection-closure hypothesis of Section 2, minimal reconstruction is the least protected Moore closure; structural, realization, and completed depths are separated before low-high elimination. Standard $SU(2)$ representation theory [5] identifies the two non-scalar second-jet types as $V_1 \oplus V_2$. The oriented normal metric places this module in the principal compact A_2 algebra, whose Coxeter plane gives the Eisenstein elliptic datum E_ω . Independently, Borel-Weil visibility and faithful central marking reconstruct the normal projective carrier. The invariant theta degrees three and six give ranks 2 and 3; their agreement with the projective ranks is retained only as an equianharmonic-projective consistency check.

The reconstructed $2 + 3$ carrier supplies the centered determinant line, the Levi stabilizer, and the even exterior package. Twistor-space Standard-Model constructions provide a representation-theoretic comparison [6]. The quotient stack $[E_\omega/\mu_6]$ has tubular type $(6, 3, 2)$, and its weight-two orbit $E_\omega[2] \setminus \{0\}$ gives the rank-three family torsor before the Clifford parity linearization is selected. Finite permutation and Clifford family models provide comparisons [7]. The Eisenstein norm spectrum and the degree-seven theta sector then organize the commensurate and neutral responses without introducing an additional continuous structural parameter.

The analysis is organized in five layers. The Lorentzian defect geometry first fixes the primitive graded source. Protected reconstruction then fixes the finite carrier and its equianharmonic and torsor data. The Alena-Codazzi collar tests realization through the isolated projector, the first two Riesz responses, intrinsic Rees geometry, and the marked Hodge bridge. After the common 2:3:5 lift and neutral relay have passed their operator tests, one universal dimensionless scalar u is fixed once and the remaining local CKM, neutral, and PMNS outputs are evaluated without retuning. Absolute normalization and global Riesz-Callias topology are treated separately. This order prevents a downstream numerical reading from changing an upstream structural selection; compatible deformation-retract data give the usual homological-transfer reading of the Schur expansion.

2 Protected Minimal Reconstruction and Gapped Transfer

The self-reconstruction loop of (1) is used with an ordered definability filtration. Structural residuals are closed before analytic attachment and completed low-energy residuals, so a later completion cannot change an earlier finite selection. Spectral projections are understood in the sense of [8]; the open Fredholm comparison uses [9] and its geometric form [10]. A retained description at depth i is $X_{\leq i}(\Phi) = (\mathcal{A}_\Phi; \mathfrak{A}_{\leq i}^{\text{rec}}, \pi_i, \mathcal{H}_i, \mathcal{M}_{\leq i})$. Observation and reconstruction satisfy $\mathcal{R} \circ \mathcal{O} = \text{id}$ on the physical quotient, and the resulting split idempotent fixes the reconstructible descriptions.

For a residual R_i with represented metric G_i , the local quadratic form at a closed stratum is

$$H_i^{\text{Gram}} = DR_i^\dagger G_i DR_i. \quad (2)$$

Closures are evaluated lexicographically on the preceding exact stratum. Within the completed Schur depth the retained order is

$$\mathfrak{E}_{\text{dir, ch}} \prec \mathfrak{E}_{\text{dir, det}} \prec \mathfrak{E}_{\text{dir, spec}} \prec \mathfrak{E}_{\text{dir, tr}} \prec \mathfrak{E}_{\text{neu}} \prec \mathfrak{E}_{\text{mix}}. \quad (3)$$

Standard variational, curvature, boundary-charge, determinant, and odd-superconnection closures are used without changing their usual content; the curvature convention is represented by [11], boundary superselection by [12], determinant families by [13], differential index refinement by [14], and odd finite insertions by [15].

For active principal types Λ_{pr} , the marked sector map is

$$\zeta_F : \bigoplus_{a \in \Lambda_{\text{pr}}} \mathbb{C}e_a \longrightarrow Z(\mathfrak{A}_F), \quad \zeta_F(e_a) = p_a, \quad (4)$$

with orthogonal central images. For equivariant coefficient maps $\iota_{F,a}$ and reconstructions $\rho_{F,a}$, the fidelity defect is

$$\delta_{\text{fid}}(F) = \dim \ker \zeta_F + \sum_{a \in \Lambda_{\text{pr}}} \left(\dim \ker \iota_{F,a} + \|\rho_{F,a} \circ \iota_{F,a} - \text{id}_{V_a}\|_{\text{HS}}^2 \right) \geq 0. \quad (5)$$

Reconstructive completeness is the commutant condition

$$\pi_i(\mathfrak{A}_{\leq i}^{\text{rec}})' = \pi_i(Z(\mathfrak{A}_{\leq i}^{\text{rec}})). \quad (6)$$

Proposition 2.1 (Protected minimality under intersection closure) *Fix an admissible ambient envelope and assume that admissible marking-preserving protected subextensions are closed under intersections. Their intersection is the least admissible extension. On the fixed poset of protected subobjects the resulting closure is extensive, monotone, and idempotent.*

The intersection-closure hypothesis is part of the admissible reconstruction class used below; all minimality statements are relative to that class.

Definition 2.2 (Finite reconstruction core) The finite reconstruction core is the faithful marked representation of the least protected extension satisfying (6), modulo marking-preserving unitary equivalence.

Proposition 2.3 (Finite-core uniqueness and protected descent) *For $\mathfrak{A} \simeq \bigoplus_a M_{n_a}(\mathbb{C})$, a faithful representation satisfying (6) has multiplicity one on every labeled simple block and is unique up to marking-preserving unitary equivalence. An irreducible copy not generated by the source or required by a protected downstream operation is absent whenever its deletion remains admissible. Presentation data may be removed after every invariant needed downstream has been exported.*

Remark 2.4 (Integral reconstruction refinement) If the protected interface carries a canonical integral or cyclotomic form, admissibility may be imposed before complexification. Exact preservation is retained when compatible with the closed residuals; a commensurate downstream corner retains the least admissible finite-index enlargement.

Lemma 2.5 (Gapped compression and transfer comparison) After protected descent, elimination of an invertible complementary block gives

$$H_i^{\text{red}} = H_{xx} - H_{xh}H_{hh}^{-1}H_{hx}, \quad (7)$$

and, for an isolated splitting $P + Q = \mathbf{1}$ with QKQ invertible,

$$\mathfrak{F}_P(K) = PKP - PKQ(QKQ)^{-1}QKP. \quad (8)$$

For a two-dimensional block $K = \begin{pmatrix} 0 & c \\ b & t \end{pmatrix}$ with P on the first coordinate, $t \neq 0$, and $\det K = n$, one has $bc = -n$ and (8) gives $\mathfrak{F}_P(K) = n/t$. In particular, the Eisenstein companion block $C_\xi = \begin{pmatrix} 0 & 1 \\ -N_E(\xi) & \text{Tr } \xi \end{pmatrix}$ gives $\mathfrak{F}_P(C_\xi) = N_E(\xi)/\text{Tr } \xi$ when $\text{Tr } \xi \neq 0$. This reading uses the marked zero-diagonal cyclic splitting of the companion block. Galois conjugacy alone does not select that splitting: in an orthonormal real presentation of multiplication by ξ , the corresponding one-step Schur value is $2N_E(\xi)/\text{Tr } \xi$ when $\text{Tr } \xi \neq 0$. If the splitting independently extends to a deformation retract whose contracting homotopy on the eliminated sector is represented by $Q(QKQ)^{-1}Q$, the perturbative expansion of (8) has the standard homological-transfer organization; the general HPL construction is represented by [16]. No differential graded structure is required for the finite Schur statements used here. The range/cokernel splitting (131), inverse-Sylvester response (183), higher-moment suppression (157), and neutral Schur complement (221) are the corresponding bundle-motion, first-response, perturbative, and symmetric-block readings.

For a graded source $Q = \bigoplus_{\ell \in \Lambda} Q_\ell$ and reconstructed level operator N_F , degree fidelity is

$$R_{\text{fl}}(F, \zeta) = (\zeta(e_\ell) - \mathbf{1}_{\{\ell\}}(N_F))_{\ell \in \Lambda} = 0. \quad (9)$$

It protects the source-grade projections and prevents a removed earlier coordinate from reappearing as an independent later channel.

Every finite core obeys the Robertson-Schrödinger bound [17]; the companion formulation is represented by [18]. For a trace-preserving conditional expectation $\mathbb{E}_X$ in the sense of [19], the relaxation $\mathbb{E}_X + e^{-\gamma t}(\text{id} - \mathbb{E}_X)$ is a standard unital completely positive semigroup [20], and data processing gives entropy monotonicity [21]. With the finite-dimensional subalgebra convention of [22],

$$\mathcal{C}_X(\rho) = S(\mathbb{E}_{X,*}\rho) - S(\rho) \geq 0, \quad (10)$$

and, for faithful σ fixed by $\mathbb{E}_{X,*}$,

$$D(\rho \parallel \sigma) = \mathcal{C}_X(\rho) + D(\mathbb{E}_{X,*}\rho \parallel \sigma). \quad (11)$$

Approximate recovery is represented by [23]; entropy decay with a nontrivial fixed algebra is developed in [24], with the von Neumann algebra setting treated in [25]. Only the expectation and its entropy/Dirichlet forms are used below; the rate γ remains dynamical.

3 A_2 -Equianharmonic Structure from a Primitive Graded Defect

The structural reconstruction is applied to a codimension-three timelike defect whose resolved link carries a primitive positive line and two nonzero principal normal grades. Three comparison directions are retained: the codimension of the spherical link, the positive-line degree, and the first higher principal-harmonic source order. For a spherical link S^{r-1} , the integral second cohomology class used by the line reconstruction is nonzero only for $r = 3$. The oriented rank-three normal metric then identifies the two active non-scalar grades with the principal decomposition of the compact A_2 algebra. Its root lattice supplies the equianharmonic elliptic datum used below, while faithful central marking retains the two source labels as distinct finite sectors.

On the primitive branch, Borel-Weil visibility and central faithfulness reconstruct the normal $2 + 3$ carrier, while the equianharmonic invariant theta spaces give an independent rank agreement. The Kronecker exceptional pair supplies the projective multiplication witness. Clifford and determinant closure are applied to the reconstructed carrier. The elliptic quotient stack $[E_\omega/\mu_6]$ supplies the structural weight-two family orbit and the canonical two-linearization order-six envelope; its physical μ_2 linearization is conditional. Conditions on an Alena-Codazzi collar which realize the source and transport data are given in Section 4. Masses, mixing matrices, Pfaffian data, running, and absolute scales belong to the transferred and completed low problem.

3.1 Primitive graded defect datum and reconstruction assumptions

Let Γ be a smooth embedded future-directed timelike worldline in an oriented and time-oriented four-dimensional Lorentzian branch. Its orthogonal normal bundle $N\Gamma = (T\Gamma)^\perp$ carries the induced orientation and Euclidean metric and has rank three. The real blow-up along Γ is $\tilde{X} = [M; \Gamma]$ with $\partial\tilde{X} = S(N\Gamma)$ and boundary fiber $S(N\Gamma)_t \simeq S_\Gamma^2$. In an optical Codazzi branch, the two-eigenvalue splitting gives the corresponding pair of shear-free geodesic null directions in the Lorentzian principal plane. The Riemannian two-eigenvalue mechanism is represented by [26], while Lorentzian Codazzi structures and the related $2 + 2$ optical geometry are represented by [27] and [28]. The projective-spinor reading identifies the link with

$$S_\Gamma^2 \simeq \mathbb{CP}_\Gamma^1. \quad (12)$$

The spinor conventions are those of [29] and [30]. The twistor comparison is supplied by [31].

The local complement has $\mathbb{R}^4 \setminus \mathbb{R} \simeq \mathbb{R} \times (\mathbb{R}^3 \setminus \{0\})$ and $H^2(\mathbb{R}^4 \setminus \mathbb{R}; \mathbb{Z}) \simeq \mathbb{Z}$. A positive resolved transverse class therefore determines

$$c_1(L_\Gamma) = n, \quad L_\Gamma \simeq \mathcal{O}(n), \quad n > 0. \quad (13)$$

Positive classes form a monoid under tensor product. Atomicity means that the class is not a sum of two nonzero positive classes. Hence the atomic positive representative is

$$c_1(L_\Gamma) = 1, \quad L_\Gamma \simeq \mathcal{O}(1). \quad (14)$$

Higher positive degrees define composite filtered link data. The regularity and removable-singularity background is represented by [32]; the local elliptic convention is the one of [33].

Let N_Γ be the oriented rank-three normal fiber. A natural scalar-sector source of transverse order at most two

has associated graded in $\text{Sym}^0(N_\Gamma^*) \oplus \text{Sym}^1(N_\Gamma^*) \oplus \text{Sym}^2(N_\Gamma^*)$. After the scalar traces have been separated, its non-scalar part is

$$\text{gr}_{>0} \mathcal{J}_{\perp, \text{ns}}^{\leq 2} \simeq V_1[1] \oplus V_2[2]. \quad (15)$$

The brackets record normal order. For a particular second-order source J , put $I(J) = \{\ell \in \{1, 2\} : Q_\ell \neq 0\}$. For the comparison branches below, I denotes the finite set of retained principal harmonic grades and $n = c_1(L_\Gamma) > 0$ the positive line degree. The marked harmonic reconstruction is therefore organized by (n, I) ; atomicity later fixes $n = 1$, while the Standard-Model branch is $I = \{1, 2\}$. The spherical tensor convention is that of [34]; the multipole terminology follows [35].

A principal boundary charge is read by a surface pairing on a linking sphere and is independent of the linking radius in a source-free annulus. In a curved collar the pairing is transported by the frozen collar connection and the corresponding adjoint link mode.

Definition 3.1 (Relative Gauss-local observable algebra) Let $\mathcal{Q}_\partial$ be the finite boundary-charge algebra generated by the retained principal charges and their quadratic Casimir $\mathcal{C}_\partial$. A relative Gauss-local observable algebra $\mathcal{A}_{\text{loc}}$ is generated by annulus-supported observables A satisfying $[A, Q] = 0$ for every $Q \in \mathcal{Q}_\partial$ and by the marked spectral projections of $\mathcal{C}_\partial$. The boundary data are fixed and no singular charge density is present in the annulus interior.

Definition 3.2 (Primitive graded defect datum) A possibly degenerate primitive graded defect datum consists of the link (12), a positive line as in (13), the graded source (15), charges $Q_1 \in V_1$ and $Q_2 \in V_2$ which may vanish, and a relative Gauss-local algebra in the sense of Definition 3.1. It is two-channel when $I(J) = \{1, 2\}$. It is primitive after finite reconstruction when the positive class is atomic and its finite structural representation is the core of Definition 2.2.

The boundary-charge superselection language is compared with [36]. In the present reconstruction, faithful central marking of the retained charge types is part of structural entry B below; Gauss locality supplies its boundary-charge realization. In Section 4, Q_1 is supplied by the dipole-type first normal current moment and Q_2 by the trace-free quadrupole-type second Codazzi-Gauss moment.

Lemma 3.3 (Second-jet reduction) For a natural scalar-sector source of transverse order at most two, removal of the scalar traces gives (15).

Proof The associated graded is contained in the first three symmetric powers of the normal cotangent fiber. Their $\text{Spin}(3) \simeq \text{SU}(2)$ types are V_0 , V_1 , and $V_0 \oplus V_2$, respectively. Removing the scalar summands gives the stated module. $\square$

Corollary 3.4 (Second-order branch exhaustion) The possible active sets are $I(J) \in \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$.

Remark 3.5 (Principal harmonic grade) For the higher-order comparison below, the principal harmonic grade at normal order ℓ means the symmetric trace-free summand $V_\ell[\ell] \subset \text{Sym}^\ell(N_\Gamma^*)$. Lower-spin trace descendants are not counted as additional principal grades in that comparison. This convention does not alter the second-order reduction of Lemma 3.3.

Proposition 3.6 (A_2 normal-source identification) Let N_Γ carry its oriented Euclidean metric and identify $\text{Sym}_0^2(N_\Gamma)$ with the trace-free symmetric endomorphisms. The map

$$\Psi_\Gamma : V_1 \oplus V_2 \longrightarrow \mathfrak{su}(N_\Gamma^\mathbb{C}), \quad \Psi_\Gamma(u, B) = [u]_\times + iB \quad (16)$$

is an $\text{SO}(3)$ -equivariant real isomorphism. Under the principal $\text{SU}(2)$ action, the right-hand side has the $V_1 \oplus V_2$ decomposition associated with the exponents 1, 2 of A_2 [37]. For the simply connected compact group $G_\Gamma = \text{SU}(N_\Gamma^\mathbb{C})$, a maximal torus is fixed up to conjugacy and its exponential lattice is the A_2 coroot lattice. With the Coxeter-compatible complex structure on the oriented Cartan plane, $Q^\vee(A_2) = Q(A_2)$ may be written

$$\Lambda_E := Q(A_2) \simeq \mathbb{Z}[\omega], \quad E_\omega := \mathbb{C}/\Lambda_E. \quad (17)$$

The metric on $\mathbb{Z}[\omega]$ is $2N_E$. The weight-lattice model is homothetic to (17) and defines the same elliptic isomorphism class.

Proof The cross product identifies N_Γ with $\mathfrak{so}(N_\Gamma)$, while multiplication by i identifies trace-free symmetric endomorphisms with the complementary skew-Hermitian trace-free subspace. This gives (16). The remaining statements are the standard simply laced A_2 root, coroot, and Coxeter-plane identifications; a change of Cartan or simple-root marking acts by conjugacy or a lattice automorphism and does not change the elliptic isomorphism class. $\square$

Lemma 3.7 (Marked Gauss-local boundary centrality) In a source-free relative Gauss-local annulus, the marked Casimir projections of Definition 3.1 are central in every represented local algebra.

Proof Gauss closure gives commutation with $\mathcal{Q}_\partial$, and functional calculus gives commutation with the marked spectral projections already retained in the local algebra. The Stokes or Green identity identifies the same boundary-charge algebra on linking spheres in the annulus. $\square$

On $V_1 \oplus V_2$, the Casimir eigenvalues are 2 and 6. The type projections are

$$\Pi_1 = \frac{6\mathbf{1} - \mathcal{C}_\partial}{4}, \quad \Pi_2 = \frac{\mathcal{C}_\partial - 2\mathbf{1}}{4}. \quad (18)$$

The source Euler operator is $\mathbf{N}_\partial = \Pi_1 + 2\Pi_2 = (\mathcal{C}_\partial + 2\mathbf{1})/4$. Under marked Gauss reconstruction, Π_a is represented by $p_a = \zeta_F(e_a)$ in (4).

Lemma 3.8 (Reconstructive charge-sector separation) Assume that both principal charges are retained and that the fidelity defect (5) vanishes. Then p_1 and p_2 are nonzero orthogonal central projections and

$$\iota_{F,a}(V_a) \subset p_a \mathfrak{A}_F p_a, \quad a = 1, 2. \quad (19)$$

Proof Centrality follows from Lemma 3.7. Orthogonality follows because ζ_F is a $*$ -homomorphism, while $\ker \zeta_F = 0$ follows from vanishing fidelity. The coefficient support is part of the marked encoding used in (5). $\square$

Corollary 3.9 (Exclusion of the one-block alias) Suppose that both retained principal labels are represented on a single E_3 block with $\mathfrak{A}_F = M_3(\mathbb{C})$. Then no faithful marked sector map (4) exists for the two labels.

Proof The center of $M_3(\mathbb{C})$ is $\mathbb{C}\mathbf{1}$ and contains no pair of nonzero orthogonal projections. Lemma 3.8 requires such a pair when both principal charges are retained with vanishing fidelity defect. Hence $\ker \zeta_F \neq 0$, and (5) gives $\delta_{\text{fid}}(F) > 0$. $\square$

The central separation used below is the faithful marked image of the distinct Casimir spectral projections in (18); its central marking belongs to entry B. Borel-Weil visibility reconstructs the least marked normal $SU(2)$ support, while the equianharmonic theta construction gives an independent rank witness.

Toeplitz visibility is understood equivariantly on the marked projective support. Since p_a reconstructs the Casimir projection Π_a , each marked sector projection is fixed by the Borel-Weil $SU(2)$ action.

The structural data used below are collected in Table 1. Minimal reconstruction is fixed by Proposition 2.1 and is not an additional structural datum. Proposition 3.6 is derived from the oriented normal metric and the active source types. Projective degree fidelity is derived from Borel-Weil reconstruction; the equianharmonic rank agreement is a separate structural check. The rank-three weight-two orbit is fixed before its physical parity linearization is selected.

Table 1 Inputs of the structural reconstruction.

Entry	Mathematical content	Role	Status
A	$\mathbb{CP}_\Gamma^1$, atomic L_Γ , oriented N_Γ , and nonzero $V_1[1] \oplus V_2[2]$	primitive link and A_2 -graded source datum	structural input
B	relative Gauss locality and faithful central label and coefficient reconstruction	marked separation and protected finite support	structural input
C	commutant-closed Clifford completion and determinant/top-form closure	chiral finite module and compact determinant carrier	structural input
D	marked identification of the spin-lift μ_2 with the weight-two family stabilizer linearization compatible with Γ_{lock}	physical parity attachment of the derived rank-three family module	conditional lock

The nearby alternatives remain useful rigidity checks. If the spherical-link codimension is changed from three, the integral line class used by the reconstruction is absent. A single E_3 block can make both source types Toeplitz-visible but fails the faithful two-label center by Corollary 3.9. The one-channel sets $\{1\}$ and $\{2\}$ retain

only the rank-two and rank-three projective blocks, respectively, while the sets $\{1, 3\}$ and $\{2, 3\}$ lie outside the second-order source exhaustion. Adding the first higher principal grade gives $E_2 \oplus E_3 \oplus E_4$ with only the order-two finite intersection of Corollary 3.19. At fixed $I = \{1, 2\}$, a composite positive degree $n > 1$ gives ranks $(2n + 1, n + 1)$ and the finite factors of (39); the W -side factor is then not exhausted by the primitive order-two parity comparison of Corollary 3.20. These comparisons are made before anomaly coefficients or physical family rank are used.

3.2 Equianharmonic and projective carrier

Let ρ be the order-three Coxeter action on the lattice in (17). In the equianharmonic model $E_\omega : y^2 = x^3 - 1$ it is represented by $\rho(x, y) = (\omega x, y)$.

Theorem 3.10 (Equianharmonic quotient and theta rank witness) *Put $L_3 := \mathcal{O}_E(3O)$ and $W_\theta := H^0(E, L_3)^\rho$. The invariant linear system gives $q : E_\omega \rightarrow \mathbb{P}(W_\theta^*)$, represented in the basis $\{1, y\}$ by $q = (1 : y)$, and identifies $E_\omega / \langle \rho \rangle \simeq \mathbb{P}(W_\theta^*)$. With $C_\theta := H^0(E, L_3^{\otimes 2})^\rho$, multiplication gives*

$$C_\theta \simeq \text{Sym}^2 W_\theta, \quad \dim W_\theta = 2, \quad \dim C_\theta = 3. \quad (20)$$

The ρ -character multiplicities in $H^0(E, 3O)$, $H^0(E, 6O)$, and $H^0(E, 7O)$ are respectively $(2, 1, 0)$, $(3, 2, 1)$, and $(3, 2, 2)$. A projective marking $\mathbb{P}(W_\theta^) \simeq \mathbb{CP}_\Gamma^1$ identifies the positive line with L_Γ and gives $q^* L_\Gamma \simeq L_3$; it does not transport the normal $SU(2)$ action to the theta section spaces.*

Proof The invariant function field is $\mathbb{C}(y)$ and y has a pole of order three at O . The bases $\{1, x, y\}$ and $\{1, x, y, x^2, xy, x^3\}$ give the invariant parts $\text{span}\{1, y\}$ and $\text{span}\{1, y, y^2\}$. Multiplication identifies the latter with $\text{Sym}^2 \text{span}\{1, y\}$, and adding $x^2 y$ gives the degree-seven multiplicities. The theta-section interpretation is standard [38]; changes of theta frame are presentation data under Proposition 2.3. $\square$

For the primitive projective link, retain independently $\mathcal{R}_\Gamma = \bigoplus_{k \geq 0} R_{\Gamma, k}$ with $R_{\Gamma, k} = H^0(\mathbb{CP}_\Gamma^1, L_\Gamma^k)$ and $E_\Gamma|_{R_{\Gamma, k}} = k\mathbf{1}$. The source and section Euler gradings remain distinct until the marked Borel-Weil carrier is fixed.

With the Borel-Weil indexing, $E_q = H^0(\mathbb{CP}_\Gamma^1, L_\Gamma^{q-1}) \simeq \text{Sym}^{q-1} \mathbb{C}^2$ and $\dim E_q = q$. The homogeneous-bundle statement is represented by [39]; the finite harmonic cutoff is $\text{End}_0(E_q) = \bigoplus_{\ell=1}^{q-1} V_\ell$, so V_ℓ first occurs on $E_{\ell+1}$, as in the $\mathbb{CP}^1$ Berezin-Toeplitz cutoff [40]. The finite-level convention follows [41]. Since $S^2 = SO(3)/SO(2)$ has isotropy Euler class two, an $SO(3)$ -homogeneous line of stabilizer weight k has even Chern class $c_1 = \pm 2k$. The primitive line $L_\Gamma \simeq \mathcal{O}(1)$ therefore requires the lift $SU(2) \rightarrow SO(3)$. On this lift, E_2 is the defining spinor, $E_3 \simeq \text{Sym}^2 E_2$ is the complexified vector representation, and $E_3 \oplus E_2 \simeq N_\Gamma^\mathbb{C} \oplus S(N_\Gamma)$.

For a finite positive support $F = \bigoplus_q m_q E_q$, put $\mathbf{N}_F = \sum_{q \geq 1} (q-1) P_q^F$ and measure source-degree fidelity by

$$R_{\text{deg}}(F, \zeta_F) = (\zeta_F(e_\ell) - \mathbf{1}_{\{\ell\}}(\mathbf{N}_F))_{\ell \in \Lambda_{\text{pr}}}. \quad (21)$$

This is the primitive-link instance of (9) together with (19).

Lemma 3.11 (Marked Borel-Weil cutoff) *If an $SU(2)$ -invariant marked block carries V_ℓ in its endomorphism algebra, its rank is at least $\ell + 1$; equality gives $E_{\ell+1}$.*

Theorem 3.12 (Minimal marked projective carrier) *Let Γ satisfy entries A and B of Table 1. The least faithful marked Borel-Weil support is*

$$V_\Gamma = C_\Gamma \oplus W_\Gamma, \quad C_\Gamma = E_3 = R_{\Gamma, 2}, \quad W_\Gamma = E_2 = R_{\Gamma, 1}. \quad (22)$$

Hence rank $V_\Gamma = 5$ and (21) vanishes.

Proof Lemma 3.11 gives the least ranks 2 and 3 carrying the retained V_1 and V_2 types. The two labels require nonzero orthogonal central sectors by Corollary 3.9, so they cannot be merged into one simple block. Initiality and Proposition 2.3 remove unmarked remainder and representation multiplicity. $\square$

The independently obtained theta ranks in (20) agree with (22); this agreement is retained as an equianharmonic-projective consistency check.

On the least separated support, the pair $\mathcal{O}(1) \oplus \mathcal{O}(2)$ is a twist of the standard exceptional pair on $\mathbb{P}^1$ [42]. Multiplication by a basis of W gives the two-arrow Kronecker witness $W \rightrightarrows C$ with dimension vector $(w, c) = (2, 3)$, Euler value $w^2 + c^2 - 2wc = 1$, and stability line generated by $(-c, w)$ in the convention of

[43]. The same primitive branch is characterized by $\text{Ext}^1(\mathcal{O}(2n), \mathcal{O}(n)) \simeq H^1(\mathbb{P}^1, \mathcal{O}(-n)) = 0$ only at positive degree $n = 1$. The Euler value is independently recovered by the tubular comparison of Lemma 3.22; the multiplication and resolution maps are retained for the low-high comparison below.

Corollary 3.13 (Equianharmonic-projective witness descent) On the minimal marked stratum, presentations inducing the same split, level operator, positive line class, order-six degree shift, and exported coefficient invariants are protected-equivalent. Section frames, coefficient embeddings, and multiplication intertwiners not retained downstream are removed by Proposition 2.3.

Proposition 3.14 (Integer-source block parity and low-block rigidity) *The mixed block $\text{Hom}(E_2, E_3) \simeq V_{1/2} \oplus V_{3/2}$ contains no integer principal type, while $\text{End}_0(E_2 \oplus E_3)$ contains no V_ℓ with $\ell \geq 3$. Retained integer principal channels therefore act block-diagonally and every higher principal moment has zero direct compression to (22).*

3.3 The centered carrier line and degree calculus

After equianharmonic-projective witness descent put $C := C_\Gamma$, $W := W_\Gamma$, $V := C \oplus W$, and $(c, w) = (3, 2)$. The retained level is $D_\Gamma = 2P_C + P_W$, and the centered degree is

$$H_\Gamma^{\text{deg}} = 5D_\Gamma - 8\mathbf{1} = 2P_C - 3P_W, \quad (23)$$

Let (s_1, s_2) be an orthonormal basis of $W = R_{\Gamma,1}$ and let $A_i : W \rightarrow C \simeq \text{Sym}^2 W$ be multiplication by s_i in $\mathcal{R}_\Gamma$. For the induced symmetric-power metric,

$$\sum_{i=1}^2 A_i A_i^\dagger = \mathbf{1}_C, \quad \sum_{i=1}^2 A_i^\dagger A_i = \frac{3}{2} \mathbf{1}_W, \quad \mu_K := P_C - \frac{3}{2} P_W = \frac{1}{2} H_\Gamma^{\text{deg}}. \quad (24)$$

Thus the determinant-centered Rees line, the Kronecker stability line, and the normalized multiplication moment direction coincide as one real line; (24) fixes the metric normalization of the last reading. Only this exported line is retained after witness descent, while the ordered multiplication arrows are presentation data. Hence

$$\widehat{R}_\Gamma(t) = \left(\det t^{D_\Gamma} \right)^{-1/5} t^{D_\Gamma} = t^{H_\Gamma^{\text{deg}}/5}. \quad (25)$$

Stabilization of the split top line gives the compact Levi stabilizer

$$\text{Aut}_{\text{u,gr,det}}(V, D_\Gamma) = S(U(C) \times U(W)), \quad (26)$$

and hence

$$S(U(C) \times U(W)) \simeq S(U(3) \times U(2)). \quad (27)$$

Proposition 3.15 (Centered marked grading calculus) *Let $V = \bigoplus_j V_j$ be a marked Hermitian carrier with $d_j = \dim V_j$ and distinct integral levels ℓ_j . Put $r = \sum_j d_j$, $T = \sum_j d_j \ell_j$, $h_j = r \ell_j - T$, and divide (h_j) by its common divisor to obtain primitive integers a_j . Then $H = \sum_j a_j P_j$ is the primitive centered cocharacter generator, the determinant-normalized Rees flow is generated by the same line, and $z \mapsto \sum_j z^{a_j} P_j$ meets $\prod_j SU(V_j)$ in a cyclic group of order $\gcd_j(d_j |a_j|)$. For a coprime two-block split of levels $(2, 1)$ and ranks (c, w) this gives $H = wP_C - cP_W$, finite order $N = cw$, $\text{Tr}_V(H^2) = N(c + w)$, $\dim \Lambda^{\text{even}} V = 2^{c+w-1}$, rational degree $-H/N$ with block weights $-1/c, 1/w$, and Abelian-to-non-Abelian trace ratio $2(c + w)/N$.*

For the selected levels $(2, 1)$ set $r = c + w$, $T = 2c + w$, $T^* = c + 2w$, and $N = cw$; hence $(r, T, T^*, N) = (5, 8, 7, 6)$. The centered line is used with the multiplication normalization (24), the determinant-centered relative-phase normalization with denominator r , and the charge normalization with denominator N . The latter two are read in (147) and (31). The level reversal $(2, 1) \mapsto (1, 2)$ sends the centered generator to its negative and exchanges T with T^* .

For the selected split the covering action gives

$$S(U(3) \times U(2)) \simeq \frac{SU(3)_c \times SU(2)_L \times U(1)_Y}{\mathbb{Z}_6}. \quad (28)$$

This is the effective global carrier form. The finite $\mathbb{Z}_6$ kernel is the global-form datum controlling the embedding of electric and magnetic line-operator lattices; it is not the full cocharacter lattice. Its line-operator sensitivity is discussed in [44], while the Standard-Model cocharacter reading is compared with [45]. The filtered-fiber-functor comparison of (25) is represented by [46]. If $t \in \mathbb{Z}_3$ is color triality and $p \in \mathbb{Z}_2$ weak parity, descent through (28) is $Y \equiv p/2 - t/3 \pmod{\mathbb{Z}}$.

3.4 Fock completion and Levi-spinor readings

Let c be an odd action of $V \oplus V^*$ on a finite graded module. The CAR residual $R_{\text{ext}}(c) = (\{c(v), c(w)\}, \{c(\alpha), c(\beta)\}, \{c(v), c(\alpha)\} - \alpha(v)\mathbf{1})$ closes to $\text{Cl}(V \oplus V^*)$.

Proposition 3.16 (Commutant-closed Clifford completion) *Every finite graded solution is $\Lambda^\bullet V \otimes \mathbb{C}^m$; (6) fixes $m = 1$.*

The convention is that of [47]. Fixing even chirality gives

$$F_{\Gamma}^{\text{even}} = \Lambda^{\text{even}} V, \quad \dim F_{\Gamma}^{\text{even}} = 16. \quad (29)$$

This is the standard $S(U(3) \times U(2)) \subset SU(5) \subset \text{Spin}(10)$ exterior-spinor package represented by [48]. Its normalized CAR trace is

$$\frac{1}{8} \text{Tr}_{F_{\Gamma}^{\text{even}}} \left(\varepsilon(e_i)^{\dagger} \varepsilon(e_j) \right) = \frac{1}{8} \text{Tr}_{F_{\Gamma}^{\text{even}}} \left(\iota(e_i^*)^{\dagger} \iota(e_j^*) \right) = \delta_{ij}, \quad (30)$$

with vanishing mixed trace.

The charge normalization of the same centered line gives

$$Y = d\Gamma \left(-\frac{1}{6} H_{\Gamma}^{\text{deg}} \right) = -\frac{1}{3} N_C + \frac{1}{2} N_W, \quad (31)$$

with $Q = T_3 + Y$. With $N_F = N_C + N_W$,

$$X_F = 5\mathbf{1} - 2N_F = 5(B - L) - 4Y, \quad (32)$$

where $B - L = 1 - 2N_C/3$.

The even bidegree character is

$$\chi_+(u, v) = \frac{1}{2} \left((1+u)^c (1+v)^w + (1-u)^c (1-v)^w \right), \quad (33)$$

with $D_C = u\partial_u$ and $D_W = v\partial_v$. On the selected split it expands as $1 + 3u^2 + 6uv + v^2 + 3u^2v^2 + 2u^3v$; the charge operators above are affine functions of D_C, D_W .

Table 2 Even exterior package and canonical Abelian gradings.

Summand	$SU(3) \times SU(2)$ type	Y	$B - L$	X_F
$\Lambda^0 V$	$(1, 1)$	0	1	5
$\Lambda^2 C$	$(\bar{3}, 1)$	$-\frac{2}{3}$	$-\frac{1}{3}$	1
$C \otimes W$	$(3, 2)$	$\frac{1}{6}$	$\frac{1}{3}$	1
$\Lambda^2 W$	$(1, 1)$	1	1	1
$\Lambda^2 C \otimes \Lambda^2 W$	$(\bar{3}, 1)$	$\frac{1}{3}$	$-\frac{1}{3}$	-3
$\Lambda^3 C \otimes W$	$(1, 2)$	$-\frac{1}{2}$	-1	-3

The separate readings are $B = N_C(3 - N_C)(3 - 2N_C)/6$ and $L = B - (B - L)$; evaluation of (33) gives

$$I_3 = I_2 = 2, \quad (34)$$

$$\text{Tr}_{\Lambda^{\text{even}} V}(Y^2) = \frac{10}{3}, \quad k_Y = \frac{5}{3} = \frac{2r}{N}. \quad (35)$$

For a coprime two-block split, Proposition 3.15 gives $k_Y = 2r/N = 2(1/c + 1/w)$; the selected (3, 2) Levi therefore has the canonical 5/3 exterior-trace normalization.

Proposition 3.17 (Finite-trace Abelian plane and determinant kernel) *The exterior character gives*

$$\text{Tr}(Y(B - L)) = \frac{8}{3}, \quad \text{Tr}(Y X_F) = \text{Tr}(X_F) = 0, \quad \text{Tr}(X_F^2) = 80, \quad (36)$$

and the common connected determinant kernel is

$$\mathfrak{g}_\Gamma = s(u(C) \oplus u(W)). \quad (37)$$

Its central line is generated by (23).

For $Y_{a,b} = aN_C + bN_W$, the top-form and local determinant conditions share the factor $3a + 2b$; the cubic obstruction has the remaining positive quadratic factor. Evaluation of the same exterior character gives vanishing $SU(3)^3$, $SU(3)^2Y$, $SU(2)^2Y$, grav^2Y , and Y^3 coefficients, as well as the corresponding $B - L$ anomalies; the locked even chirality gives the vanishing $SU(2)$ global parity class. The cubic non-Abelian cancellation follows from exterior duality and is not a rank discriminator; the one-block $I(J) = \{2\}$ alternative has nonzero cubic $SU(3)^3$ coefficient.

Proposition 3.18 (Determinant trivialization) *A gauge-invariant connection-compatible trivialization of the chiral determinant line removes its local vertical curvature obstruction and gauge-loop holonomy.*

The determinant-family interpretation is that of [13]; characteristic and residual flat/torsion conventions are represented by [49] and [50].

3.5 Elliptic quotient stacks and algebraic family module

For comparison with nonprimitive positive-degree branches, let $L_\Gamma \simeq \mathcal{O}(n)$ and let I be the active principal-harmonic set. Put $m = |I|$, $S_1 = \sum_{\ell \in I} \ell$, and $S_2 = \sum_{\ell \in I} \ell^2$. Proposition 3.15 gives $d_\ell = n\ell + 1$, $r_{n,I} = nS_1 + m$, $T_{n,I} = nS_2 + S_1$, and raw centered weights $h_\ell = r_{n,I}\ell - T_{n,I}$. For a two-grade set $I = \{p, q\}$ with $p < q$, the primitive cocharacter is $-(d_q/g_0)P_p + (d_p/g_0)P_q$, where $g_0 = \gcd(d_p, d_q)$, and its finite intersection has order $\text{lcm}(d_p, d_q)$.

Corollary 3.19 (Harmonic cocharacter comparisons) For $I = \{1, 2\}$ one obtains

$$D_\Gamma^{(n)} = 2P_C + P_W, \quad H_{\Gamma, (n)}^{\deg} = (n+1)P_C - (2n+1)P_W, \quad (38)$$

$$\mu_{N_n} \simeq \mu_{2n+1} \times \mu_{n+1}, \quad N_n = (2n+1)(n+1). \quad (39)$$

For $(n, I) = (1, \{1, 2, 3\})$ the primitive weights are $(-11, -2, 7)$ on $E_2 \oplus E_3 \oplus E_4$, and the finite intersection has order $\gcd(22, 6, 28) = 2$.

Corollary 3.20 (Parity-exhausted degree comparison) If, within the positive-degree two-channel comparison, the W -side factor in (39) is exhausted by an order-two parity factor, then $n = 1$, $N_n = 6$, and the two factors are μ_3 and μ_2 .

The Eisenstein scalar unit group is $\mathbb{Z}[\omega]^\times \simeq \mu_6$. Its action gives the quotient stacks $\mathbb{X}_6 := [E_\omega/\mu_6]$ and $\mathbb{X}_3 := [E_\omega/\mu_3]$. The stabilizers of the μ_6 action are μ_6 at O , μ_3 on $E[\varpi] \setminus \{O\}$, and μ_2 on $E[2] \setminus \{O\}$; hence $\mathbb{X}_6$ has tubular type $(6, 3, 2)$. The μ_3 action fixes the three points of $E[\varpi]$, so $\mathbb{X}_3$ has type $(3, 3, 3)$. These elliptic quotient descriptions agree with the tubular equivariantization of [51]. The Smith forms give $\text{Tor } L(6, 3, 2) \simeq \mathbb{Z}_6$ and $\text{Tor } L(3, 3, 3) \simeq \mathbb{Z}_3^2$; no symplectic identification of the latter with the level-three theta phase space is assumed. The weight-two orbit $E[2] \setminus \{O\}$ and the ramified fixed subgroup $E[\varpi]$ are retained separately below.

Reduction of Eisenstein units modulo $\varpi = 1 - \omega$ gives $1 \rightarrow \mu_3 \rightarrow \mu_6 \rightarrow \mathbb{F}_3^\times \simeq \mu_2 \rightarrow 1$. Its unique primary subgroups give $\mu_6 \simeq \mu_3 \times \mu_2$, matching $Z(SU(C)) \times Z(SU(W))$ and the primary decomposition of $\text{Tor } L(6, 3, 2)$. If $x_{(6)}, x_{(3)}, x_{(2)}$ are its exceptional generators, their degrees are $(1, 2, 3)$. The degree-zero elements $u_2 = x_{(2)} - 3x_{(6)}$ and $u_3 = x_{(3)} - 2x_{(6)}$ have orders 2 and 3, and $\vec{\omega}_\mathbb{X} = -u_2 - u_3$. Thus $\text{ord}(\vec{\omega}_\mathbb{X}) = 6$; for the Auslander-Reiten translation $\tau_\mathbb{X}$ and derived Serre functor $\mathbb{S}_\mathbb{X} = \tau_\mathbb{X}[1]$, one has $\tau_\mathbb{X}^6 = \text{id}$ and $\mathbb{S}_\mathbb{X}^6 = [6]$. The color and parity factors, and the primary shifts used below, are readings of the same order-six datum.

Lemma 3.21 (Tubular exceptional-degree relation) In $L(6, 3, 2)$ one has

$$x_{(6)} + x_{(3)} = x_{(2)} - \vec{\omega}_\mathbb{X}. \quad (40)$$

Hence $\mathcal{O}(x_{(6)} + x_{(3)}) \simeq \tau_\mathbb{X}^{-1}\mathcal{O}(x_{(2)})$, whose degree shadow is $1 + 2 = 3$.

Proof The definitions above give $\vec{\omega}_\mathbb{X} = 5x_{(6)} - x_{(2)} - x_{(3)}$. Using the string-group relation $2x_{(2)} = 6x_{(6)}$ gives (40); the line-bundle statement follows from $\tau_\mathbb{X}E = E(\vec{\omega}_\mathbb{X})$. $\square$

The centered carrier gives the same order through $\lambda_\Gamma(z) := z^{H_\Gamma^{\deg}}$, since Proposition 3.15 gives $\lambda_\Gamma(U(1)) \cap (SU(C) \times SU(W)) \simeq \mu_6$. The projective reduction $\text{Pic}(\mathbb{CP}_\Gamma^1)/6\text{Pic}(\mathbb{CP}_\Gamma^1)$ is cyclic with generator $[L_\Gamma]$; after the primitive degree orientation is fixed, $[L_\Gamma] \mapsto \vec{\omega}_\mathbb{X}$ gives the full six-cycle presentation $\mathcal{T}_\Gamma^{(6)}$. Let S_6 denote its oriented shift and put $\mathcal{A}_6 := \text{End}(\ell^2(\mathcal{T}_\Gamma^{(6)}))$.

Lemma 3.22 (Tubular two-block rigidity comparison) Let a coprime two-block split with ranks $c > w \geq 2$ have centered finite shadow of order $N = cw$ as in Proposition 3.15. If this cyclic shadow is required to agree with an elliptic scalar automorphism group and the associated signature (N, c, w) is tubular, then $(N, c, w) = (6, 3, 2)$ and the two-arrow Kronecker Euler value is 1.

Proof Tubularity gives $N^{-1} + c^{-1} + w^{-1} = 1$. Substitution of $N = cw$ gives $(c-1)(w-1) = 2$, hence $(c, w) = (3, 2)$ and $N = 6$. The two-arrow Kronecker Euler form is then $(c-w)^2 = 1$. $\square$

For the principal A_2 datum, the exponents $(1, 2)$, the Weyl invariant degrees $(2, 3)$, the Borel-Weil ranks $(2, 3)$, and $|W(A_2)| = 6$ therefore form the same rank-degree-order package. Conversely, the rank-two reflection degrees $(2, m)$ with $|I_2(m)| = 2m$ satisfy the tubular relation only for $m = 3$. The same signature has the Coxeter-star reading $T(2, 3, 6) = \tilde{E}_8$: its arm lengths are $(w-1, c-1, N-1) = (1, 2, 5)$, and deletion of the central node gives $A_1 \oplus A_2 \oplus A_5$. This is used only as an affine root-system comparison; no gauge extension is inferred.

Proposition 3.23 (Weight-two family orbit and two-linearization envelope) *Since 2 is inert in $\mathbb{Z}[\omega]$, $E[2] \simeq \mathbb{F}_4$ and the Coxeter action is simply transitive on $E[2]^\times := E[2] \setminus \{O\}$. Under the full μ_6 action, every point of $E[2]^\times$ has stabilizer μ_2 , so $E[2]^\times \simeq \mu_6/\mu_2$. For a parity character ε of the stabilizer,*

$$\mathcal{H}_{\text{fam}}^{(\varepsilon)} := \text{Ind}_{\mu_2}^{\mu_6} \varepsilon \simeq \ell^2(E[2]^\times; \varepsilon), \quad \dim \mathcal{H}_{\text{fam}}^{(\varepsilon)} = 3. \quad (41)$$

Here $\ell^2(E[2]^\times; \varepsilon)$ denotes sections of the equivariant line associated with the stabilizer character. Retaining both stabilizer characters gives the canonical two-linearization envelope

$$\mathcal{H}_\Gamma^{(6)} := \mathcal{H}_{\text{fam}}^{(+)} \oplus \mathcal{H}_{\text{fam}}^{(-)} \simeq \ell^2(\mathcal{T}_\Gamma^{(6)}) \simeq \mathbb{C}[\mathbb{Z}_6], \quad (42)$$

with covariant algebra $\mathcal{A}_6 \simeq M_6(\mathbb{C})$. For the canonical primary shifts $S_3 = S_6^2$ and $S_2 = S_6^3$,

$$\mathcal{H}_\Gamma^{(6)} \simeq \mathbb{C}[\mathbb{Z}_3] \otimes \mathbb{C}[\mathbb{Z}_2], \quad (43)$$

and $P_\varepsilon = (\mathbf{1} + \varepsilon S_2)/2$ gives

$$P_\varepsilon \mathcal{A}_6 P_\varepsilon \simeq M_3(\mathbb{C}), \quad P_\varepsilon \mathcal{H}_\Gamma^{(6)} \simeq \mathcal{H}_{\text{fam}}^{(\varepsilon)}. \quad (44)$$

The orbit and each rank-three module in (41) are structural. The six-dimensional envelope (42) retains both stabilizer linearizations as a canonical presentation object and is not the physical family core before entry D. Entry D selects the μ_2 linearization compatible with the physical Clifford parity; light-sector attachment additionally requires finite-shadow covariance and the Callias-Schur gap below. For parity-factor phase θ_2 and edge weight $\nu_2 > 0$, the parity Laplacian has eigenvalues $2\nu_2(1 \mp \cos \theta_2)$; at $e^{i\theta_2} = \varepsilon$ the selected character has eigenvalue 0 and the complementary character has eigenvalue $4\nu_2$.

For the selected split, Proposition 3.15 and (29) show that r, T, T^*, d_F, k_Y are dependent readings: $r = N-1$, $T^* = N+1$, $T = N+2$, $d_F = 2^{N-2}$, and $k_Y = 2(N-1)/N$. The family rank is fixed independently by (41). For later completed readings their distinct realizations are retained in the structural dictionary

$$(c, w, r, T, T^*, N, f, d_F, k_Y) = (3, 2, 5, 8, 7, 6, 3, 16, 5/3), \quad f := \dim \mathcal{H}_{\text{fam}}^{(\varepsilon)}, \quad d_F := \dim F_\Gamma^{\text{even}}. \quad (45)$$

3.6 Primitive graded Standard-Model reconstruction theorem

Theorem 3.24 (Primitive graded Standard-Model reconstruction) *On the primitive two-grade branch, let Γ satisfy the structural inputs A and B of Table 1. Proposition 3.6 fixes the A_2 -equianharmonic datum. Theorem 3.12 independently reconstructs the rank-five marked projective carrier (22) and its degree fidelity, while Theorem 3.10 gives the matching theta ranks. Under the structural closure C, the determinant-preserving carrier group is (27), the chiral finite module is (29), and the canonical Abelian gradings are (31) and (32). The global form is (28). Proposition 3.23 gives the structural rank-three family torsor $E[2]^\times$ and the canonical two-linearization envelope (42). Under entry D, one μ_2 parity linearization is selected for physical attachment.*

Proof The projective carrier and its minimality are Theorem 3.12; the theta-rank agreement is Theorem 3.10. The carrier group, Clifford module, and determinant statements follow from (26), Proposition 3.16, and Proposition 3.17. Proposition 3.23 gives the weight-two family orbit and its two parity linearizations; entry D fixes the physical one. $\square$

Proposition 3.25 (Rational elliptic-surface and Mordell-Weil realization) *Over $[u : v] \in \mathbb{P}^1$, let S_{632} be the relatively minimal elliptic surface associated with*

$$y^2 = x^3 + uv^2(u - v)^3. \quad (46)$$

Its singular fibers are II , IV , and I_0^ at $u = 0$, $v = 0$, and $u = v$, respectively, so $e(S_{632}) = 12$ and $T_{\text{fib}} \simeq A_2 \oplus D_4$. The surface is rational and is the relatively minimal desingularization of a diagonal order-six product quotient of $E_\omega \times E_\omega$. Moreover, $\text{MW}(S_{632}) \simeq \mathbb{Z}[\omega]P_0$, $\text{MWL}(S_{632}) \simeq \frac{1}{6}A_2$, and a minimal section satisfies $h(P_0) = 1/3$.*

Proof The stated Kodaira fibers follow from the vanishing orders in (46) by the standard elliptic-surface calculus [52]; their Euler numbers sum to 12. On the chart $v = 1$, with $t = u/v$, the cyclic base change $s^6 = t(t - 1)^3$, followed by $x = s^2X$ and $y = s^3Y$, reduces (46) to $Y^2 = X^3 + 1$. The smooth compactification of the covering curve has genus one and an order-six deck transformation with a fixed point, hence is equianharmonic. For a deck generator $s \mapsto \zeta s$, the constant-fiber coordinates transform as $(X, Y) \mapsto (\zeta^{-2}X, \zeta^{-3}Y)$, which is multiplication by ζ in the uniformization of E_ω . Thus S_{632} is the relatively minimal desingularization of the diagonal order-six product quotient, and sections are μ_6 -equivariant maps $E_\omega \rightarrow E_\omega$. Equivariance fixes O , and $\text{End}(E_\omega) = \mathbb{Z}[\omega]$, giving $\text{MW}(S_{632}) \simeq \mathbb{Z}[\omega]P_0$. The $D_4 \oplus A_2$ entry of the rational elliptic-surface classification [53] independently gives trivial torsion and $\text{MWL}(S_{632}) \simeq \frac{1}{6}A_2$, fixing $h(P_0) = 1/3$; the height scales by the Eisenstein norm. $\square$

Lemma 3.26 (Split I_0^* family torsor) For the split I_0^* fiber of S_{632} , the four outer components are indexed by $E_\omega[2]$, with the zero-section component corresponding to O . The remaining three are therefore indexed by $E_\omega[2]^\times$, and the order-three CM action ρ induces the triality cycle on them.

Proof The local quotient uses the order-two subgroup $\{\pm 1\} \subset \mu_6$, whose fixed set on the constant equianharmonic fiber is $E_\omega[2]$. Resolution gives the four outer D_4 components, while ρ acts simply transitively on the three nonzero two-torsion points. $\square$

Remark 3.27 (Isotropy parity and triality reflection) The isotropy subgroup μ_2 fixes $E_\omega[2]$ pointwise and acts through the character ε on the equivariant line in (41). Complex conjugation κ instead induces a nontrivial reflection of the three nonidentity arms; together with ρ its permutation action generates $\text{Out}(D_4) \simeq S_3$. Entry D therefore remains the independent conditional parity linearization.

Remark 3.28 (Three A_2 readings) The A_2 datum of Proposition 3.6 is the principal root datum reconstructed from the normal source, the A_2 summand of T_{fib} is the root lattice of the IV fiber, and $\text{MWL}(S_{632}) \simeq \frac{1}{6}A_2 \simeq (A_2(2))^\vee$ is the scaled dual height lattice. In the standard E_8 frame of the rational elliptic-surface classification [53], $(A_2 \oplus D_4)^\perp \simeq A_2(2)$; hence $A_2 \oplus D_4 \oplus A_2(2) \subset E_8$ has index 12 and quotient $\mathbb{Z}_2 \oplus \mathbb{Z}_6$. This quotient contains two cyclic order-six subgroups exchanged by an automorphism, so it does not canonically select the physical μ_6 without an additional marking. The fiber and narrow Mordell-Weil lattices remain complementary pieces of the same E_8 frame, while the principal A_2 is reconstructed independently.

3.7 Finite torsor and Heisenberg response geometry

The response geometry below is applied after Theorem 3.24 has fixed the carrier, global form, structural rank-three family torsor, and canonical two-linearization order-six envelope. It supplies finite transport and spectral response data without changing the structural selection.

The projective-color torsor has the intrinsic elliptic realization $\mathcal{T}_\Gamma := E[2]^\times$. After the retained orientation marking, the Picard projection $q_3(\mathcal{T}_\Gamma^{(6)})$ gives the same $\mathbb{Z}_3$ torsor presentation, with translation induced by the Coxeter action:

$$\mathcal{T}_\Gamma = E[2]^\times \simeq q_3(\mathcal{T}_\Gamma^{(6)}), \quad \tau_\Gamma = \rho|_{E[2]^\times}. \quad (47)$$

Its finite-core representation is multiplicity free by Proposition 2.3, hence

$$\mathcal{H}_{\text{cen}} = \ell^2(E[2]^\times) \simeq \mathbb{C}[\mathbb{Z}_3] \simeq \mathcal{H}_{\text{fam}}^{(\varepsilon)} \quad (48)$$

up to the one-dimensional parity linearization. Here central refers to commutation with the retained charge and exterior algebra before completed family blocks are inserted.

The underlying real regular module splits as

$$\mathbb{R}[\mathbb{Z}_3] = \mathbb{R}\mathbf{1} \oplus A_{2,\mathbb{R}}, \quad A_{2,\mathbb{R}} := \left\{ (x_0, x_1, x_2) \in \mathbb{R}^3 : \sum_a x_a = 0 \right\}. \quad (49)$$

The second summand is the real Coxeter plane of the A_2 datum fixed in Proposition 3.6; its integral lift is $I \simeq A_2 \simeq \mathbb{Z}[\omega]$, the augmentation lattice in $L_{\text{fam}} := \mathbb{Z}[C_3]$. The projections P_1 and P_T , the Hodge quotient, the factor $\sqrt{2}$, and the real Fourier frame are readings of this decomposition; the low-symbol map (75) fixes its complex finite response without a microscopic transport normalization.

Proposition 3.29 (Regular family lattice and central cycle lift) *The extension of the invariant line by I in L_{fam} is the non-split unimodular class: for $t = e_0 + e_1 + e_2$, $a = e_0 - e_1$, $b = e_1 - e_2$, the split lattice $L_0 = \mathbb{Z}t \oplus \mathbb{Z}a \oplus \mathbb{Z}b$ has index three and $L_{\text{fam}} = L_0 + \mathbb{Z}(t + 2a + b)/3$. After complexification every unitary lift of the marked projective regular cycle is $T_\mu = \mu S_{\text{reg}}$, $\mu \in U(1)$, with $T_\mu^3 = (\det T_\mu)\mathbf{1}$; the fixed discrete Fourier transform sends it to $\mu Z^{\pm 1}$ in the character frame. Thus the integral C_3 cycle and a non-torsion central determinant phase are compatible parts of one lift.*

For a Hermitian coefficient system $\mathcal{E}_\Gamma$, the edge complex is

$$0 \longrightarrow C^0(\mathcal{T}_\Gamma, \mathcal{E}_\Gamma) \xrightarrow{d_\Gamma} C^1(\mathcal{T}_\Gamma, \mathcal{E}_\Gamma) \longrightarrow 0, \quad (50)$$

with

$$(d_\Gamma \xi)_{a+1} = U_a \xi_a - e^{i\theta_\Gamma} \xi_{a+1}. \quad (51)$$

The associated finite Hodge module is

$$\mathfrak{H}_\Gamma^{\text{tor}} = (C^\bullet(\mathcal{T}_\Gamma, \mathcal{E}_\Gamma), d_\Gamma, D_\Gamma^{\text{tor}}, \Delta_\Gamma, \mathcal{A}_{\text{fam}}), \quad \mathcal{A}_{\text{fam}} = \text{End}(\mathcal{H}_{\text{cen}}). \quad (52)$$

The graph-Hodge convention is represented by [54], the unitary bundle-Laplacian comparison by [55], and finite spectral-module comparisons by [56].

The transported Wilson defect is

$$\mathcal{W}_\Gamma = e^{-3i\theta_\Gamma} U_2 U_1 U_0. \quad (53)$$

With $\mathbb{E}_{Z,\Gamma}$ the trace-preserving expectation onto the center of the transported charge algebra, torsor closure is

$$R_{\text{tor}}^\Gamma = \mathcal{W}_\Gamma - \mathbb{E}_{Z,\Gamma}(\mathcal{W}_\Gamma). \quad (54)$$

Its flat subbranch is

$$R_{\text{flat}}^\Gamma = \mathcal{W}_\Gamma - \mathbf{1}. \quad (55)$$

On a simple central summand and in a cycle-trivial coefficient frame, $d_\theta = S - e^{i\theta_\Gamma} \mathbf{1}$. The scalar Laplacian is

$$\Delta_\theta = 2\mathbf{1} - e^{-i\theta_\Gamma} S - e^{i\theta_\Gamma} S^\dagger. \quad (56)$$

For the scalar eigenvalues $2\nu_\Gamma(1 - \cos(2\pi k/3 - \theta_\Gamma))$, the unscaled elementary symmetric functions are

$$\sum_k \mu_k = 6, \quad \sum_{i < j} \mu_i \mu_j = 9, \quad \prod_k \mu_k = |1 - \mathcal{W}_\Gamma|^2, \quad (57)$$

and

$$\text{Disc}_\mu \chi_{\Delta_\theta} = 27w_\Gamma(4 - w_\Gamma), \quad w_\Gamma := |1 - \mathcal{W}_\Gamma|^2. \quad (58)$$

Thus the scalar spectrum is simple away from $\mathcal{W}_\Gamma = \pm 1$, with endpoint spectra $(0, 3, 3)$ and $(1, 1, 4)$. On the scalar cycle-trivial branch, $\det d_\theta = 1 - e^{3i\theta_\Gamma}$, whereas $\det \Delta_\theta = |\det d_\theta|^2$, so the Gram determinant retains the magnitude and loses the oriented Wilson phase. For vector-bundle coefficients, let $w_j = e^{i\phi_j}$ be the eigenvalues of $\mathcal{W}_\Gamma$. A unitary vertex gauge reduces the complex to scalar twisted three-cycles, giving

$$\chi_{\Delta_\Gamma}(\mu) = \prod_j \left(\mu^3 - 6\mu^2 + 9\mu - |1 - w_j|^2 \right), \quad \det \Delta_\Gamma = |\det(\mathbf{1} - \mathcal{W}_\Gamma)|^2. \quad (59)$$

Hence the Gram spectrum determines the multiset $\{|1 - w_j|^2\}$ and therefore the Wilson eigenvalues up to permutation and independent conjugations $w_j \leftrightarrow \bar{w}_j$; these orientations remain pre-Gram data. If $\delta_j = \text{Arg } w_j \in [-\pi, \pi]$, then $\lambda_{\min}(\Delta_\Gamma) = 4 \min_j \sin^2(|\delta_j|/6)$, so the gap closes exactly when $1 \in \text{Spec } \mathcal{W}_\Gamma$.

Moreover $\dim H^0 = \dim H^1 = \dim \ker(1 - \mathcal{W}_\Gamma)$. On a central rank- r summand write $\mathcal{W}_\Gamma = e^{i\chi_\Gamma} \mathbf{1}$. For a closed microscopic loop γ , put $n_a := \text{wind}_\gamma \det U_a$ and $n_\theta := (2\pi)^{-1} \oint_\gamma d\theta_\Gamma$. Taking determinants of the Wilson product gives

$$n_W := \frac{1}{2\pi} \oint_\gamma d\chi_\Gamma = -3n_\theta + \frac{1}{r} \sum_{a=0}^2 n_a \in \mathbb{Z}. \quad (60)$$

Hence $\sum_a n_a \equiv 0 \pmod{r}$ whenever $n_\theta \in \mathbb{Z}$; for the physical rank-three block the summed edge-determinant winding is divisible by three. The low-symbol channel in (75) is traceless and therefore fixes only the projective family transport. Proposition 3.29 separates this integral projective cycle from its central determinant coefficient, which may be supplied by determinant-line/Berry or global transport without requiring the full non-torsion operator to preserve $\mathbb{Z}[C_3]$ literally.

3.8 Eisenstein norms and finite Heisenberg coefficient calculus

Let Λ_E be the A_2 lattice fixed in (17). In an oriented Eisenstein basis its Gram matrix for $2N_E$ is $\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$. Put $\varpi = 1 - \omega$. Since $\varpi^2 = -3\omega$, the mod-three coefficient algebra is $\Lambda_E/3\Lambda_E \simeq \mathbb{F}_3[\varpi]/(\varpi^2)$ and has the canonical nonzero proper ideal $\mathfrak{r}_3 := \varpi\Lambda_E/3\Lambda_E$. In the oriented basis $(1, \omega)$, the alternating pairing of $a + b\omega$ and $c + d\omega$ is $ad - bc \pmod{3}$.

Lemma 3.30 (Ramified Jordan line) Multiplication by ϖ on $\Lambda_E/3\Lambda_E$ is a nonzero rank-one nilpotent N_ϖ with $N_\varpi^2 = 0$. Hence $\text{im } N_\varpi = \ker N_\varpi = \mathfrak{r}_3$. For $\xi = 1 + \varpi$, multiplication is $1 + N_\varpi$, has order three, and fixes $\mathfrak{r}_3$ pointwise.

Commensurability of the squared singular values of d_θ , equivalently of the eigenvalues of $\Delta_\theta = d_\theta^\dagger d_\theta$, is the additional arithmetic condition

$$\mu_1 = \frac{p^2}{n}, \quad \mu_2 = \frac{q^2}{n}, \quad \mu_3 = \frac{r^2}{n}, \quad n \in \mathbb{Z}_{>0}, \quad 0 < p \leq q \leq r, \quad \gcd(p, q, r) = 1. \quad (61)$$

It is imposed after reconstruction of the finite cycle. Torsor closure alone does not discretize θ_Γ : in the cycle-trivial coefficient frame $U_0 = U_1 = U_2 = \mathbf{1}$, (54) vanishes for every θ_Γ , while the eigenvalues of (56) vary continuously. A pre-carrier derivation of (61) therefore requires an additional integral holonomy condition. The identification $\mathbb{Z}[\mathbb{Z}_3]/(1 + S + S^2) \simeq \Lambda_E$ is the integral lift used here; over $\mathbb{Q}$ it is the nontrivial factor of $\mathbb{Q}[\mathbb{Z}_3] \simeq \mathbb{Q} \oplus \mathbb{Q}(\omega)$, while integrally the two factors remain glued at the prime above 3.

Proposition 3.31 (Primitive commensurate torsor spectrum) *Under (61),*

$$r = p + q, \quad n = \frac{p^2 + pq + q^2}{3}. \quad (62)$$

With $\omega = e^{2\pi i/3}$,

$$\xi = \frac{p - q\omega}{1 - \omega} \in \mathbb{Z}[\omega], \quad N_E(\xi) = \xi \bar{\xi} = n, \quad (63)$$

and, after cyclic labeling and orientation,

$$e^{i\theta_\Gamma} = -\omega^2 \frac{\xi}{\bar{\xi}}, \quad |1 - e^{i\theta_\Gamma}|^2 = \frac{p^2}{n}, \quad |\omega - e^{i\theta_\Gamma}|^2 = \frac{q^2}{n}, \quad |\omega^2 - e^{i\theta_\Gamma}|^2 = \frac{r^2}{n}. \quad (64)$$

Writing $\tau_k := \text{Tr}_{\mathbb{Q}(\omega)/\mathbb{Q}}(\xi \omega^k)$, the same orientation gives $(\tau_0, \tau_1, \tau_2) = (r, -p, -q)$. Hence $\sum_k \tau_k = 0$, $\sum_k \tau_k^2 = 6N_E(\xi)$, and the commensurate packet is $(|\tau_1|, |\tau_2|, |\tau_0|; N_E(\xi))$; in this parametrization $r = p + q$ is the trace shadow of $1 + \omega + \omega^2 = 0$. The ratio $\xi/\bar{\xi}$ is the Hilbert-90 parametrization of the rational norm-one torus of $\mathbb{Q}(\omega)/\mathbb{Q}$. Primitive denominators are norms of primitive Eisenstein elements modulo units and conjugation. Inert rational factors force common divisibility, while the ramified prime above 3 is excluded by the primitive congruence in this parametrization. The first simple class therefore has norm 7, with

$$\text{spec } \Delta_\theta = \left\{ \frac{1}{7}, \frac{16}{7}, \frac{25}{7} \right\}. \quad (65)$$

Proof Substitution in (57) gives the vanishing Heron product; the ordering yields $r = p + q$, and then (62). Since $n \in \mathbb{Z}_{>0}$, (62) gives $(p - q)^2 \equiv 0 \pmod{3}$, hence $p \equiv q \pmod{3}$; this is exactly the condition $\xi \in \mathbb{Z}[\omega]$, whose Eisenstein norm is n . The phase formula is direct. The nonsimple primitive endpoint is $(1, 1, 2; 1)$. For a simple spectrum $p < q$; the first split rational prime in the Eisenstein norm form is 7, represented by $2 - \omega$, and the corresponding primitive pair is $(1, 4)$. This gives (65) and $w_\Gamma = 400/343$. $\square$

Corollary 3.32 (Mordell-Weil height and commensurate denominator) For a primitive commensurate class ξ of Proposition 3.31, put $P_\xi := \xi P_0$ in $\text{MW}(S_{632})$. Then

$$n = N_E(\xi) = 3h(P_\xi). \quad (66)$$

Proof The first equality is Proposition 3.31. The second follows from $h(P_0) = 1/3$ in Proposition 3.25 and the Eisenstein-norm scaling of the Mordell-Weil height. $\square$

Corollary 3.33 (Rank-one Hodge-Eisenstein lift) The primitive rank-one lifts form the affine Eisenstein tower $\xi_m = 1 + m\varpi = (m+1) - m\omega$, $m \in \mathbb{Z}_{\geq 0}$, with $N_E(\xi_m) = 3m^2 + 3m + 1$. Their Mordell-Weil realization is the specialization of (66) to $P_m := \xi_m P_0$. The corresponding ambient rank is $R = 3m + 2$ and the packet is $(1, 3m+1, 3m+2; 3m^2 + 3m + 1)$. The endpoint $m = 0$ is nonsimple, so the first nontrivial simple element is $\xi_1 = 2 - \omega$ and gives

$$(p_H, q_H, r_H; n_H) = (1, 4, 5; 7). \quad (67)$$

Its oriented cubic phase is $e^{i\delta_{\xi_m}} = -(\xi_m/\bar{\xi}_m)^3$, invariant under $\xi_m \mapsto \epsilon\xi_m$ for $\epsilon \in \mu_6$; the selected $m = 1$ value is used in (209).

Corollary 3.34 (Galois-rank compatibility and rigidity) Suppose a commensurate Eisenstein class is compatible with the selected rank readings in the sense $\text{Tr } \xi = r$ and $N_E(\xi) = T^*$. For $(r, T^*) = (5, 7)$ it is unique up to conjugation, $\xi \in \{2 - \omega, 3 + \omega\}$. More generally, if $r = c + w$, $T^* = c + 2w$ with integers $c > w \geq 2$, existence of such a class forces $(c, w) = (3, 2)$ and hence $(c - 1)(w - 1) = 2$.

Proof For $\xi = a + b\omega$, the trace condition gives $b = 2a - r$ and the norm condition gives $3a^2 - 3ar + r^2 = T^*$; at $(r, T^*) = (5, 7)$ this gives $a = 2, 3$. In general, $(\text{Tr } \xi)^2 \leq 4N_E(\xi)$ gives $(c + w)^2 \leq 4(c + 2w)$. If $w \geq 3$, then $c \geq w + 1$ makes the left-hand side minus the right-hand side at least $4w^2 - 8w - 3 > 0$. Thus $w = 2$, and then $c^2 \leq 12$ with $c > 2$, so $c = 3$. $\square$

For an odd trace $t = \text{Tr}(a + b\omega)$ one has $N_E(a + b\omega) = (t^2 + 3b^2)/4$, so the minimal primitive norm is $(t^2 + 3)/4$, unique up to conjugation. The selected trace sequence therefore begins $5 \mapsto 7 \mapsto 13$; in particular, 13 is the unique minimal primitive norm at trace 7 and on the selected split satisfies $13 = r^2 - 2cw = c^2 + w^2$.

Lemma 3.35 (Minimal-trace cyclic companion) Let $A \in M_2(\mathbb{Z})$ have odd trace t and determinant $n = (t^2 + 3)/4$. Then there is a primitive $v \in \mathbb{Z}^2$ for which (v, Av) is an integral basis. In that basis A is, up to orientation and transpose convention, the companion matrix $\begin{pmatrix} 0 & -n \\ 1 & t \end{pmatrix}$. Thus an integral minimal-trace cell supplies the marked zero-diagonal cyclic splitting used in Lemma 2.5.

Proof For $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $v = (x, y)^T$, the form $q_A(v) := \det(v, Av) = cx^2 + (d - a)xy - by^2$ has discriminant $(\text{Tr } A)^2 - 4\det A = -3$. It is primitive, and the primitive definite integral form of discriminant -3 is, up to sign and $SL_2(\mathbb{Z})$ equivalence, $x^2 + xy + y^2$; hence q_A represents ± 1 . The resulting cyclic basis gives the stated matrix by Cayley-Hamilton. $\square$

The arithmetic rigidity is independent of the normal-carrier reconstruction. Corollary 3.36 supplies the coefficient-side rank witness $(3, 2)$ directly from the Galois/theta datum; no identification of its character spaces with the normal $SU(2)$ modules is used. For the selected orientation $\xi = 2 - \omega = 1 + \varpi$, Lemma 3.30 gives the order-three fixed line directly, while the same corollary gives the corresponding theta and ramified Weil readings. Torsor closure fixes the central Wilson defect without identifying this coefficient-lattice symmetry with the microscopic U_a . The integral status still requires preservation of the protected commensurability class in Remark 2.4.

For $H_\Gamma = \nu_\Gamma \Delta_\theta$, the spectral entropy starts as $S_{\beta, \Gamma} = \log 3 - (\beta\nu_\Gamma)^2 + (\beta\nu_\Gamma)^3(w_\Gamma - 2)/3 + O((\beta\nu_\Gamma)^4)$, tends to 0 at the flat endpoint and to $\log 2$ at $\mathcal{W}_\Gamma = -1$.

The mod-three Heisenberg reduction is obtained by $Ze_a = \omega^a e_a$ and $ZS = \omega SZ$. In the oriented basis above, the monomial commutator is ω^{ad-bc} , and

$$\mathcal{W}_3 := \mathbb{C}_\omega[\Lambda_E/3\Lambda_E] \simeq \mathbb{C}_\omega[\mathbb{Z}_3^2] \simeq \mathcal{A}_{\text{fam}} = M_3(\mathbb{C}), \quad T = \sum_{a, b \in \mathbb{Z}_3} t_{ab} Z^a S^b. \quad (68)$$

This is the standard irreducible finite-Heisenberg representation. Put $\mathcal{A}_a = \text{span}\{Z^a S^b : b \in \mathbb{Z}_3\}$, so $\mathcal{W}_3 = \mathcal{A}_0 \oplus \mathcal{A}_1 \oplus \mathcal{A}_2$ and $\mathcal{A}_a \mathcal{A}_b \subset \mathcal{A}_{a+b}$. The vertex and transport MASAs are the two complementary finite polarizations and the discrete Fourier transform is their Weil intertwiner.

The degree-three theta group on E_ω gives the irreducible finite-Heisenberg module used below, with ramified polarization $E[\varpi]$. A cyclic level-six theta structure presents the full coefficient module (42) and its two parity linearizations; the theta-group representation is standard [38].

Corollary 3.36 (Norm-seven CM and ramified Weil reading) For the least simple class $\xi = 2 - \omega$ of Proposition 3.31, one has $N_E(\xi) = 7$ and $[\xi]^* \mathcal{O}_E(O) \simeq L_7 := \mathcal{O}_E(7O)$, and

$$q_* L_7 \simeq \mathcal{O}(2) \oplus \mathcal{O}(1)_\omega \oplus \mathcal{O}(1)_{\omega^2}. \quad (69)$$

where the subscripts denote the ρ -characters. Hence $H^0(E, L_7)$ has the marked decomposition $C \oplus W_\sigma \oplus W_{-\sigma}$ with section ranks $3|2|2$. On the coefficient side this gives the ordered rank witness $(c_\xi, w_\xi) := (3, 2)$, satisfying $\text{Tr } \xi = c_\xi + w_\xi$ and $N_E(\xi) = c_\xi + 2w_\xi$ for the selected class. After one orientation is retained, (69) leaves projective degrees 2 and 1; under the marking of Theorem 3.10 these agree with the levels $R_{\Gamma,2} = E_3$ and $R_{\Gamma,1} = E_2$ in (22). This is a level compatibility and does not identify the theta character spaces with the normal $SU(2)$ modules. Modulo three, multiplication by ξ is the order-three symplectic transvection fixing $\mathfrak{r}_3$. In the symmetrized Weyl frame $W(a, b) := \omega^{ab} Z^a S^b$, the oriented fixed vector $(1, 2)$ and any Weil lift U_ξ satisfy

$$W(1, 2) = \omega^2 Z S^2, \quad Z U_\xi Z^{-1} U_\xi^{-1} = \omega Z^2 S. \quad (70)$$

Thus the commensurate phase of Proposition 3.31 and the discrete neutral Weyl factor in (70) are the archimedean and ramified readings of the same Eisenstein element.

Proof The degree-seven CM class has $h^0(L_7) = 7$. A basis of $H^0(E, L_7)$ is $\{1, x, y, x^2, xy, x^3, x^2 y\}$; the ρ -character multiplicities are 3, 2, 2. The three eigensheaves of $q_* L_7$ have rank one and total degree four, so their section dimensions give (69). In the basis $(1, \omega)$, multiplication by ξ modulo three is $\begin{pmatrix} 2 & 1 \\ 2 & 0 \end{pmatrix}$, whose fixed line is generated by $(1, 2)$. The two identities in (70) follow from the Weyl product rule and Weil covariance; the projective scalar of U_ξ cancels in the commutator. $\square$

Proposition 3.37 (Finite Weyl-Fourier projection calculus) *The trace-preserving expectations onto the complementary transport and vertex MASAs are*

$$\mathbb{E}_{\text{circ}}(T) = \frac{1}{3} \sum_{r=0}^2 S^r T S^{-r}, \quad \mathbb{E}_{\text{vert}}(T) = \frac{1}{3} \sum_{r=0}^2 Z^r T Z^{-r}. \quad (71)$$

Their non-circulant Dirichlet form is

$$\Pi_{\text{noncirc}}(T) = T - \mathbb{E}_{\text{circ}}(T) = \sum_{a=1,2} \sum_b t_{ab} Z^a S^b, \quad \text{Tr}_{\text{cen}}([d_\theta, T]^\dagger [d_\theta, T]) = 3 \|\Pi_{\text{noncirc}}(T)\|_{\text{HS}}^2. \quad (72)$$

The dual identity uses Z and $\mathbb{E}_{\text{vert}}$. The discrete Fourier transform intertwines the two MASAs; its canonical real frame is $f_0, \sqrt{2} \text{Re } f_1, \sqrt{2} \text{Im } f_1$ for $f_k = 3^{-1/2} (\omega^{kr})_r$. For real x , $P_1 = (\mathbf{1} + S + S^2)/3$, $P_\Gamma = \mathbf{1} - P_1$, and $\Sigma_{\text{H}} = \log(\|P_1 x\|^2 / \|P_\Gamma x\|^2)$. Parseval gives $\mathcal{Q}_{\text{H}}(x) := \|x\|^2 / |\sum_a x_a|^2 = (1 + e^{-\Sigma_{\text{H}}})/3$. The transport and vertex seminorm kernels are $\mathbb{C}[S]$ and $\mathbb{C}[Z]$, respectively. Weyl monomials have zero trace unless both degrees vanish modulo three; adjunction reverses both degrees up to phase, while transposition preserves the $\mathcal{A}_a$ grading and reverses the shift exponent.

The vertex frame is used for the charged-lepton baseline, the transport frame for the common quark baseline and circulant neutral denominator, and the canonical real Fourier frame for the leading resolved neutral basis.

The Hermitian generators used later are

$$K_Z = \frac{Z - Z^\dagger}{2i}, \quad H_S(\vartheta) = e^{-i\vartheta} S + e^{i\vartheta} S^\dagger, \quad G = \frac{S - S^\dagger}{\sqrt{3}}. \quad (73)$$

Their commutator is

$$[K_Z, H_S(\vartheta)] = \frac{\sqrt{3}}{2} \left(-\omega e^{-i\vartheta} Z S + \omega^2 e^{i\vartheta} Z S^2 - \omega^2 e^{-i\vartheta} Z^2 S + \omega e^{i\vartheta} Z^2 S^2 \right). \quad (74)$$

Choose an order-three element $g_3 \in SU(2)$ whose action on E_3 has characters $1, \omega, \omega^2$. The induced conjugation grading on $V_1 \oplus V_2 \simeq \text{End}_0(E_3)$ is written $V_1 \oplus V_2 = \mathcal{H}_0 \oplus \mathcal{H}_1 \oplus \mathcal{H}_2$, with $\mathcal{H}_r$ of degree r . Let $\sigma_{\text{low}} : V_1 \oplus V_2 \oplus V_3 \rightarrow V_1 \oplus V_2$ be the $SU(2)$ -equivariant projection and let $\mathcal{J}_\Gamma : E_3 \rightarrow \mathcal{H}_{\text{cen}}$ be the calibrated μ_3 -equivariant unitary identification of the regular character modules.

Proposition 3.38 (V_1/V_2 low-symbol decomposition) *The low-symbol map satisfies*

$$\kappa_\Gamma = \text{Ad}_{\mathcal{J}_\Gamma} \circ \sigma_{\text{low}}, \quad \ker \kappa_\Gamma = V_3, \quad (75)$$

with

$$\kappa_\Gamma(\mathcal{H}_0) = \mathcal{A}_0 \cap \text{End}_0(\mathcal{H}_{\text{cen}}), \quad \kappa_\Gamma(\mathcal{H}_r) = \mathcal{A}_r, \quad r = 1, 2. \quad (76)$$

This follows from the Clebsch-Gordan kernel V_3 and the regular order-three character decomposition of E_3 . Thus (75) is the retained low-symbol bridge between the principal $V_1 \oplus V_2$ decomposition and the Weyl-graded finite response. The integral A_2 lattice is already fixed by (17); the calibration of J_Γ remains presentation data.

Multiplication by S^b preserves every $\mathcal{A}_r$; for invertible $X \in \mathcal{A}_1$, the Weyl relation gives $XSX^{-1}S^{-1} = \omega \mathbf{1}$. The normalized torsor trace gives

$$\mathrm{tr}_{\mathrm{cen}}(\Delta_\theta) \mathrm{Tr}_{\Lambda^{\mathrm{even}} V}(Y^2) = \frac{20}{3}. \quad (77)$$

The family entropy is $\mathcal{C}_\Gamma(\rho) = D(\rho \| \mathbb{E}_{\mathrm{circ}} \rho)$ with the finite-dimensional convention of [22]. Proposition 3.37 and (10) give $0 \leq \mathcal{C}_\Gamma \leq \log 3$ and

$$\mathcal{C}_\Gamma(\mathbf{1}/3 + \varepsilon X) = \frac{\varepsilon^2}{2} \mathcal{L}_\Gamma(X)^2 + O(\varepsilon^3), \quad (78)$$

where $\mathcal{L}_\Gamma^2$ is the left-hand side of (72). The corresponding Dirichlet form is $\gamma \mathcal{L}_\Gamma^2/3$. On the full $\mathbb{Z}_6$ shadow, the commuting parity and projective-color expectations obey (11); parity compression removes the active $\log 2$ contribution and leaves the projective-color capacity.

4 Alena-Codazzi Realization of the A_2 -Graded Reconstruction Data

The structural theorem of Section 3 starts from a primitive projective link, a degree-resolved two-channel normal source, relative Gauss-local charges, and the reconstruction inputs of Table 1. Proposition 3.6 is algebraic once the oriented normal metric and the two active source types are present; the present section supplies the geometric source, charges, and conditional transport from a current-residual Alena-Codazzi collar. The faithful central coefficient reconstruction in entry B and the Clifford/determinant closure in entry C remain reconstruction inputs.

Split conservation, scalar nondegeneracy, multiplier closure, thin-core regularity, torsor admissibility, and the spectral gap are hypotheses of the realization layer. Under them, the compact-leaf construction gives the source and thin-core worldline, the moment expansion gives the Euler-graded $V_1 \oplus V_2$ pair, the Gauss identities give its two boundary charges, and torsor-admissible transport gives the finite edge data. When the realization criterion is satisfied, the source Euler grading is preserved by the moment map and agrees with the projective carrier level on the minimal marked support by Theorem 3.12. The transferred matrix elements are treated in Section 5.

4.1 Current-Codazzi collar and multiplier closure

The residual part of the Alena-type current branch is written as

$$\mathcal{L}_{\mathrm{cr}} = \varphi p_\Lambda, \quad \varphi = 1 - \zeta^2 - \mu_\zeta R_\omega. \quad (79)$$

Here ζ is the translational-current amplitude, $\mu_\zeta = \mu_\zeta(\rho_\zeta)$ is positive, and R_ω is the normalized vorticity response. The branch-stress interpretation is supplied by the Alena Tensor identification [57]. The current and vorticity slots are those of [58]. The continuum, variational, and branch-potential inputs are represented by [59], [60], and [61]. Only the current, vorticity, stress, and Codazzi coefficients are used in the realization below.

The translational current is

$$J_{\mathrm{tr}}^\mu = p_\Lambda \zeta^2 U^\mu, \quad \nabla_\mu^{(k)} J_{\mathrm{tr}}^\mu = 0. \quad (80)$$

The frozen scalar block is assumed non-degenerate:

$$V_\zeta''(\rho_\zeta) + R_\omega \mu_\zeta''(\rho_\zeta) > 0. \quad (81)$$

This condition gives local persistence of the scalar collar inside the selected current branch.

The product-rule Hilbert response of (79) is decomposed as $T_{\mu\nu}^{\mathrm{cr}} = \Xi_{\mu\nu} + \varphi Y_{\mu\nu}$. The first term records the response of the residual scalar density. The second carries the branch-response coefficient associated with p_Λ . The punctured collar is split-conserved when

$$\nabla_\mu^{(k)} \Xi^{\mu\nu} = 0, \quad \nabla_\mu^{(k)} (\varphi Y^{\mu\nu}) = 0. \quad (82)$$

The split is used before the normal two-jet is extracted.

Let $\tau = k^{\mu\nu} Y_{\mu\nu}$ and put $B_{\mu\nu} = Y_{\mu\nu} - \tau k_{\mu\nu}/3$. The indices in this subsection range over the four-dimensional collar. The trace-adjusted B is regarded as a symmetric tensor with coefficients in a real multiplier line. In a local trivialization $B \mapsto e^f B$ and $\theta \mapsto \theta - df$; for the multiplier below $\theta = d \log |\varphi|$, so this coefficient-line

connection is locally flat. A nonzero scalar multiplier gives $A_{\mu\nu} = \varphi B_{\mu\nu}$, with $C_{\alpha\mu\nu}^B = \nabla_\alpha^{(k)} B_{\mu\nu} - \nabla_\mu^{(k)} B_{\alpha\nu}$ and $\theta = d \log |\varphi|$. The corresponding line-valued Codazzi equation is

$$C_{\alpha\mu\nu}^B + \theta_\alpha B_{\mu\nu} - \theta_\mu B_{\alpha\nu} = 0. \quad (83)$$

If B is invertible, contraction fixes the local connection form

$$\theta_\alpha^B = -\frac{1}{3}\beta^{\mu\nu} C_{\alpha\mu\nu}^B, \quad (84)$$

where the coefficient is $(\dim \Omega - 1)^{-1}$ on the four-dimensional collar.

Proposition 4.1 (Current-Codazzi closure) *On a connected non-degenerate source-free collar, a scalar Codazzi multiplier exists exactly when (83) holds with $\theta = \theta^B$ and the multiplier-line connection is exact. If $\theta^B = df_B$, then*

$$\varphi = C_\varphi e^{f_B}, \quad \zeta^2 = 1 - \mu_\zeta R_\omega - C_\varphi e^{f_B}, \quad (85)$$

and current conservation reduces to

$$\begin{aligned} U(\mu_\zeta R_\omega) + C_\varphi e^{f_B} \theta^B(U) &= \left(1 - \mu_\zeta R_\omega - C_\varphi e^{f_B}\right) \\ &\quad \times \left(\nabla_\mu^{(k)} U^\mu + U(\log p_\Lambda)\right). \end{aligned} \quad (86)$$

At principal normal level, current conservation contributes the scalar symbol $u \mapsto \langle \xi, u \rangle$ from V_1 to V_0 . The tensor-valued $V_1 \rightarrow V_1 \oplus V_2$ completion required by the finite Clifford response must therefore be supplied by the tensorial Hilbert/stress residual; its constitutive identification is not assumed here.

Proposition 4.2 (Principal Clifford rigidity) *Use the normal metric to identify $N_\Gamma^* \simeq N_\Gamma$, identify $V_1 \simeq \mathfrak{so}(N_\Gamma)$ by $u \mapsto [u]_\times$, and put $V_2 = \text{Sym}_0^2(N_\Gamma)$ with the Frobenius metrics. For $0 \neq \xi \in N_\Gamma$, let the Hodge-dual principal Codazzi block be $A_\xi b = [\xi]_\times b$. Every bilinear $SO(3)$ -equivariant tensor-valued completion, linear in ξ and u , has the form $C_\xi u = a[\xi \times u]_\times + c(\xi \odot u)_0$, where $\xi \odot u = (\xi \otimes u + u \otimes \xi)/2$. If the target metric is $G_{h,\beta} = hP_{V_1} + \beta P_{V_2}$ with $h, \beta > 0$, isotropy of the Codazzi Gram is equivalent to $h = 3\beta$. Under this condition, $C_\xi(V_1) \perp A_\xi(V_2)$ is equivalent to $6a + c = 0$. Requiring the same Clifford constant on the standard Frobenius source metrics then gives $c^2 = 3$. Hence, up to the common sign,*

$$C_\xi(u) = \pm\sqrt{3} \left[(\xi \odot u)_0 - \frac{1}{6}[\xi \times u]_\times \right], \quad \Sigma_\xi(u, b) := C_\xi(u) + A_\xi b, \quad (87)$$

and

$$\Sigma_\xi^\dagger G_{3\beta, \beta} \Sigma_\xi = \beta |\xi|^2 \mathbf{1}, \quad \Sigma_\xi^\dagger G_{3\beta, \beta} \Sigma_\eta + \Sigma_\eta^\dagger G_{3\beta, \beta} \Sigma_\xi = 2\beta \langle \xi, \eta \rangle \mathbf{1}. \quad (88)$$

Proof For $b \in V_2$,

$$\|A_\xi b\|_{h,\beta}^2 = \beta |\xi|^2 \|b\|_{\mathbb{F}}^2 + \frac{h-3\beta}{2} |b\xi|^2, \quad (89)$$

so isotropy gives $h = 3\beta$. In that metric, $\langle C_\xi u, A_\xi b \rangle = -\frac{\beta}{2}(6a+c)\langle \xi \times u, b\xi \rangle$, while $6a+c=0$ gives $\|C_\xi u\|_{3\beta, \beta}^2 = (\beta c^2/3)|\xi|^2 \|u\|_{\mathbb{F}}^2$. Matching (89) gives $c^2 = 3$, and polarization gives (88). $\square$

Corollary 4.3 (Principal Clifford spectrum and conditioning) For $0 \neq \xi$, write $R_i = P_{V_i} \Sigma_\xi$. An adapted orthonormal frame gives orthogonal source projections $E_1 + E_2 = \mathbf{1}$ with $\text{rank } E_1 = 3$, $\text{rank } E_2 = 5$, $R_1^\dagger R_1 = |\xi|^2 E_1/3$, and $R_2^\dagger R_2 = |\xi|^2 E_2$. Hence, for arbitrary $h, \beta > 0$,

$$N_{\text{Pr}}(h, \beta) = |\xi|^2 \left(\frac{h}{3} E_1 + \beta E_2 \right), \quad \kappa(G_{h,\beta}^{1/2} \Sigma_\xi) = \sqrt{\frac{\max\{h/3, \beta\}}{\min\{h/3, \beta\}}}. \quad (90)$$

Thus the principal tensor symbol is invertible for $\xi \neq 0$, and the Clifford ratio $h = 3\beta$ is exactly the isotropic, optimally conditioned point with the single singular value $\sqrt{\beta}|\xi|$.

Corollary 4.4 (Scalar-current Clifford obstruction) Suppose the tensor block of Proposition 4.2 is enlarged by an independent scalar residual $j_\xi(u) = \kappa \langle \xi, u \rangle$ with positive weight g_J . Its source quadratic form acquires the term $g_J |\kappa|^2 |\langle \xi, u \rangle|^2$. For $\kappa \neq 0$ this changes only the longitudinal eigenvalue and therefore cannot belong to the same isotropic Clifford Gram. The scalar current equation may instead be imposed in defining the physical slice.

Lemma 4.5 (Self-adjoint constitutive first-jet normal form) Let $m \in V_1 \simeq N_\Gamma$ and $M \in V_2 = \text{Sym}_0^2(N_\Gamma)$. Put $S_mu = (m \odot u)_0$, $B_M u = Mu$, $C_M b = (Mb + bM)_0$, and $R_M u = [M, [u]_\times]$; their adjoints are $S_m^\dagger b = bm$ and $R_M^\dagger b = 2 \text{ax}([M, b])$, where ax is inverse to $u \mapsto [u]_\times$. Every $SO(3)$ -equivariant formally self-adjoint zero/first frozen-moment endomorphism of $V_1 \oplus V_2$ is uniquely of the form

$$Z_H(m, M) = \begin{pmatrix} z_1 \mathbf{1} + b_{11} B_M & a S_m^\dagger + b_{12} R_M^\dagger \\ a S_m + b_{12} R_M & z_2 \mathbf{1} + b_{22} C_M \end{pmatrix}, \quad (91)$$

with six real coefficients $z_1, z_2, a, b_{11}, b_{12}, b_{22}$.

Proof The decomposition $\text{Sym}^2(V_1 \oplus V_2) = 2V_0 \oplus V_1 \oplus 3V_2 \oplus V_3 \oplus V_4$ leaves two scalar blocks, one m -linear cross block, and three M -linear V_2 copies. The maps displayed above are nonzero representatives of these multiplicity spaces, and self-adjointness pairs the two cross blocks. $\square$

This normal form fixes the allowed tensor contractions but not their coefficients. Current conservation and exact multiplier Codazzi closure do not determine them: in a frozen flat tangent model, $A_{\mu\nu} := \partial_\mu \partial_\nu f$ is Codazzi for arbitrary smooth f , while $\varphi Y_{\mu\nu} := A_{\mu\nu} - (\text{tr } A) k_{\mu\nu}$ is divergence-free. The same pointwise freedom remains on a curved collar after choosing $df = 0$ at the base point. A constitutive Hilbert/stress relation, or its mixed variation, is therefore required.

Proposition 4.6 (Cross-block isotropy rigidity) Let $m \neq 0$, put $r = |m|$, $e = m/r$, $Q_e = e \otimes e - \frac{1}{3} \mathbf{1}$, and $X = a S_m + b_{12} R_M$ with $ab_{12} \neq 0$. Then $X^\dagger X = \sigma^2 \mathbf{1}_{V_1}$ holds if and only if $M = \nu Q_e$ for some $\nu \neq 0$ and

$$\frac{b_{12}\nu}{ar} = \pm \frac{\sqrt{3}}{6}, \quad \sigma^2 = \frac{2}{3} a^2 r^2. \quad (92)$$

Proof By $SO(3)$ covariance set $e = e_3$ and absorb the relative factor into $N = (b_{12}/ar)M$. Writing N as a general traceless symmetric 3×3 matrix, the five traceless entries of $(S_{e_3} + R_N)^\dagger (S_{e_3} + R_N)$ reduce to the conditions that all off-diagonal entries vanish, the first two diagonal entries agree, and their squared value is $1/108$. Thus $N = \pm(6\sqrt{3})^{-1} \text{diag}(1, 1, -2)$, which is equivalent to (92). $\square$

Corollary 4.7 (Two-level constitutive selector and centered grading) On the branch of Proposition 4.6, the canonical two-level representative is selected by requiring (91) to have two global eigenvalues. Then

$$b_{22} = -3b_{11}, \quad z_2 - z_1 = \frac{5}{3} b_{11} \nu, \quad a^2 r^2 = 18 b_{11}^2 \nu^2, \quad \left| \frac{b_{12}}{b_{11}} \right| = \sqrt{\frac{3}{2}}. \quad (93)$$

The eigenvalue multiplicities are 3 and 5. If P_3 is the multiplicity-three spectral projection and $P_5 = \mathbf{1} - P_3$, then

$$P_{V_1} P_3 P_{V_1}|_{V_1} = \frac{4}{7} \mathbf{1}_{V_1} - \frac{1}{7} e \otimes e, \quad \text{Spec}(P_{V_1} P_3 P_{V_1}|_{V_1}) = \left\{ \frac{4}{7}, \frac{4}{7}, \frac{3}{7} \right\}. \quad (94)$$

Writing $p = b_{11} \nu$, the two eigenvalues are $\lambda_3 = z_1 - 10p/3$ and $\lambda_5 = z_1 + 11p/3$, and hence

$$Z_H - \frac{\text{Tr } Z_H}{8} \mathbf{1} = \frac{7p}{8} (3P_5 - 5P_3). \quad (95)$$

Thus $(\lambda_5 - \lambda_3)/p = 7$. On a fixed sign branch, $e \mapsto P_3(e)$ is the oriented S^2 lift of the quadratic $e \mapsto Q_e$ orbit.

Proof The axial $SO(2)$ decomposition consists of one weight-zero mixed block, two identical weight-one mixed blocks, and a pure weight-two block of multiplicity two. Equality of their two-level spectra gives (93); direct spectral projection gives (94). Centering the resulting 3|5 spectrum gives (95). $\square$

Proposition 4.8 (Projector-first selector certificate) Let P be a rank-three orthogonal projection and put $A_P = P_{V_1} P P_{V_1}|_{V_1}$. The selected overlap packet is characterized by

$$\mathcal{R}_{\text{CM}}(P) := 49A_P^2 - 49A_P + 12\mathbf{1} = 0, \quad \text{Tr } A_P = \frac{11}{7}. \quad (96)$$

These conditions are equivalent to $\text{Spec } A_P = \{3/7, 4/7, 4/7\}$. If a microscopic self-adjoint response has P as an invariant rank-three subspace and its complementary diagonal spectra are separated by a positive gap, the selected projector is realized without requiring degeneracy inside either block. Conversely, for $\delta = \|\mathcal{R}_{\text{CM}}(P)\| < 1/4$, every eigenvalue lies within $r(\delta) = (1 - \sqrt{1 - 4\delta})/14$ of $\{3/7, 4/7\}$; if $|\text{Tr } A_P - 11/7| + 3r(\delta) < 1/7$, the multiplicities are 1|2. For off-diagonal norm ε and inter-block separation g , $\varepsilon < g/2$, the actual isolated projector differs from the reference projector by at most $\varepsilon/(g - \varepsilon)$.

Corollary 4.9 (Selector packet and norm-seven two-projection rigidity) Put $\Gamma_{\text{sel}} = P_5 - P_3$, $\Gamma_{\text{gr}} = P_{V_1} - P_{V_2}$, and $U_{\text{rel}} = \Gamma_{\text{sel}}\Gamma_{\text{gr}}$. Equation (94) gives

$$\{\delta_{\text{ov}}, |\text{Tr}(\Gamma_{\text{sel}}\Gamma_{\text{gr}})|, \text{Tr}(P_5 P_{V_2})\} = \left\{ \frac{1}{7}, \frac{16}{7}, \frac{25}{7} \right\}, \quad \delta_{\text{ov}} = \frac{1}{7}. \quad (97)$$

Thus $(p_{\text{sel}}, q_{\text{sel}}, r_{\text{sel}}; n_{\text{sel}}) = (1, 4, 5; 7)$ and Corollary 3.34 gives

$$\xi_{\text{sel}} \in \{2 - \omega, 3 + \omega\}. \quad (98)$$

The relative operator satisfies

$$\chi_{U_{\text{rel}}}(x) = \frac{(x+1)^2(7x^2 - 2x + 7)(7x^2 + 2x + 7)^2}{343}. \quad (99)$$

Its nontrivial eigenvalues have real parts $\pm 1/7$, have infinite order, and are not integral lattice automorphisms. Clearing the denominator gives the conductor-eight order $\mathbb{Z} + 8\mathbb{Z}\omega$ with integral closure $\mathbb{Z}[\omega]$; the positively oriented simple numerator is $\omega(2 - \omega)^2$. Hence the projector contains a canonical CM arithmetic image, while the finite torsor map remains marked. In the reconstructed Eisenstein field, 7 is also the least denominator of a primitive non-torsion rational point on the norm-one torus.

Lemma 4.10 (Unique minimal filtered A_2 constitutive pencil) Use the orthonormal principal embedding $\iota(u, b) = i[u]_{\times}/\sqrt{2} + b$, put $G_{\text{Cl}} = 3P_{V_1} + P_{V_2}$ and $\Gamma_{\text{gr}} = P_{V_1} - P_{V_2}$, and for $H = \iota(m, M)$ set $\mathcal{J}_H := G_{\text{Cl}}^{-1/2}\{H, \cdot\}_0 G_{\text{Cl}}^{-1/2}$ and $\mathcal{K}_M := G_{\text{Cl}}^{-1/2}\Gamma_{\text{gr}}i[M, \cdot]G_{\text{Cl}}^{-1/2}$. In the pencil $c_J\mathcal{J}_H + c_K\mathcal{K}_M$ the noncentral coefficients satisfy $b_{11} = -c_J/3$, $b_{22} = c_J$, $a = -c_J/\sqrt{3}$, and $b_{12} = c_K/\sqrt{6}$. The selector (93) therefore gives $|c_K| = |c_J|$. Modulo a common scale, the noncentral selected pencil is $\mathcal{J}_H \pm \mathcal{K}_M$; no continuous Jordan/Lie mixing parameter remains. For $|m| = 1$, compatibility fixes $|\nu| = 1/\sqrt{6}$ and the relative grade shift is fixed by the same two-level condition.

Proof The anticommutator gives the B_M , C_M , and S_m blocks, while the grading-twisted commutator gives the R_M block. Clifford whitening fixes their relative coefficients, and (93) fixes the remaining ratio. $\square$

On the literal degree-fidelity subbranch $M = \mu(m \odot m)_0$, the selected relation $M = \nu Q_e$ gives $|\nu| = |\mu||m|^2$. Homogeneity of the filtered selector gives $|\nu| = |m|/\sqrt{6}$, so every nonzero selected source lies on the normalized shell $|m| = (\sqrt{6}|\mu|)^{-1}$. This removes one source-amplitude modulus without fixing an absolute action unit.

Proposition 4.11 (Topological separation of the constitutive projector) Let $\mathcal{E}_3 = \text{Ran } P_3 \rightarrow S^2$. The map $J_e = P_{V_1}|_{\text{Ran } P_3(e)}$ has squared singular values $3/7, 4/7, 4/7$ by (94), and is invertible for every e . Hence $\mathcal{E}_3 \simeq_{SO(3)} S^2 \times V_1$; after complexification its determinant has $c_1 = 0$. Since P_3 and dP_3 are real symmetric, $\text{Tr}(P_3 dP_3 \wedge dP_3) = 0$ pointwise. Consequently a rank-preserving complex gap homotopy on S^2 cannot identify the complexified constitutive bundle with a rank-three Callias bundle satisfying (164). Moreover, an $SO(3)$ -equivariant integral rank-two local system over $S^2 = SO(3)/SO(2)$ is trivial, since every continuous homomorphism $SO(2) \rightarrow GL_2(\mathbb{Z})$ is trivial. Integral CM monodromy therefore requires discrete boundary, torsor, theta, or singular-monodromy data.

For an oriented unit source axis e , $g_e^{\pm} = \exp(\pm 2\pi[e]_{\times}/3)$ has spectrum $\{1, \omega, \omega^2\}$ and order three. An orientation-preserving marked identification with the family cycle selects the branch in (98). After the trivial-character line and determinant are fixed, a C_3 -equivariant unitary identification is $\text{diag}(1, e^{i\phi}, e^{-i\phi})$, exactly the spin-one action of the axial stabilizer $SO(2)_e$. If the calibration extends to the protected regular lattice $L_{\text{fam}} \simeq \mathbb{Z}[C_3]$, then $\mathbb{Z}[C_3]^{\times} = \{\pm 1, \pm S, \pm S^2\} \simeq C_6$; fixing the positive trivial line leaves only C_3 translation, i.e. a torsor-origin change. Existence and preservation of the corresponding source-side regular lattice remain part of the boundary/Riesz realization, as does physical parity attachment. At cubic order, $\text{Sym}^2 V_2 = V_0 \oplus V_2 \oplus V_4$ contains no V_1 . A same-field $SO(3)$ -invariant scalar Hessian therefore has $b_{12} = 0$, as does the pure symmetric-cone Jordan Hessian. The completed adjoint high carrier has $\mathfrak{a}_W = \text{End}_0(E_2) \simeq V_1$ and $\mathfrak{a}_C = \text{End}_0(E_3) \simeq V_1 \oplus V_2$, so Schur-Feshbach elimination can generate the unique M -linear $V_1 \leftrightarrow V_2$ tensor line. Its scalar coefficient remains dependent on the microscopic couplings and denominators; the ratio in (93) is not fixed by mediator multiplicity alone.

The covariance $R_B(e^f B, \theta - df) = e^f R_B(B, \theta)$ is the change-of-trivialization law of this flat coefficient line. The residual is affine-linear in B plus the bilinear term $\theta \wedge B$, so its third differential vanishes. Consequently this principal multiplier sector has no independent intrinsic cubic vertex; cubic reduced terms arise through low-high elimination, path curvature, residual-metric variation, or other nonlinear residual components. On the principal normal fiber, Hodge duality sends the mixed second differential to $(\eta, b) \mapsto [\eta]_{\times} b$ from $V_1 \otimes V_2$ to $\text{End}_0(V_1) \simeq V_1 \oplus V_2$. The map is nonzero on the V_1 and V_2 summands; multiplicity one in the Clebsch-Gordan decomposition therefore gives kernel exactly V_3 , the same quotient as (75). Thus $(\delta B, \delta \theta) = (fB, -df)$ is the presentation-rescaling direction removed from the stationary normal block. Global multiplier existence is the trivial-holonomy condition, equivalently the vanishing of the periods of θ^B .

Proposition 4.12 (Principal multiplier source-Hodge sequence) *Let $B_0 \in \text{Sym}_0^2(N_\Gamma)$ be non-degenerate and $0 \neq \xi \in N_\Gamma^*$. Define*

$$(\mathcal{A}_\xi b)_{ijk} = \xi_i b_{jk} - \xi_j b_{ik}, \quad (\mathcal{M}_{B_0} \eta)_{ijk} = \eta_i (B_0)_{jk} - \eta_j (B_0)_{ik}. \quad (100)$$

Both maps are injective and

$$\text{Im } \mathcal{A}_\xi \cap \text{Im } \mathcal{M}_{B_0} = \text{span}\{\mathcal{A}_\xi B_0\} = \text{span}\{\mathcal{M}_{B_0} \xi\}. \quad (101)$$

Put $\mathcal{S}_{\xi, B_0} = \text{Im } \mathcal{A}_\xi + \text{Im } \mathcal{M}_{B_0}$ and $\Phi_{\xi, B_0}(b, \eta) = \mathcal{A}_\xi b + \mathcal{M}_{B_0} \eta$. Then $\dim \mathcal{S}_{\xi, B_0} = 7$, $\ker \Phi_{\xi, B_0} = \mathbb{R}(B_0, -\xi)$, and

$$\begin{aligned} 0 \longrightarrow \mathbb{R} \longrightarrow V_2 \oplus V_1 \xrightarrow{\Phi_{\xi, B_0}} \mathcal{S}_{\xi, B_0} \longrightarrow 0, \\ 0 \longrightarrow \mathbb{R} B_0 \longrightarrow V_2 \longrightarrow \mathcal{S}_{\xi, B_0} / \text{Im } \mathcal{M}_{B_0} \longrightarrow 0. \end{aligned} \quad (102)$$

For every positive residual metric, orthogonal reduction modulo $\text{Im } \mathcal{M}_{B_0}$ gives

$$\text{rank}(\Pi_{B_0}^\perp \mathcal{A}_\xi) = 4, \quad \ker(\Pi_{B_0}^\perp \mathcal{A}_\xi) = \mathbb{R} B_0. \quad (103)$$

Thus the source-level lost/surviving/ambient ranks are $(1, 4, 5)$ inside a seven-dimensional combined symbol space, with lost-to-surviving ratio $1 : 4$. The quotient is real rank four; comparison with a complex rank-four finite Hodge flag is therefore made through its complexification and must preserve the induced real structure. If $B_0 = \beta Q_e$ on the selected branch, $\beta \neq 0$, then the quotient is $\mathcal{P}_{H,e}^{(2)} = V_2 / \mathbb{R} Q_e$. With $P_\perp = \mathbf{1} - e \otimes e$, it has the stabilizer decomposition

$$V_2 = \mathbb{R} Q_e \oplus W_1(e) \oplus W_2(e), \quad \dim W_1 = \dim W_2 = 2, \quad (104)$$

where $W_1(e) = \{e \otimes v + v \otimes e : v \perp e\}$ and $W_2(e) = \{B : Be = 0, \text{tr}_{e^\perp} B = 0\}$. Rotations about e act on these planes with weights one and two.

Proof Injectivity follows directly from the alternating first index pair and trace-freeness. If $\mathcal{A}_\xi b = \mathcal{M}_{B_0} \eta$, multiplication by B_0^{-1} reduces the equality to $[\xi]_\times (b B_0^{-1}) = [\eta]_\times$; symmetry and trace-freeness then give $b = a B_0$ and $\eta = a \xi$. This proves (101), and the two exact sequences and (103) follow by rank. The stabilizer decomposition is the standard axial splitting of $\text{Sym}_0^2(N_\Gamma)$ after the Q_e line is removed. $\square$

Corollary 4.13 (Axial marking required for the source-Hodge bridge) *On the aligned branch, complexification of (104) has weights $\pm 1, \pm 2$ and restricts to $2\mathbb{C}_\omega \oplus 2\mathbb{C}_{\omega^2}$ under C_3 , so*

$$\text{End}_{C_3}(\mathcal{P}_{H,e}^{(2)} \otimes \mathbb{C}) \simeq M_2(\mathbb{C}) \oplus M_2(\mathbb{C}). \quad (105)$$

The real structure exchanges the character blocks. Full $SO(2)_e$ equivariance removes mixing between weights one and two and leaves two complex line amplitudes, reduced to $U(1)^2$ in the unitary case. A C_3 marking alone therefore cannot determine the source-Hodge amplitude ratio.

Proposition 4.14 (Marked multiplicative source-to-finite descent) *Writing $\iota_1(m) = i[m]_\times / \sqrt{2}$ and $\iota_2(M) = M$ for the two components of the embedding in Lemma 4.10, one has $(\iota_1(m)^2)_0 = -\iota_2((m \odot m)_0)/2$. Hence the aligned static harmonic-square line is generated by the first source. A finite Rees square is identified with this line only if the actual first- and second-response maps T_1, T_2 satisfy the projective multiplicative compatibility*

$$\begin{array}{ccc} V_1 \otimes V_1 & \xrightarrow{q_{\text{src}}} & V_2 \\ \downarrow T_1 \otimes T_1 & & \downarrow T_2 \\ \mathcal{A}_1 \otimes \mathcal{A}_1 & \xrightarrow{\mu_{\text{fin}}} & \mathcal{A}_2. \end{array} \quad (106)$$

For $m \neq 0$, the projective defect is

$$\Delta_{\text{sq}}(m)^2 = \|T_2 q_{\text{src}}(m)\|^2 - \frac{|\langle T_1(m)^2, T_2 q_{\text{src}}(m) \rangle|^2}{\|T_1(m)^2\|^2}. \quad (107)$$

Thus $\Delta_{\text{sq}}(m) = 0$ is the square-line compatibility test; direction-independence of the minimizing proportionality factor is the corresponding degree-normalization test. Source Euler degree, finite Weyl character, and branch Taylor order are independent gradings. On a common marked axial branch the aligned Hodge lost line has character zero, whereas a finite first Rees generator $X \in \mathcal{A}_1$ has $X^2 \in \mathcal{A}_2$. A grade-preserving descent therefore cannot identify these quotient kernels. A nonzero primitive finite second response also cannot be obtained solely from the linear image of the aligned generated source square, since its finite Rees subtraction would vanish.

The multiplier criterion is local on the connected non-degenerate collar. Global existence additionally requires the vanishing of the periods of θ^B . The realization theorem below uses exact multiplier closure on the selected punctured collar.

4.2 Compact-leaf source and conditional thin-core extraction

An explicit compact-leaf source model is supplied by a warped collar. Let Σ_g be a compact hyperbolic surface of genus $g \geq 2$, with metric γ . On a frozen collar interval, let $s = \tanh \chi \in (0, 1)$ be the leaf anisotropy and let $a(s) > 0$ be the warp factor. The two-eigenvalue Codazzi first integrals contain a constant F_0 . The warped-product curvature convention is the one of [11].

The normalized vorticity closure has the leaf equation

$$\Delta_\gamma \alpha = \sigma_\omega \frac{2a(s)^2 D_o^2}{c^2 s}, \quad \sigma_\omega = \pm 1, \quad (108)$$

where $D_o \neq 0$ is the frozen vorticity-flux scale. The right-hand side has nonzero mean. A one-core regularization is obtained by choosing a non-negative mollifier ρ_ε with unit integral and setting

$$\Delta_\gamma \alpha_\varepsilon = \sigma_\omega \frac{2a(s)^2 D_o^2}{c^2 s} + q_\varepsilon(s) \rho_\varepsilon, \quad (109)$$

where

$$q_\varepsilon(s) = -\sigma_\omega \frac{2a(s)^2 D_o^2}{c^2 s} \text{Area}_\gamma(\Sigma_g). \quad (110)$$

The right-hand side of (109) has zero integral. The resulting compact Poisson problem is therefore solvable, uniquely after the additive constant has been fixed, by the standard elliptic result represented by [33]. If $\rho_\varepsilon \rightarrow \delta_p$, the limit has one distributional leaf source at p . Transport of p along a timelike integral curve gives the local core.

Source-free will mean singular-source-free. The smooth zero-mode-compensating term in (109) is retained as part of the frozen leaf geometry. The principal boundary data are read after subtraction of a smooth local particular solution.

The same frozen sector has Codazzi gap

$$\Delta_C = -\frac{2F_0}{3-s}. \quad (111)$$

For $F_0 \neq 0$ and $0 < s < 1$, the collar lies in the non-degenerate two-eigenvalue sector used by the projective-link construction.

Proposition 4.15 (Resolved compact-leaf source collar) *Let Σ_g be a compact hyperbolic leaf with $g \geq 2$. For every frozen range with $0 < s < 1$, $D_o \neq 0$, and $F_0 \neq 0$, equation (109) has a unique smooth solution with zero mean. As $\rho_\varepsilon \rightarrow \delta_p$, its source converges to a one-core distributional closure. Away from the core, after subtraction of the smooth background, the collar is singular-source-free and has the nonzero gap (111).*

Proof The zero-mean condition follows from (110). Compact elliptic solvability gives the smooth solution. Weak convergence of the mollifier gives the distributional limit, and the gap statement is immediate from (111). $\square$

Smooth frozen-background terms do not alter the finite carrier when the singular associated-graded charges, their source grading, and the nonzero gap are unchanged. The support selection depends on the primitive link and the degree-resolved non-scalar source.

The compact-leaf model is calibrated to the current residual on a connected non-degenerate punctured collar. Let a smooth nonzero scalar ϕ_C make $\phi_C B_{\mu\nu}$ trace-adjusted Codazzi. Proposition 4.1 then gives $\theta^B = d \log |\phi_C|$ and $\phi_C = C_\phi e^{f_B}$. For $p_\Lambda > 0$ and a smooth U , choose positive initial data for

$$U(y) + y \left(\nabla_\mu^{(k)} U^\mu + U(\log p_\Lambda) \right) = 0 \quad (112)$$

on a flow collar, with the required sign of $1 - C_\phi e^{f_B} - y$. Setting $\zeta^2 = y$ and $\mu_\zeta R_\omega := 1 - C_\phi e^{f_B} - y$ gives

$$\phi_C = 1 - \zeta^2 - \mu_\zeta R_\omega, \quad \zeta^2 > 0, \quad \nabla_\mu^{(k)} \left(p_\Lambda \zeta^2 U^\mu \right) = 0, \quad (113)$$

and hence the current compatibility (86). This calibration is used below as part of the resolved compact-leaf realization.

Under the regularity hypothesis stated below, the thin-core limit identifies the primitive worldline component. The compactness input is supplied by integral-current compactness [62], in the geometric-measure form represented by [63]. Vortex concentration gives the comparison with [64]; the Jacobian-current formulation

is represented by [65]. A smooth auxiliary Riemannian metric h is fixed on the collar. All mass bounds and local flat compactness statements below are taken with respect to h , while timelikeness and orthogonal normal bundles are defined by the Lorentzian metric.

Theorem 4.16 (Conditional thin-core extraction) *Let $\{\mathcal{C}_\varepsilon\}$ be regularized Alena-Codazzi collars and let T_ε be the integral 1-current carried by their regularized cores. Assume:*

- (i) *the local masses of T_ε and ∂T_ε are uniformly bounded;*
- (ii) *$\partial T_\varepsilon = 0$ on the singular-source-free part of the collar;*
- (iii) *the integral linking charge converges to a primitive degree-one charge;*
- (iv) *the trace-adjusted Codazzi residual converges to zero in H_{loc}^{-1} away from every subsequential current limit;*
- (v) *the absolute frozen Codazzi gap is uniformly bounded below on compact subsets disjoint from the cores;*
- (vi) *the component selected by the primitive linking charge is a regular timelike multiplicity-one component of the limiting current.*

After passage to a subsequence, T_ε converges locally to an integral 1-current T without interior boundary on the source-free collar. The selected regular component is a timelike worldline Γ . Its oriented normal bundle gives (12), and the limiting charge gives (14). The trace-adjusted Codazzi equation holds distributionally away from $\text{spt } T$.

Proof Integral-current compactness gives a locally convergent subsequence. Boundary continuity gives the absence of interior boundary. The residual convergence gives the distributional Codazzi equation away from the limit. The selected regular multiplicity-one component is represented by a timelike worldline. The limiting primitive charge gives (14), and the uniform gap supplies the two-eigenvalue splitting on the punctured collar. $\square$

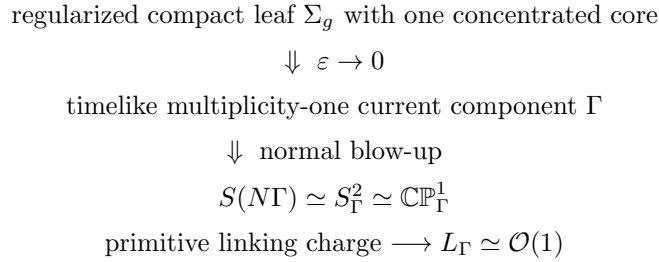


Fig. 1 Compact-leaf concentration, the thin-core worldline, and the resolved projective link.

4.3 Normal Rees source and Euler grading

The normal source filtration of Section 3 is realized geometrically by the associated-graded normal moments. Up to order two, its non-scalar pieces are a first moment $m \in N_\Gamma$ and a trace-free second moment $M_0 \in \text{Sym}_0^2(N_\Gamma)$; the scalar trace remains in the separated singlet channel.

Definition 4.17 (Second-order moment-resolved source) A defect source is second-order moment-resolved when a normalized core profile realizes the degree-one and degree-two normal Rees components by m and M_0 , and the singular degree-two coefficient of the principal trace-free Codazzi potential is $\kappa_2 M_0$ for fixed $\kappa_2 \neq 0$.

The harmonic realization of the two graded pieces is

$$\mathcal{M}(m, M_0) = (m \cdot \omega, \omega^T M_0 \omega) \in V_1[1] \oplus V_2[2]. \quad (114)$$

If $\mathbf{N}_{\text{mom}}$ has weights 1, 2 on $N_\Gamma \oplus \text{Sym}_0^2(N_\Gamma)$, equivariance of the homogeneous harmonic map gives

$$\mathcal{M} \circ \mathbf{N}_{\text{mom}} = \mathbf{N}_\partial \circ \mathcal{M}. \quad (115)$$

Hence the normal moment filtration realizes the same Euler grading used by the Rees reconstruction; frozen transport preserves this homogeneous degree on a source-free annulus. The two-channel locus $m \neq 0$, $M_0 \neq 0$ is open and dense, and a prescribed trace-free second moment is compatible with a positive core profile after adding a sufficiently large separated scalar trace. The resulting non-scalar two-jet is

$$\left(\text{gr } J_\perp^{\leq 2} \right)_{\text{ns}} = \mathcal{M}(m, M_0) \in V_1[1] \oplus V_2[2]. \quad (116)$$

Thus (115), faithful central reconstruction, and the Borel-Weil cutoff identify the realized normal Rees degrees with the marked section levels of Theorem 3.12.

4.4 Gauss-local charges and transported boundary data

Let j_{tr}^i be the principal normal reduction of (80). Extend the first moment m in the frozen normal frame by $\nabla^\perp m = 0$ and put $\mathbf{j}_1^i = m j_{\text{tr}}^i$. Let A_{ij} be the trace-free principal Codazzi representative.

Lemma 4.18 (Principal current and Codazzi-Gauss charges) *On a source-free principal normal annulus, $\nabla_i^\perp \mathbf{j}_1^i = 0$. The V_1 current-flux charge is*

$$Q_1(r) = \int_{S_r^2} \mathbf{j}_1^i n_i dS \in N_\Gamma \simeq V_1. \quad (117)$$

Writing $q_{\text{tr}}(r) = \int_{S_r^2} j_{\text{tr}}^i n_i dS$, one has $Q_1(r) = m q_{\text{tr}}(r)$ and

$$Q_1(r_1) - Q_1(r_0) = \int_{\Omega_{r_0, r_1}} \nabla_i^\perp \mathbf{j}_1^i dV = 0. \quad (118)$$

The trace-free Codazzi representative has a harmonic Hessian potential,

$$A_{ij} = \partial_i \partial_j u, \quad \Delta u = 0. \quad (119)$$

For $M \in \text{Sym}_0^2(N_\Gamma)$, let $H_M(x) = M_{ij} x^i x^j$. The V_2 charge is determined by

$$\langle Q_2(r), M \rangle = \int_{S_r^2} (u \partial_n H_M - H_M \partial_n u) dS. \quad (120)$$

It is independent of r , and $Q_2 = 0$ if and only if $M_0 = 0$.

Proof The principal current equation gives $\partial_i j_{\text{tr}}^i = 0$. Parallellity of m gives $\nabla_i^\perp \mathbf{j}_1^i = 0$, and Stokes' theorem gives (118).

For fixed k , the principal Codazzi equation makes $A_{ik} dx^i$ closed. Since $H^1(\mathbb{R}^3 \setminus \{0\}) = 0$, one obtains $A_{ik} = \partial_i v_k$. Symmetry gives $v_k = \partial_k u$, and trace-freeness gives $\Delta u = 0$. Green's identity gives the radius independence of (120). By Definition 4.17, the singular degree-two coefficient is $\kappa_2 M_0$ with $\kappa_2 \neq 0$, and the pairing on this term is a non-degenerate multiple of the standard V_2 inner product. $\square$

The first moment is computed with origin on the regular component selected by Theorem 4.16. A normal translation which changes this component changes the defect datum. On the locus $q_{\text{tr}} \neq 0$, the two charges are nonzero exactly when $m \neq 0$ and $M_0 \neq 0$.

On a curved collar, Q_1 is compared by normal parallel transport. The V_2 charge is compared by the Green identity for the first-order Codazzi operator $\mathfrak{C}$ and its formal adjoint:

$$\int_{\Omega_{r_0, r_1}} (\langle \mathfrak{C}A, \Psi \rangle - \langle A, \mathfrak{C}^* \Psi \rangle) = \mathcal{B}_{S_{r_1}}(A, \Psi) - \mathcal{B}_{S_{r_0}}(A, \Psi). \quad (121)$$

If $\mathfrak{C}A = 0$ and $\mathfrak{C}^* \Psi = 0$, the boundary pairing is independent of the linking radius. The transported current law is $\mathbf{P}_{r_0 \leftarrow r_1}^\perp Q_1(r_1) = Q_1(r_0)$.

The fixed charge pair generates the boundary-charge algebra used in Definition 3.1. Lemma 3.7 then gives central type projections in every finite sector representation. Thus the current flux realizes the V_1 sector and the Codazzi-Gauss pairing realizes the V_2 sector.

Theorem 4.19 (Dual Veronese source-jet quotients) *On the aligned selected branch, let $e(t) \subset S^2$ be unit speed, with $e = e(0)$, $v = \dot{e}(0)$, $n = e \times v$, and signed geodesic curvature κ_g , so $\dot{v} = -e + \kappa_g n$. For $Q(e) = e \otimes e - \mathbf{1}/3$, write $Q(e(t)) = Q_e + tA + t^2 B + O(t^3)$. Then $A = e \otimes v + v \otimes e$ and $B = v \otimes v - e \otimes e + \kappa_g(e \otimes n + n \otimes e)/2$. The source-Hodge quotient removes the Q_e component, giving $B_H = B + 3Q_e/2$. The generated tangent-square line is $G = (A^2)_0$ and direct contraction gives $\langle B, G \rangle = 0$, so the norm of B is unchanged by its removal. With $\chi_H = \sqrt{3}\|B_H\|/\|A\|^2$ and $\chi_R = \sqrt{3}\|B\|/\|A\|^2$,*

$$\chi_H^2 = \frac{3}{8}(1 + \kappa_g^2), \quad \chi_R^2 = \frac{3}{8}(4 + \kappa_g^2), \quad \chi_R^2 - \chi_H^2 = \frac{9}{8}. \quad (122)$$

Moreover,

$$B_H = \left(v \otimes v - \frac{1}{2} P_\perp \right) + \frac{\kappa_g}{2} (e \otimes n + n \otimes e), \quad p_{\text{src}} = \frac{1}{1 + \kappa_g^2}, \quad \Delta_{\text{src}} = \frac{3\kappa_g^2}{(1 + \kappa_g^2)^2}, \quad (123)$$

where the two terms lie in $W_2(e)$ and $W_1(e)$ of (104). The third jet satisfies

$$Q^{(3)} = -(4 + \kappa_g^2)A + 3\kappa_g(v \otimes n + n \otimes v) + \dot{\kappa}_g(e \otimes n + n \otimes e), \quad (124)$$

and, after removal of the A direction, $\|Q_{\perp A}^{(3)}/6\|^2 = \kappa_g^2/2 + \dot{\kappa}_g^2/18$. All Taylor jets in this theorem remain in the source module $V_2[2]$; the quantities χ_H , χ_R , and Δ_{src} are therefore source branch-motion diagnostics rather than finite Rees-degree assignments.

Proof The formulas follow by differentiating $Q(e(t))$ and using the Frenet equation on S^2 . Orthogonal projection onto $Q_e^\perp$ gives B_H . Direct contraction gives $\|Q_e\|^2 = 2/3$, $\|A\|^2 = 2$, $\|B_H\|^2 = (1 + \kappa_g^2)/2$, and $\|B\|^2 = 2 + \kappa_g^2/2$, which yields (122); the stabilizer and cubic statements follow from the same frame decomposition. $\square$

For a geodesic source-axis path, (122) gives $\chi_H = \sqrt{3/8}$ and $\chi_R = \sqrt{3/2}$, while (124) reduces to $Q^{(3)} = -4Q'$. Hence the pure geodesic selector orbit has no independent cubic source direction. A finite Picard polarization, finite Rees norm, or ordered cubic coefficient still requires the corresponding marked channel map.

4.5 Torsor-admissible transport and spectral admissibility

The charge pair must be transported compatibly with the affine finite shadow before the torsor complex of (50) is obtained. This is an additional boundary condition on the realized source data.

Definition 4.20 (Torsor-compatible boundary transport) An Alena-Codazzi collar is torsor-admissible when:

- (i) the V_1 charge (117) has a compact unitary phase calibration on linking spheres;
- (ii) the V_2 charge (120) is represented in a transported unitary Green-adjoint frame;
- (iii) the transported order-three normal-frame action, including the selected-axis representative when used, is marked to the translation in (47);
- (iv) the relative charge transport descends to the unitaries U_a in (51);
- (v) its completed low compression is compatible with (75);
- (vi) the edge transport and the low-high coupling are subcritical with respect to the separating Callias-Schur gap.

The transport is torsor-compatible when (54) vanishes. The flat subbranch additionally satisfies (55).

The first two entries fix unitary boundary normalizations. The third and fourth give the finite edge descent. The fifth controls the Schur-visible low symbol. Centrality of the Wilson defect gives a well-defined structural torsor cycle; its value remains a finite holonomy datum.

For torsor-admissible data, (118) and (121) induce the transports in (51). The scalar and mixed responses are then those of (56) and (72). The structural closure fixes centrality, while flatness selects the subbranch (55).

The analytic attachment requires a boundary-admissible normal representative with an isolated primitive cluster. The spectral projection is defined by a Riesz contour and persists under perturbations smaller than the spectral separation, in the standard sense of [8]. The open normal problem is compared with the Callias mechanism [9]; its geometric Fredholm form is represented by [10].

The primitive linking degree is locally constant under deformations which preserve the resolved transverse frame. The conditions $m \neq 0$ and $M_0 \neq 0$ are open in the finite moment space. A nonzero Codazzi gap remains nonzero on a fixed compact subcollar under sufficiently small constrained perturbations. If the low-high coupling has norm below one half of the Callias-Schur separation, the rank of the isolated low projection remains fixed.

Persistence is taken inside the current-Codazzi class. An unrestricted neighborhood theorem for the multiplier would require solvability of the nonlinear map $(\varphi, B) \mapsto C(\varphi B)$ together with the period condition for (84). Torsor admissibility remains a finite boundary-transport condition.

4.6 Microscopic normal family and marked Schur descent

Let $\mathcal{R}_{\text{mic}}$ collect the differentiable realization residuals preceding the torsor, completed Schur, and running closures. On a physical slice put $R_b := D\mathcal{R}_{\text{mic}}(\Phi_b)\Pi_{\text{ret}}$ and

$$L_b = G_{\text{mic}}^{1/2} R_b, \quad N_+(b) = L_b^\dagger L_b = R_b^\dagger G_{\text{mic}} R_b. \quad (125)$$

Lemma 4.21 (Equivariant residual metrics and exact zero modes) Let the unitary residual representation decompose as $\mathcal{W}_{\text{res}} = \bigoplus_{\alpha} V_{\alpha} \otimes M_{\alpha}$ with pairwise inequivalent irreducible V_{α} . Every positive equivariant Hermitian residual metric has the form

$$G_{\text{mic}} = \bigoplus_{\alpha} \mathbf{1}_{V_{\alpha}} \otimes g_{\alpha}, \quad g_{\alpha} > 0. \quad (126)$$

In particular, a multiplicity-one channel carries an independent positive scalar normalization. For every $G_{\text{mic}} > 0$ one has

$$\ker N_+(b) = \ker R_b. \quad (127)$$

Hence, on a constant-rank exact zero-mode branch and with the retained-domain Hermitian structure fixed, $P_b = P_{\ker R_b}$ and the connection $P_b dP_b$ are independent of the target residual metric. For a general isolated near-zero cluster this independence is absent: if the Riesz contour is fixed,

$$\dot{P} = \frac{1}{2\pi i} \oint_{\Gamma} (z - N_+)^{-1} \dot{N}_+ (z - N_+)^{-1} dz, \quad \dot{N}_+ = \dot{R}^{\dagger} G_{\text{mic}} R + R^{\dagger} \dot{G}_{\text{mic}} R + R^{\dagger} G_{\text{mic}} \dot{R}. \quad (128)$$

For multiplicity-one channels, put $A_{\alpha} = R_{\alpha}^{\dagger} R_{\alpha}$. If the A_{α} commute pairwise, simultaneous diagonalization shows that varying the positive channel weights changes eigenvalues but not the isolated eigenspaces away from crossings. The principal tensor block is more rigid, since Proposition 4.2 fixes the relative weight $h : \beta = 3 : 1$. Under a common scaling $G_{\text{mic}} \mapsto \lambda G_{\text{mic}}$ with R_b fixed, N_+ and G_H have homogeneity degree 1, $\mathcal{J}_X$ and the reduced pre-Gram jets have degree 1/2, while P , $\mathcal{B}_X$, normalized spectra, and mixing shapes have degree 0.

Proof Equation (126) is Schur decomposition on the multiplicity spaces. Positivity of G_{mic} gives $\langle v, N_+ v \rangle = \|G_{\text{mic}}^{1/2} R_b v\|^2$, which proves (127). The last formula is the differentiated Riesz projection. $\square$

Thus symmetry fixes the metric shape on each irreducible channel but does not fix a common normalization across inequivalent residual channels.

Let P be an isolated zero projection of $N_+(0)$ with $L_0 P = 0$, set $Q = \mathbf{1} - P$, $M = L_0 Q$, and $G_H = M^{\dagger} M$. The Callias-Schur gap is invertibility of G_H on the active high subspace. On the residual/retained sum, the canonical odd doubling is

$$\mathbb{Q}_b = \begin{pmatrix} 0 & L_b \\ L_b^{\dagger} & 0 \end{pmatrix}, \quad \mathbb{Q}_b^2 = \begin{pmatrix} L_b L_b^{\dagger} & 0 \\ 0 & N_+(b) \end{pmatrix}. \quad (129)$$

Thus P belongs to the retained chiral block; the full kernel of $\mathbb{Q}_b$ is $\ker L_b^{\dagger} \oplus \ker L_b$. This stable doubling is kept separate from any rank-one Callias factor with positive asymptotic line L_{Γ} .

Proposition 4.22 (Microscopic Moore-Penrose-Riesz splitting) For a tangent direction X , put $A_X = (\partial_X L)_0 P$. Since M is injective on the active high subspace, $M^+ = G_H^{-1} M^{\dagger}$ and

$$\Pi_{\text{ran}} = M M^+, \quad \Pi_{\text{cok}} = \mathbf{1} - M M^+, \quad \mathcal{J}_X = \Pi_{\text{cok}} A_X, \quad \mathcal{B}_X = Q(\partial_X P)P = -M^+ A_X. \quad (130)$$

Thus

$$A_X = \mathcal{J}_X - M \mathcal{B}_X, \quad A_X^{\dagger} A_Y = \mathcal{J}_X^{\dagger} \mathcal{J}_Y + \mathcal{B}_X^{\dagger} G_H \mathcal{B}_Y. \quad (131)$$

The first term is the stationary Kuranishi-Schur form and $\mathcal{B}_X$ is the second fundamental form of the moving Riesz bundle.

Proposition 4.23 (Pre-Gram Kuranishi two-jet) Along a one-parameter branch write $L(t) = L_0 + t L_1 + \frac{1}{2} t^2 L_2 + O(t^3)$. On a first-order unobstructed kernel branch, the reduced pre-Gram jets are

$$K_1 = \Pi_{\text{cok}} L_1 P, \quad K_2 = \Pi_{\text{cok}} (L_2 P - 2 L_1 M^+ L_1 P), \quad (132)$$

with $K_1 = 0$ for the second formula. Direct differentiation of (125) gives

$$P N_+'(0) P = 0, \quad P N_+''(0) P = 2(L_1 P)^{\dagger} (L_1 P). \quad (133)$$

Hence the Gram family retains the Hermitian quadratic combination of the first pre-Gram jet, while its oriented complex representative is retained only at the level of L .

Proof In the Moore-Penrose high gauge, the first-order high correction is $-M^+ L_1 P$; substitution in the second-order cokernel equation gives (132). Equation (133) follows from $L_0 P = 0$. $\square$

Proposition 4.24 (Riesz-graph response and persistent-kernel closure) *Let O_b be an endomorphism of the retained domain and let $G_b : \text{Ran } P \rightarrow \text{Ran } Q$ be the local graph of $\text{Ran } P_b$. With $W_G = (P + G)(1 + G^\dagger G)^{-1/2}$,*

$$O_{\text{low}}(b) = W_G^\dagger O_b W_G. \quad (134)$$

For a tangent direction X , $D_X G|_0 = \mathcal{B}_X$ and

$$D_X O_{\text{low}}|_0 = P(D_X O)|_0 P + \mathcal{B}_X^\dagger Q O_0 P + P O_0 Q \mathcal{B}_X. \quad (135)$$

If P reduces O_0 , the motion terms vanish and $R_1 = P O_1 P$; hence Riesz motion alone cannot generate the first finite response from a block-diagonal baseline, and a vanishing grade-one component of $P O_1 P$ excludes that branch. Along one parameter write $G(t) = t\mathcal{B} + \frac{1}{2}t^2\mathcal{C} + O(t^3)$ and $O(t) = O_0 + tO_1 + \frac{1}{2}t^2O_2 + O(t^3)$. Direct expansion gives

$$\begin{aligned} O_{\text{low}}''(0) &= P O_2 P + 2\mathcal{B}^\dagger Q O_1 P + 2P O_1 Q \mathcal{B} + 2\mathcal{B}^\dagger Q O_0 Q \mathcal{B} \\ &\quad + \mathcal{C}^\dagger Q O_0 P + P O_0 Q \mathcal{C} - \{\mathcal{B}^\dagger \mathcal{B}, P O_0 P\}. \end{aligned} \quad (136)$$

If the exact kernel persists through second order, $K_1 = K_2 = 0$ in (132) and

$$\mathcal{B} = -M^+ L_1 P, \quad \mathcal{C} = -M^+ (L_2 P - 2L_1 M^+ L_1 P). \quad (137)$$

Thus the first two compressed responses contain no independent graph-motion coefficient on this branch. If $O_0 = c\mathbf{1}$ and $Q O_1 P = P O_1 Q = 0$, the graph terms in (136) cancel and $O_{\text{low}}''(0) = P O_2 P$; a grade-two $P O_2 P$ confined to $\mathbb{C}X^2$ then gives $Y_\perp = 0$ after Rees subtraction.

Proof Expanding $(1 + G^\dagger G)^{-1/2} = \mathbf{1} - t^2 \mathcal{B}^\dagger \mathcal{B}/2 + O(t^3)$ gives the first two response formulas. The kernel equation $L(t)(P + G(t)) = O(t^3)$ gives (137) after projection to the active high space. $\square$

Proposition 4.25 (Finite first-response certificate) *Let $R_1 = D_X O_{\text{low}}|_0$ be a marked family first response and put $\Pi_1^S(T) = \frac{1}{3} \sum_{j=0}^2 \omega^j S^j T S^{-j}$ and $a_1^2 = \text{Tr}(R_1^\dagger R_1)/3$. The two basis-independent gates are*

$$\mathfrak{r}_{\text{gr}1} = \|R_1 - \Pi_1^S(R_1)\|_{\text{HS}}, \quad \mathfrak{r}_{\text{uni}} = \|R_1^\dagger R_1 - a_1^2 \mathbf{1}\|_{\text{HS}}. \quad (138)$$

If $a_1 > 0$ and both vanish, $X = R_1/a_1$ is unitary in $\mathcal{A}_1$, $X^3 = (\det X)\mathbf{1}$, and

$$h_1 := \det X = \frac{\det R_1}{a_1^3} = \frac{\text{Tr}(R_1^3)}{3a_1^3}. \quad (139)$$

Modulo unitary $\mathcal{A}_0$ conjugation, h_1 is the only continuous invariant of such a grade-one response.

The Moore-Penrose splitting and Proposition 4.23 are the stationary pre-Gram and spectral forms of the same low/high elimination. In the associated Lyapunov-Schmidt chart write the reduced map as $K_{\text{red}}(t) = \sum_{m \geq 1} t^m K_m/m!$. On an analytic high-gap branch, persistence of the full initial kernel rank is equivalent to $K_{\text{red}}(t) \equiv 0$; if K_m is the first nonzero reduced jet, the reduced Gram begins as $t^{2m} K_m^\dagger K_m/(m!)^2 + O(t^{2m+1})$, so the lifted low eigenvalues start at order t^{2m} on the nonzero spectral support of $K_m^\dagger K_m$. CP-sensitive and general Takagi data protected downstream therefore retain their pre-Hermitian representative before passage to N_+ .

Theorem 4.26 (Equivariant metric Gram-Riesz descent) *Let a marking-preserving frame transition satisfy*

$$L_{b'} U_g = V_g L_b. \quad (140)$$

For a common isolated Riesz contour,

$$N_+(b') = U_g N_+(b) U_g^{-1}, \quad P_{b'} = U_g P_b U_g^{-1}, \quad G_H(b') = U_g G_H(b) U_g^{-1}. \quad (141)$$

With compatible Hermitian covariant derivatives,

$$\mathcal{J}_{g_* X} = V_g \mathcal{J}_X U_g^{-1}, \quad \mathcal{B}_{g_* X} = U_g \mathcal{B}_X U_g^{-1}. \quad (142)$$

Hence the reduced quadratic form, projected connection, rank, spectrum, gap, trace invariants, and determinant holonomy descend up to marking-preserving unitary conjugation.

This follows from unitary covariance of L_b , functional calculus for the Riesz projection, and functoriality of the Moore-Penrose range/cokernel projections. The channelwise scalar statement is the multiplicity-one case of Lemma 4.21; $\mathcal{J}_X$ controls stationary obstruction and $\mathcal{B}_X$ controls projected bundle motion.

Definition 4.27 (Common Clifford-Picard high-gap completion) Let $D_{\text{ad}} = P_{\text{a}_W} + 2P_{\text{a}_C}$. Using the structural dictionary (45), the normalized high-gap operator is

$$\widehat{\Omega}_{\text{adj}} = \frac{1}{k_Y d_F} P_{\text{a}_W} + \frac{1}{k_Y} P_{\text{a}_C} = \frac{3}{80} P_{\text{a}_W} + \frac{3}{5} P_{\text{a}_C}. \quad (143)$$

The completed high branch imposes

$$G_H|_{\mathcal{H}_{H,\text{adj}}} = M_H^2 \mathbf{1}_{\text{cen}} \otimes \mathbf{1}_{\text{par}} \otimes \widehat{\Omega}_{\text{adj}}^2, \quad \mathcal{D}_{H,\text{adj}}^F = M_H \mathbf{1}_{\text{cen}} \otimes J_{\text{par}} \otimes \widehat{\Omega}_{\text{adj}}, \quad (144)$$

where J_{par} exchanges the matched parity/chirality lines of Proposition 4.33. The fermionic threshold subbranch additionally requires the standard four-dimensional Dirac kinetic lift. The absolute normalization, kinetic lift, and common scale M_H remain completed data.

Any unitary polarization action compatible with automorphisms of the reconstructed $\mathfrak{su}(2) \oplus \mathfrak{su}(3)$ preserves the two nonisomorphic simple ideals and therefore commutes with $\widehat{\Omega}_{\text{adj}}$ and the high resolvent. Hence a dependence of the form $B_b = BR_b$ alone is Schur-invisible on this common high branch. A nonzero polarization discriminator requires a polarization-dependent coupling or denominator, an additional high sector, or interference of several paths.

For a principal source Hessian $H^{\text{pr}} = (H_{ab})$ and faithful coefficient maps, put

$$\widehat{H}_{ab} = \iota_{F,a} H_{ab} \rho_{F,b}. \quad (145)$$

At the algebraic fidelity level, $\iota_{F,a} \mapsto \lambda_a \iota_{F,a}$ and $\rho_{F,a} \mapsto \lambda_a^{-1} \rho_{F,a}$ preserve $\rho_{F,a} \iota_{F,a} = \text{id}_{V_a}$ and send $\widehat{H}_{ab} \mapsto (\lambda_a/\lambda_b) \widehat{H}_{ab}$. If the exported operator is required to remain self-adjoint in a fixed downstream Hermitian structure and its $V_1 \leftrightarrow V_2$ block is nonzero, conjugation by $T = \lambda_1 P_{V_1} + \lambda_2 P_{V_2}$ is self-adjoint only when $|\lambda_1| = |\lambda_2|$. Positive fidelity rescalings therefore leave only a common scale; a complex relative phase is unitary presentation data. If $\rho_{F,a} \iota_{F,a} = \text{id}_{V_a}$, Schur functoriality gives $\widehat{H}_{22}^{\text{red}} = \iota_{F,2} H_{22}^{\text{red}} \rho_{F,2}$.

Definition 4.28 (Marked microscopic Schur realization) A microscopic family has marked Schur realization when its principal source Hessian is exported blockwise by (145); the Euler assignment and local fidelity are already fixed by (115), (5), and Theorem 3.12.

Proposition 4.29 (Marked current-Codazzi support chain) *Under marked Schur realization and Proposition 4.12, the principal support descends as*

$$V_2|V_1|V_2 \mapsto C|W_\sigma|C, \quad 3|2|3. \quad (146)$$

The reverse weak block is the Hermitian adjoint of the oriented leg and introduces no second incidence.

Proposition 4.30 (Determinant-centered weak-leg connection and global principal lift) *For the phase coordinate $\vartheta = \vartheta_\Gamma$ with relative projected rate $r_\vartheta = a_W - a_C$, Proposition 3.15 gives the infinitesimal normalized rates*

$$(a_C, a_W) = r_\vartheta \left(-\frac{2}{5}, \frac{3}{5} \right). \quad (147)$$

For a global principal $U(1)$ with primitive angle t , determinant preservation requires $3a_C + 2a_W = 0$; up to common sign its primitive integer solution is $(-2, 3)$. The relative character has weight five, so $\vartheta_\Gamma = 5t$ and the minimal common character system is

$$\rho_C(e^{it}) = e^{-2it} \mathbf{1}_3, \quad \rho_W(e^{it}) = e^{3it} \mathbf{1}_2, \quad X_{\text{fam}}(t) \sim e^{5it} Z, \quad \vartheta_\Gamma = 5t. \quad (148)$$

The carrier determinant is one and a family first-response representative carrying this character obeys $\det X = e^{3i\vartheta_\Gamma}$. For the anti-Hermitian projected coefficient,

$$r_\vartheta = \frac{i}{6} \text{Tr}_V \left(H_\Gamma^{\text{deg}} A_\vartheta \right), \quad (149)$$

since $\text{Tr}_V((H_\Gamma^{\text{deg}})^2) = 30$. With $A_\vartheta^{\text{cen}} = -(ir_\vartheta/5) H_\Gamma^{\text{deg}}$ and $\Delta_\vartheta = \|A_\vartheta - A_\vartheta^{\text{cen}}\|_{\text{HS}}$, the centered unit branch is equivalently $r_\vartheta = 1$ and $\|A_\vartheta\|_{\text{HS}}^2 = 6/5$. If F_C, F_W, F_F are the marked central curvatures, a common $(-2, 3, 5)$ connection must satisfy

$$5F_C + 2F_F = 0, \quad 5F_W - 3F_F = 0. \quad (150)$$

On the boundary sphere these curvature residuals are also sufficient after the marked central transports are required to be integral line-bundle connections with a common primitive Chern class; their periods are then $(-2n, 3n, 5n)$. The determinant-centered C/W relation is dependent. A fixed family phase leaves only a μ_5 lift differing by a common central scalar on the rank-five carrier, hence invisible to local conjugation-valued gates.

The microscopic component $\mathcal{B}_\vartheta$ tests the common lift after torsor-compatible calibration. On a constant-rank exact zero-mode branch, Lemma 4.21 makes the projected rate independent of the target residual metric. The oriented outer- C coefficient obeys

$$\partial_\vartheta \arg c_{ZS^2}^{\text{neu}} = -\frac{2}{5} r_\vartheta. \quad (151)$$

On the physical common-lift branch $r_\vartheta = 1$ is therefore fixed together with the determinant character.

4.7 Alena-Codazzi realization theorem

Definition 4.31 (Layered Alena-Codazzi realization criterion) Write $\mathfrak{R}_{AC} = (\mathfrak{R}_{src}, \mathfrak{R}_{tr}, \mathfrak{R}_{sp})$. The source layer $\mathfrak{R}_{src}$ consists of (79), (80), (81), the split conservation law (82), Proposition 4.1, the resolved compact-leaf and one-core thin-limit hypotheses, $mM_0q_{tr} \neq 0$, and relative Gauss locality with the two boundary charges fixed. The transport layer $\mathfrak{R}_{tr}$ is torsor-compatible boundary transport with closure of (54). The spectral layer $\mathfrak{R}_{sp}$ consists of an open Callias-Schur gap with subcritical low-high coupling, the covariance (140), marked Schur realization, and a non-degenerate frozen principal multiplier coefficient.

Theorem 4.32 (Alena-Codazzi realization of the primitive graded defect) *If $\mathfrak{R}_{src}$ holds, the limiting collar supplies a primitive graded defect datum of Definition 3.2, realizes entry A and the relative Gauss-local part of entry B, and preserves the source Euler grading by (115). After faithful marked visibility in entry B, minimal reconstruction gives (22) with degree fidelity. If $\mathfrak{R}_{tr}$ also holds, the transported charge pair realizes the boundary Hodge-spectral module (52); the flat subbranch is (55). If $\mathfrak{R}_{sp}$ holds, Theorem 4.26 supplies the isolated low bundle and its descended Riesz data, while Proposition 4.29 gives the support chain (146).*

Proof The source statement combines Proposition 4.1, Proposition 4.15, Theorem 4.16, (116), and Lemma 4.18. The transport statement follows from (118), (121), and Definition 4.20. The spectral statement is Theorem 4.26 followed by Proposition 4.29. $\square$

The geometric realization supplies entry A, the relative Gauss-local part of entry B, the admissible finite transport, and the isolated Gram-Riesz bundle. Faithful central coefficient reconstruction remains the structural input in entry B. The finite conclusions are then those of Theorem 3.24; under marked Schur realization the principal multiplier Hessian exports the support chain (146). Completed matrix elements are evaluated only after this protected descent.

4.8 Conditional low-sector attachment and Standard-Model channel structure

The finite objects above precede the choice of a low spectral representative. Physical attachment requires a boundary-admissible family, an isolated primitive window, exact finite-shadow covariance, and parity closure. On an isolated Borel-Weil block,

$$\nabla_{B,q} = P_q d P_q \quad (152)$$

is the Berry-Wilczek-Zee connection [66] in the non-Abelian form of [67]. Riesz stability is understood as in [8]; the open normal comparison uses [9] and [10]. The equivariant Gram-Riesz descent of Theorem 4.26 makes the isolated projection, gap, projected curvature, and determinant holonomy basic on the protected quotient. For the odd completion the full graded Clifford module $\mathcal{F}_\Gamma := \Lambda^\bullet V = F_\Gamma^{\text{even}} \oplus F_\Gamma^{\text{odd}}$ is retained before the locked projection; F_Γ^{even} in (29) is the selected internal chiral package on the even endpoint. Let $\Gamma_V = (-1)^{N_C + N_W}$ act on $\mathcal{F}_\Gamma$ and put

$$\Gamma_{\text{lock}} = \gamma_{\text{br}}^5 \otimes \Gamma_V. \quad (153)$$

Let $\mathcal{H}_{\text{lock}}^\pm$ be the two eigenspaces of Γ_{lock} and $P_\pm$ their projections. The completed finite odd operator is

$$Q_F = P_+ \left(D_E + c(\Phi + \Phi^\dagger) + Q_{\text{hol}} + Q_C \right) P_-. \quad (154)$$

Odd insertions admit the superconnection organization of [15]. On the retained ε eigenspace the parity residual is

$$R_{\text{par}} = (S_2 \otimes \Gamma_{\text{lock}} - \mathbf{1})|_{\mathcal{H}_{\text{par}} \otimes \mathcal{H}_{\text{lock}}^\varepsilon}. \quad (155)$$

Its closure selects one μ_2 stabilizer character of the family module.

Proposition 4.33 (Parity-paired odd high lift) *If the retained high lift has block-adjoint content $\mathfrak{a}_W \oplus \mathfrak{a}_C$ before parity compression, is odd for Γ_{lock} , and preserves (155), then*

$$\mathcal{H}_{H,\text{adj}} \simeq \mathcal{H}_{\text{cen}} \otimes \mathbb{C}_{\text{par-lock}}^2 \otimes (\mathfrak{a}_W \oplus \mathfrak{a}_C), \quad (156)$$

with $\mathfrak{a}_W = \text{End}_0(W)$ and $\mathfrak{a}_C = \text{End}_0(C)$. Thus the three points of the family torsor (47) give three parity-paired Dirac-form adjoint lines in each channel.

Higher integer principal moments have zero direct compression by Proposition 3.14; applying the reduced operator (8) therefore starts them at second order.

Proposition 4.34 (Schur suppression of higher integer moments) *Let P be an isolated projection of $K_0 = K_0^\dagger$, put $Q = \mathbf{1} - P$, and assume that QK_0Q is invertible. For $T_\ell = T_\ell^\dagger$ with $PT_\ell P = 0$ and $\ell \geq 3$, set $K(\varepsilon) = K_0 + \varepsilon T_\ell$ and $g_{\text{CSch}} = \|(QK_0Q)^{-1}\|^{-1}$. If $\|T_\ell\| < g_{\text{CSch}}/2$ and $|\varepsilon| \leq 1$, then*

$$\delta K_{\text{eff}}^{(\ell)}(\varepsilon) = -\varepsilon^2 PT_\ell Q(QK_0Q)^{-1}QT_\ell P + O\left(\frac{|\varepsilon|^3 \|T_\ell\|^3}{g_{\text{CSch}}^2}\right), \quad \|\delta K_{\text{eff}}^{(\ell)}\| = O\left(\frac{|\varepsilon|^2 \|T_\ell\|^2}{g_{\text{CSch}}}\right). \quad (157)$$

For $\ell = 3$, let T_3^{cat} be the adjoint form of the top multiplication $R_{\Gamma,2} \otimes R_{\Gamma,4} \rightarrow R_{\Gamma,6}$, viewed as a map from $V_3 \simeq R_{\Gamma,6}$ to $\text{Hom}(C, R_{\Gamma,4})$. For a marked microscopic V_3 perturbation, let $\mathcal{T}_3^{\text{mic}} : V_3 \rightarrow \text{Hom}(C, R_{\Gamma,4})$ denote its $C \rightarrow R_{\Gamma,4}$ low-high component. Since $C \simeq V_1$ and $R_{\Gamma,4} \simeq V_2$, Clebsch-Gordan gives

$$\dim \text{Hom}_{SU(2)}(V_3, \text{Hom}(C, R_{\Gamma,4})) = 1, \quad \mathcal{T}_3^{\text{mic}} = a_3 T_3^{\text{cat}}. \quad (158)$$

The scalar a_3 is not fixed by the structural reconstruction. With a fixed Hilbert-Schmidt normalization, its microscopic line test is $a_{3,*} = \langle T_3^{\text{cat}}, \mathcal{T}_3^{\text{mic}} \rangle / \|T_3^{\text{cat}}\|^2$ and $\Delta_3^2 = \|\mathcal{T}_3^{\text{mic}}\|^2 - |\langle T_3^{\text{cat}}, \mathcal{T}_3^{\text{mic}} \rangle|^2 / \|T_3^{\text{cat}}\|^2$; $\Delta_3 = 0$ is exactly the protected-line condition. More generally, if a retained one-dimensional intertwiner line is protected only under $m \sim \lambda m$ with $\lambda \in \mathbb{C}^\times$ and this scale is not downstream observable, the protected descent canonically fixes only the projective class $[m]$; a nonzero linear normalization requires an additional norm, unitary frame, determinant trivialization, or oriented primitive integral generator. Residual-metric rescaling realizes this ambiguity on the microscopic V_3 block while leaving its equivariant line, kernel, and protected carrier unchanged. The symmetric-power metric gives 7/5 on the high V_2 partial trace and 7/3 on the low C trace; the Codazzi quotient and low Feshbach contraction therefore do not export 7/5 as a physical 12 numerator. Let P_{prim} be the isolated primitive locked projection before the finite-shadow action is attached and write $\text{Ran } P_{\text{prim}} \simeq \mathcal{H}_{\text{ch}} \otimes M_{\text{prim}}$. Finite-core closure gives

$$\dim M_{\text{prim}} = 1. \quad (159)$$

For the pre-torsor representative K_{pre} , exact finite-shadow covariance is

$$R_{\text{cov}} = ([p_x, B], [S_6, B], [p_x, K_{\text{pre}}], [S_6, K_{\text{pre}}])_{\substack{x \in \mathcal{T}_\Gamma^{(6)} \\ B \in \mathfrak{A}_{\text{ch}}}}. \quad (160)$$

Proposition 4.35 (Locked central low-sector factorization) *Assume exact finite-shadow covariance, (155), and a finite-core primitive projection isolated by Δ_{CSch} . Then*

$$\mathfrak{A}_{\text{low}}^{(0)} = \mathfrak{A}_{\text{ch}} \otimes P_\varepsilon \mathcal{A}_6 P_\varepsilon, \quad \text{Ran } P_{\text{low}}^{(0)} \simeq \mathcal{H}_{\text{ch}} \otimes \mathcal{H}_{\text{cen}}, \quad (161)$$

In the fixed pre-torsor frame, the family matrix units are generated by the point projections p_x and the shift S_6 . Their commutant is scalar on the six-cycle factor, so the restriction of K_{pre} to the isolated primitive channel has the form $K_{\text{prim}} \otimes \mathbf{1}_6$ for a unique primitive operator K_{prim} . After parity compression, functional calculus gives $P_{\text{low}}^{(0)} = P_{\text{prim}} \otimes \mathbf{1}_{\mathcal{H}_{\text{cen}}}$. A central perturbation below half the isolating gap preserves

$$\text{rank } P_{\text{low}} = \text{rank } P_{\text{prim}} \dim \mathcal{H}_{\text{cen}}. \quad (162)$$

The factorization follows from the parity corner (44), exact covariance, and finite-core uniqueness.

Corollary 4.36 (Weak-parity closure) The retained parity factor is one-dimensional. Under Proposition 4.35, each simple primitive channel therefore has

$$\text{rank } P_{\text{low},\text{ch}} = 3. \quad (163)$$

On the exact locked branch with a primitive line factor, $E_{\text{low}} \simeq E_{\text{prim}} \otimes \mathbb{C}^3$ and $\det E_{\text{low}} \simeq E_{\text{prim}}^{\otimes 3}$. If edge unitaries are written $U_a = h_a W_a$ with $\det W_a = 1$, determinant data determine the scalar cycle only through $(h_2 h_1 h_0)^3$. A determinant-only reconstruction therefore retains a μ_3 ambiguity unless the primitive line and its orientation are kept.

The statement concerns the rank and spectral transport class of the isolated bundle; its determinant-line reading is represented by [13]. Let $E_+ \rightarrow S_\Gamma^2$ be a rank-three positive asymptotic eigenbundle of a normal Dirac-Callias lift. Boundary reduction [9] in the geometric form of [10] gives the index from $c_1(E_+)$. Smooth complex rank-three bundles over S^2 are classified by their clutching degree, and the determinant map induces the corresponding isomorphism on π_1 . Hence the primitive determinant condition gives

$$\det E_+ \simeq L_\Gamma^{\otimes 3}, \quad L_\Gamma \simeq \mathcal{O}(1) \implies E_+ \simeq_{\text{smooth}} L_\Gamma^{\oplus 3}, \quad [E_+] = 3[L_\Gamma], \quad |\text{Ind } D_{\text{Cal}}| = 3. \quad (164)$$

The smooth isomorphism is noncanonical and does not fix the connection or finite-shadow action. A C_3 -equivariant comparison is $E_+^{\text{reg}} = L_\Gamma \otimes \mathbb{C}[C_3]$, realizable by the standard Hopf/hedgehog asymptotic mass; it has $\text{Ind}_{C_3} D_{\text{Cal}}^{\text{model}} = [1] \oplus [\chi] \oplus [\chi^2]$ and ordinary index three. The microscopic target is therefore the regular equivariant index, together with a common-base gapped Riesz-Callias identification. An invertible C_3 -equivariant heavy neutral denominator preserves this rank and character content.

Corollary 4.37 (No additional light family on the isolated branch) Under the fixed Callias-Schur stratum and an invertible denominator in (221), the rank-three active factor is unchanged by Schur elimination.

The exterior bidegree (N_C, N_W) gives a compact incidence calculus for the remaining finite channels. Elementary C, C^*, W, W^* insertions are the linear Clifford-odd arrows; Majorana and determinant-contact classes are the corresponding even composite incidences. The four-fermion baryon-violating components of

Table 3 Exterior-incidence classes of the separated low channels.

Class	Incidence	Charges	Reading
color odd	$C \oplus C^*$	$\Delta Y = \mp 1/3$	colored odd channel
weak direct	$W \oplus W^*$	$\Delta Y = \pm 1/2$	quark, charged-lepton, and neutral Dirac blocks
Majorana	neutral singlet pair or two weak factors	$\Delta(B - L) = \pm 2$	heavy denominator or active correction
doubled top form	determinant class $2(3, 2)$	$\Delta Y = 0$ $\Delta(B - L) = 0$	separated contact sector

the doubled top-form class are the $QQQL$ and $u^c u^c d^c e^c$ types and satisfy

$$\Delta B = \Delta L = \pm 1. \quad (165)$$

Thus $\Delta B = 2$ is outside this dimension-six class. The unique color-singlet, $B - L$ -preserving linear odd class is $W \oplus W^*$; a second independent weak-doublet channel would require an additional protected source or downstream datum by Proposition 2.3. The standard Dirac/Majorana/top-form distinction is represented by [48].

At this stage the carrier, global representation, structural rank-three family module, parity linearization, and channel grading are separated. Physical rank-three attachment remains conditional on covariance and the isolated gap; matrix elements of (154), singular values, neutral denominator, and contact coefficients remain completed data. The conditional low-sector factorization above supplies the analytic attachment of the parity-linearized module of Theorem 3.24; Table 3 records the corresponding structural incidences. The status separation is summarized below.

The structural/conditional separation above is retained throughout the transferred problem; the consolidated output-status and falsification ledger is Table 4.

The realization above supplies the geometric and source data in entry A and the relative Gauss-local part of entry B, together with the conditional transported and isolated low data used in the physical attachment. The preserved moment grading is identified with the projective carrier level by Theorem 3.12. The transferred spectral and quantitative problem is recorded in Section 5.

5 Schur-Feshbach Low-Energy Response, Quantitative Branches, and Falsification

The primitive carrier, exterior module, family torsor, and canonical two-linearization order-six envelope are fixed in Section 3. Section 4 supplies the current-Codazzi realization, the microscopic Gram family (125), and the conditional physical family factor. The isolated splitting supplies the Schur-Feshbach data used by Lemma 2.5; all low/high eliminations below are its operator-theoretic specializations. A homological-transfer organization is available only on branches carrying the additional deformation-retract data stated there. The ordered interface (3) and the $B - L$ and exterior-bidegree filtration of Table 3 determine which transferred terms are retained.

Marked-Schur admissibility fixes the support before completed matrix coefficients are introduced. Equation (158) fixes the V_3 low-high direction up to its reduced microscopic scalar. Unless a later step changes status explicitly, all transferred objects inherit the provenance of their inputs as recorded in Table 4.

5.1 Completed low operator and protected Schur descent

The low operator is (154) on the isolated module of Proposition 4.35. Lemma 2.5 supplies its high correction and higher-source reduction, while (131) separates stationary obstruction from projected bundle motion. Masses are read from the resulting singular values, gaps, or reduced Hessians.

The finite Yukawa blocks are compressed matrix elements of the zero-order part of the locked operator:

$$(Y_{AB})_{ij} = \langle \psi_i^A, P_{\text{low}} \left(c(\Phi + \Phi^\dagger) + Q_{\text{hol}} + Q_C \right) P_{\text{low}} \psi_j^B \rangle. \quad (166)$$

The exterior package fixes the possible pairs (A, B) ; the entries in (166) are completed-branch data. In a local orthonormal low-mode frame, (152) has matrix coefficients $A_{ij} = \langle \psi_i, d\psi_j \rangle$. These are the projected gauge coefficients of the selected carrier. The determinant reduction of the projected connection is the projection to (37).

On the determinant-nondegenerate quark branch, the pre-Hermitian invariant is

$$\Xi_{\text{str}} := e^{i\theta_{\text{QCD}}} \frac{\det(Y_u Y_d)}{|\det(Y_u Y_d)|} = e^{i\bar{\theta}}. \quad (167)$$

Hermitian direct data do not reconstruct this phase: Rees-Picard variables use the relative basis response, while (167) depends on the sectorwise determinant sum.

Proposition 5.1 (Generated strong determinant line) *Let $\mathcal{L}_{\text{str}}^{\text{gen}}$ be the determinant line generated by the completed quark family-index datum and its canonical differential-index transgressions, represented by [14]. An external flat translate gives $\mathcal{L}_{\text{str}}^{(\alpha)} = \mathcal{F}_\alpha \otimes \mathcal{L}_{\text{str}}^{\text{gen}}$ and changes the reconstruction class. If $\mathcal{L}_{\text{str}}^{\text{gen}}$ has a connection-preserving real structure, its holonomy reduces to $\{\pm 1\}$; the remaining sign is the mod-two determinant/Pfaffian orientation class. If it has a gauge-invariant parallel trivialization, Proposition 3.18 gives $\bar{\theta} = 0 \pmod{2\pi}$. The chirality orientation of the finite Clifford module does not by itself provide either global structure. The neutral Pfaffian package below is only an algebraic comparison and does not compute the color-family orientation sign.*

After the central factor (48) has been isolated, a direct Hermitian family block $H_f = Y_f Y_f^\dagger$ has first family-motion part $H_f^{(1)} = H_f^{\text{circ}} + \Gamma_f$ with $\Gamma_f = \Pi_{\text{noncirc}}(H_f^{(1)})$. Here H_f^{circ} is the transported degree-zero part, while $H_f - H_f^{(1)}$ contains higher Schur terms not used in the first family-motion test. The projection is the one in (72). Thus only Γ_f is tested by the mixed torsor curvature (72) at first order. The vertex-resolved charged spectral representative is reduced before this transport decomposition.

The completed branch is represented by the finite Schur-Kuranishi map $\mathcal{F}_{\text{fin}}(b) = P_{\text{obs}}(K_{\text{eff}}(b)) = 0$ after gauge, diffeomorphism, and unitary redundancies have been removed. Its filtered protected interface is

$$\mathcal{P}_r(b) = (j_b^* \mathcal{F}_{\text{fin}}, \mathcal{M}_{\text{pre-Schur}}, \mathcal{D}_r(b), \Delta_{\text{low}}(b)), \quad \mathcal{D}_{r_{\text{spec}}} = \mathcal{S}_{\text{dir}}, \quad \mathcal{D}_{r_{\text{Sch}}} = [P_{\text{low}}, \nabla_{\text{low}}]. \quad (168)$$

Here $\mathcal{M}_{\text{pre-Schur}}$ collects the boundary-sector, $B - L$, exterior-bidegree, finite-shadow, and torsor markings; Δ_{low} is the isolating Riesz gap. The earlier spectral depth retains no independent family orientation.

Corollary 5.2 (Protected Schur descent) *At a zero of the finite Schur-Kuranishi map in a fixed reconstruction class, Proposition 2.3 retains the channels generated by source transport or (8) and required by (168); presentation coordinates outside that closure are removed.*

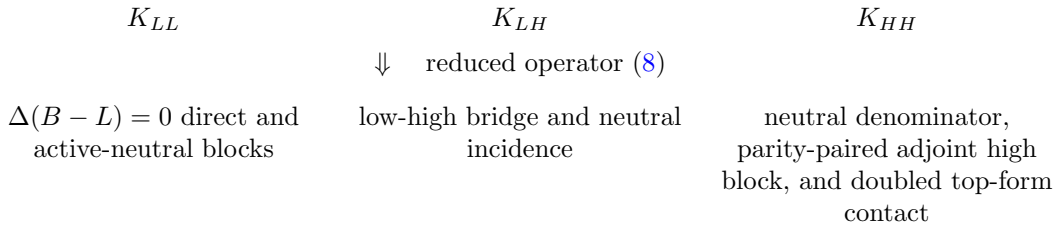


Fig. 2 Completed reduced-resolvent architecture on the structurally fixed low carrier. The common adjoint high block is normalized by Definition 4.27.

The reduced completed branch contains the channels forced by the retained observables, the finite filtration, and the gap-mediated compression; auxiliary coordinates outside the least protected Schur interface are removed by Proposition 5.2.

5.2 Marked neutral affine cell and determinant coordinate

The finite neutral Hessian is the Gram form (2) after the reduction (7), with the normalized Clifford trace (30). In the Hermitian marked neutral realization, let W_σ be the selected weak orientation and $W_{-\sigma} = W_\sigma^*$. The angular trace reservoir is $\mathcal{R}_\angle = C \oplus W_\sigma \oplus W_{-\sigma}$ and the radial reservoir is $\mathcal{R}_r = C \oplus W_\sigma$. The angular support is the marked neutral reading of the full degree-seven decomposition (69); the radial reservoir is the selected rank-five carrier with one weak projections and is not a full theta section space. Let $P_C^\angle$, $P_{W_\sigma}^\angle$, and P_C^r denote the corresponding block projections. Marked Schur realization fixes the incidence

$$\mathfrak{J}_{\text{neu}} = \left(P_C^\angle, P_C^\angle \otimes P_{W_\sigma}^\angle, P_C^\angle \otimes P_C^r \right). \quad (169)$$

The reservoir ranks are $\text{rank } \mathcal{R}_\angle = c + 2w = T^*$ and $\text{rank } \mathcal{R}_r = c + w = r$. This reservoir identification is part of the marked neutral realization and carries no independent scalar coefficient. Their normalized traces therefore give

$$A_0 = \frac{c}{T^*}, \quad B_0 = \frac{cw}{(T^*)^2}, \quad C_0 = \frac{c^2}{T^{*r}}; \quad (c, w) = (3, 2): \quad (A_0, B_0, C_0) = \left(\frac{3}{7}, \frac{6}{49}, \frac{9}{35} \right). \quad (170)$$

Thus (169) gives the three factors in (170) without an additional trace coefficient. The independent rank identity obtained from Corollary 3.19 and Corollary 3.33 remains $n_H - T^* = (n - 1)r$; at primitive degree, $n_H = T^* = 7$ is the rank-side compatibility of Corollary 3.34.

By (24), the scalar determinant tangent is the normalized multiplication moment direction of the centered line:

$$\delta S_{\text{sc}} = uP_C - \frac{3}{2}uP_W = \frac{u}{2}H_\Gamma^{\text{deg}}. \quad (171)$$

Accordingly, before an absolute normalization of u is chosen, the neutral affine cell is

$$A = A_0 - \frac{3}{2}u, \quad B = B_0, \quad C = C_0 + u. \quad (172)$$

Lemma 5.3 (Affine-cell elimination) The cell (172) is one-dimensional affine. Linear functionals annihilating its tangent are constant, while ratios of affine functionals are fractional-linear in u ; eliminating u between two such ratios therefore gives a Möbius relation.

The zero-tangent determinant coordinate is fixed by the rank dictionary,

$$s_{\text{det}}(0) = \frac{B_0}{A_0 + B_0} = \frac{w}{c + 3w} = \frac{2}{9}, \quad (173)$$

Remark 5.4 (Constitutive spectral witness) Corollary 4.7 supplies the independent spectral data $c_{\text{sp}} := \text{rank } P_3 = 3$, $w_{\text{sp}} := \text{mult}(4/7) = 2$, $r_{\text{sp}} := \text{rank}(\mathbf{1} - P_3) = 5$, and $T_{\text{sp}}^* := (4/7 - 3/7)^{-1} = 7$. Substitution of this packet into (170) and (173) reproduces the same zero-tangent neutral values obtained from marked incidence. The comparison is a constitutive-spectral witness; microscopic provenance requires a downstream map from (P_3, P_{V_1}) to the marked incidence (169).

The finite determinant-shadow coordinate along the line is

$$s_{\text{det}} = \frac{B}{A + B}. \quad (174)$$

Because (171) is scalar on the family factor and has no torsor clock degree, a source-generated first-order family-shape response requires an additional noncentral direct component; otherwise the family-shape coordinates start at quadratic Schur order.

At an unspecified positive neutral-cell scale $v_{\text{EW}}^{\text{prim}}$, the zero-remainder masses are read as

$$\begin{aligned} m_\gamma^{\text{prim}} &= 0, & m_W^{\text{prim}} &= \frac{v_{\text{EW}}^{\text{prim}}}{2} \sqrt{A}, \\ m_Z^{\text{prim}} &= \frac{v_{\text{EW}}^{\text{prim}}}{2} \sqrt{A+B}, & m_h^{\text{prim}} &= v_{\text{EW}}^{\text{prim}} \sqrt{C}. \end{aligned} \quad (175)$$

Comparing (175) with the standard tree-level mass convention gives $A = g_{\text{cell}}^2$, $B = g'_{\text{cell}}{}^2$, and $C = 2\lambda_{\text{cell}}$. At $u = 0$, (170) therefore gives $g_{\text{cell}}^2 = 3/7$, $g'_{\text{cell}}{}^2 = 6/49$, $\lambda_{\text{cell}} = 9/70$, $\sin^2 \theta_{\text{cell}} = 2/9$, and $\lambda_{\text{cell}}/g_{\text{cell}}^2 = 3/10$. These are neutral-cell matching parameters before the completed gauge kinetic normalization; relations of gauge and Higgs couplings provide the spectral-action comparison [68]. By Lemma 5.3, B and $A + \frac{3}{2}C$ are the independent linear invariants of the tangent in (172). Eliminating u and the common scale from (175) gives

$$15(m_h^{\text{prim}})^2 = 266(m_Z^{\text{prim}})^2 - 306(m_W^{\text{prim}})^2. \quad (176)$$

This relation inherits marked neutral descent and the tracial affine cell, but no absolute normalization of u , $v_{\text{EW}}^{\text{prim}}$, or S_{prim} .

To keep the neutral oscillation shape on the same one-parameter dimensionless branch, impose

$$S_{\text{prim}}^{-1} = 16u = d_F u. \quad (177)$$

The coefficient $1/d_F$ has a concrete sufficient microscopic mechanism: the trace-preserving scalar expectation on F_Γ^{even} sends a rank-one full-gauge-singlet projector to $1/d_F$; factorization of the primitive Schur transfer through the unique trivial line $\Lambda^0 V$ then yields (177). This rank-one factorization remains a realization test. Writing $S_{\text{prim}} = 4\pi^2 \kappa_\Omega$, the benchmark $u_{\text{prim}} = 1/(64\pi^2)$ is equivalent to the additional microscopic action normalization $\kappa_\Omega = 1$; topology and the Eisenstein lattice do not fix this coefficient.

On this fixed benchmark subbranch the rank-split determinant contribution is $1/(144\pi^2)$ and

$$v_{\text{EW}}^{\text{prim}} = 2\sqrt{2} M_{\text{Pl}} \exp\left(-S_{\text{prim}} - \frac{1}{144\pi^2}\right) \sqrt{\frac{C_0}{C_0 + u_{\text{prim}}}}. \quad (178)$$

The leading exponential is $2\sqrt{2} M_{\text{Pl}} e^{-4\pi^2}$ on this subbranch; absolute scale remains a separate normalization problem. The tree-level electromagnetic reading is

$$\alpha_{\text{cell}} = \frac{AB}{4\pi(A+B)} = \frac{\sqrt{2}G_F^{\text{prim}}(m_W^{\text{prim}})^2}{\pi} \left(1 - \frac{(m_W^{\text{prim}})^2}{(m_Z^{\text{prim}})^2}\right). \quad (179)$$

At $u = 0$, (170) gives the exact matching identity $\alpha_{\text{cell}}(0)^{-1} = 42\pi = NT^*\pi$. With $G_F^{\text{prim}} = [\sqrt{2}(v_{\text{EW}}^{\text{prim}})^2]^{-1}$, (179) is a G_F -scheme identity; the electroweak convention is summarized in [69], while conversion to $\alpha(0)$ belongs to radiative matching at μ_{cell} .

5.3 Direct spectral response, Rees-Picard paths, and family polarization

5.3.1 Direct Hodge rank ladder

By (31) and $Q = T_3 + Y$, the weak factor splits as $W = W_0 \oplus W_+$ with charges 0, 1. For a normalized partial isometry $T_+ : W_0 \rightarrow W_+$ and $T_- = T_+^\dagger$,

$$F_0 = \mathbf{1}, \quad F_1 = \mathbf{1} - T_- T_+, \quad F_2 = \mathbf{1} - T_- T_+ - T_+ T_-, \quad (\text{rank } F_0, \text{rank } F_1, \text{rank } F_2) = (5, 4, 3). \quad (180)$$

The normalized surviving trace is $\rho_k = (5 - k)/5$. For a positive three-family mass vector, use the scale-free shape quotient $Q_f := \sum_i m_{f,i}/(\sum_i \sqrt{m_{f,i}})^2$. The direct Hodge rank completion identifies the Hodge ratio in Proposition 3.37 by $e^{-\Sigma_{\text{H}}} = \rho_{k_f}^{-1}$, and hence

$$Q_f = \frac{10 - k_f}{3(5 - k_f)}. \quad (181)$$

Equivalently, with $R = \text{rank } F_0 = 5$, (181) is $Q_f = \frac{1}{3}(1 + R/\text{rank } F_{k_f})$. For a positive three-family mass vector, $1/3 \leq Q_f \leq 1$; hence the surviving integer rank must satisfy $3 \leq \text{rank } F_{k_f} \leq 5$, so the admissible positive rank readings are exactly 3, 4, 5. The sector readout $k_\ell = 0$, $k_d = 1$, $k_u = 2$ exhausts them and gives $Q_\ell = 2/3$, $Q_d = 3/4$, and $Q_u = 8/9 = \frac{1}{3}(1 + k_Y)$; this rank-to-shape identification remains a motivated

direct-sector completion.

The same lost/surviving/ambient ranks $(1, 4, 5)$ occur upstream in the real source sequence (102), while Corollary 4.9 reconstructs the full numerical packet $(1, 4, 5; 7)$ independently from the constitutive projector and fixes the compatible norm-seven Eisenstein class up to orientation. Corollary 3.33 gives the matching finite commensurate packet and $\mathfrak{q} := q_H/p_H = 4$. The source quotient is compared with the complex finite flag only after complexification. The remaining marked problem is the amplitude/operator incidence that realizes the structural ratio as a common-to-relative first-jet coefficient.

5.3.2 Source-generated quark transport and Schur-Berry motion

The direct quark blocks are two compressions of the same graded source class. Their common leading symbol lies in $\mathcal{A}_0 = \mathbb{C}[S]$, while relative response lies in $\mathcal{A}_1 \oplus \mathcal{A}_2$ by (76); Picard translation has degree zero by the Weyl product rule above. For a branch direction X , the microscopic detector $V_X = Q_b(\partial_X N_+)P_b$ splits by Proposition 4.22: its Moore-Penrose range component moves the Riesz bundle and the Berry-Wilczek-Zee connection [67], while the cokernel component enters the stationary penalty. Its finite family reading is the non-circulant projection (72).

After removal of the common circulant baseline, let Γ_u, Γ_d be the first Hermitian non-circulant corrections and put $J_{\text{cen}} := \Im \text{Tr}([\Gamma_u, \Gamma_d]^3)$. Its normalization to the physical invariant follows the convention of [70]. On the canonical clock-shift representative, Proposition 3.37 and (58) give the oriented-discriminant identity $J_{\text{cen}}^2 = (9/16) \text{Disc } \chi_{\Delta_\theta}$; the spectrum fixes the magnitude and the torsor orientation fixes the sign. The physical one-phase closure is

$$J_q = s_{12}s_{23}s_{13}c_{12}c_{23}c_{13}^2 \sin(3\vartheta_\Gamma). \quad (182)$$

The torsor phase is fixed after the three magnitudes; (54) is used here, while (55) remains a separate subbranch. The zero-total-degree rule also gives the general cubic dependence on $\Im \det A$ for $A \in \mathcal{A}_1$.

For a simple-spectrum circulant $H_f^{(0)}$ and first correction Γ_f , the first reduced-resolvent/Sylvester basis response is

$$(\Theta_f)_{ij} = \frac{(\Gamma_f)_{ij}}{h_{f,j}^{(0)} - h_{f,i}^{(0)}}, \quad i \neq j, \quad (\Theta_f)_{ii} = 0. \quad (183)$$

Higher basis jets are obtained by iterating the same inverse Sylvester map; an independent source second jet enters linearly at its own Rees order. Since $H_{f,0} \in \mathcal{A}_0$, this map preserves the Weyl grading, while commutators add it. Simple denominators therefore change magnitudes and possibly the orientation sign without introducing a continuous phase.

If all simple-spectrum sectoral blocks move along a source orbit by the same marking-preserving unitary covariance, $H_f(gx) = U(g)H_f(x)U(g)^\dagger$, the common factor cancels from every relative basis. Mixing moduli and Jarlskog invariants are then constant on that orbit. Physical source-axis dependence therefore requires a fixed marking not co-rotated with the source or sector-dependent susceptibilities. Conversely, independent up/down source phases are unnecessary: if $\Gamma_d = \lambda\Gamma_u = \lambda\Gamma$, the leading relative generator has entries $K_{ij} = c_{ij}\Gamma_{ij}$ with $c_{ij} = \lambda/\Delta_{ij}^d - 1/\Delta_{ij}^u$, and the leading CP invariant is proportional to $c_{12}c_{23}c_{13}\Im(\Gamma_{12}\Gamma_{23}\Gamma_{13}^*)$. A shared source can therefore carry CP response through one rephasing-invariant triangle phase and unequal sectoral susceptibilities. After a common finite marking is fixed, the common-to-relative first-jet hypothesis is tested without a basis choice by $q_* = \langle J_{\text{rel}}, J_{\text{com}} \rangle_{\text{HS}} / \|J_{\text{rel}}\|_{\text{HS}}^2$ and $\Delta_q^2 = \|J_{\text{com}}\|_{\text{HS}}^2 - |\langle J_{\text{rel}}, J_{\text{com}} \rangle_{\text{HS}}|^2 / \|J_{\text{rel}}\|_{\text{HS}}^2$. The condition $\Delta_q = 0$ is required before q_* is compared with the finite ratio $q_H/p_H = 4$.

The relative basis is $V_{\text{CKM}} = \mathbf{1} + \Theta_d - \Theta_u + O(\Gamma^2)$ with the standard conventions of [71], [72], and [73]. If δ_f is the smallest leading gap, (183) gives

$$\|V_{\text{CKM}} - \mathbf{1}\|_F \leq \frac{\|\Gamma_d\|_F}{\delta_d} + \frac{\|\Gamma_u\|_F}{\delta_u} + O_{\delta_u, \delta_d}((\|\Gamma_u\|_F + \|\Gamma_d\|_F)^2). \quad (184)$$

The reduced source-generated class is fixed before minimization, so a lower-valuation unsourced edge is absent by Proposition 2.3. Equations (72) and (78) read the same non-circulant Frobenius norm, while the physical basis susceptibility retains the gap factors in (184). Entanglement-based flavor selection gives an independent comparison [74].

5.3.3 Rees-Picard hierarchy and arithmetic CP completion

With no independently sourced lower-valuation $1 \leftrightarrow 3$ edge, the positive direct valuations are

$$v_{12} = 1, \quad v_{23} = 2, \quad v_{13} = 3. \quad (185)$$

To retained order, the associated-graded Lie algebra is the three-dimensional Heisenberg algebra: the weighted generators ϵL_{12} , $\epsilon^2 L_{23}$, and $\epsilon^3 L_{13}$ have the single visible bracket $[\epsilon L_{12}, \epsilon^2 L_{23}] = \epsilon^3 L_{13}$, while the remaining brackets start beyond order three. Hence $s_{12} = O(\epsilon)$, $s_{23} = O(\epsilon^2)$, $s_{13} = O(\epsilon^3)$ and, by (182), $J_q = O(\epsilon^6)$. Degree-zero Picard insertions preserve this filtration. This is Froggatt-Nielsen-type power counting [75], with

Rees valuation replacing the additional flavor charge and without a separate messenger sector.

Let $R_1 = a_1 X \in \mathcal{A}_1$ with $X^\dagger X = \mathbf{1}$ and let $Y \in \mathcal{A}_2$. Since $\text{Ad}_X^3 = \mathbf{1}$, put $P_r^X = \frac{1}{3} \sum_{j=0}^2 \omega^{-rj} \text{Ad}_X^j$.

Theorem 5.5 (Intrinsic finite Rees-Picard geometry) *Every unitary $X \in \mathcal{A}_1$ is conjugate by a unitary in $\mathcal{A}_0$ to ζZ , with $\zeta^3 = \det X$; changing the cube root is a degree-zero C_3 gauge. The generated line is $P_0^X(\mathcal{A}_2) = \mathbb{C}X^2$ and*

$$Y_\perp = (P_1^X + P_2^X)Y = \frac{1}{3}[X^\dagger, [X, Y]], \quad \rho_{\text{RP}} = \frac{\| [X, Y] \|_{\text{HS}}}{\| R_1 \|_{\text{HS}}^2}. \quad (186)$$

For $Y_\perp \neq 0$, put $\mathcal{C}_X = (\text{Ad}_X - \text{Ad}_X^{-1})/(i\sqrt{3})$, $p = \frac{1}{2}(1 + \langle Y_\perp, \mathcal{C}_X Y_\perp \rangle / \| Y_\perp \|^2)$, and $\Delta_{\text{pol}} = 3p(1-p)$. Moreover,

$$\rho_{\text{RP}} = 0 \iff C^*(X, Y) \simeq \mathbb{C}^3, \quad \rho_{\text{RP}} > 0 \iff C^*(X, Y) = M_3(\mathbb{C}). \quad (187)$$

Weyl orthogonality gives

$$c_1 = 1, \quad c_2 = \sqrt{\frac{1 + \rho_{\text{RP}}^2}{3}}, \quad c_3 = \frac{\rho_{\text{RP}}}{3}, \quad (188)$$

For a tangent scale $\sigma_{12} > 0$, write $\sigma_{23} = c_2 \sigma_{12}^2$ and $\sigma_{13} = c_3 \sigma_{12}^3$. Eliminating ρ_{RP} gives

$$\left(\frac{\sigma_{23}}{\sigma_{12}^2} \right)^2 - 3 \left(\frac{\sigma_{13}}{\sigma_{12}^3} \right)^2 = \frac{1}{3}. \quad (189)$$

If $Y_r = P_r^X Y \neq 0$, then $X^{-2} Y_r$ reconstructs the corresponding Weyl shift up to central phase. In particular, $Y_\perp \neq 0$, $\rho_{\text{RP}} = 1$, and $\Delta_{\text{pol}} = 0$ force, after degree-zero gauge,

$$Y_\perp = a_1^2 e^{i\theta_2} X^2 S^\varepsilon, \quad \varepsilon \in \{1, 2\}. \quad (190)$$

Consequently every primitive second response must satisfy

$$Y_\perp^\dagger Y_\perp = a_1^4 \mathbf{1}, \quad \text{Tr } Y_\perp = \text{Tr}(Y_\perp^2) = 0, \quad Y_\perp^3 = (\det Y_\perp) \mathbf{1}, \quad [X, Y_\perp]^\dagger [X, Y_\perp] = 3a_1^4 \mathbf{1}. \quad (191)$$

These basis-free identities are necessary fast falsifiers before full polarization; the projective primitive line and its modulus require no integral input, while the remaining phase is a marked refinement.

Corollary 5.6 (Integral primitive Rees refinement) If the primitive line in (190) preserves the protected Eisenstein lattice, then $e^{i\theta_2} \in \mu_6$ and

$$\frac{\det[X, Y]}{a_1^6 (\det X)^3} \in \{+3\sqrt{3}i, -3\sqrt{3}i\}. \quad (192)$$

Integrality therefore quantizes the residual orientation; cycle-trivial local CP remains zero.

Source provenance is separate. Under a marked multiplicative map with line amplitudes λ_1, λ_2 ,

$$\rho_{\text{RP}} = \frac{|\lambda_2|}{|\lambda_1|^2} \chi_R = \frac{|\lambda_2|}{|\lambda_1|^2} \sqrt{\frac{3}{8}(4 + \kappa_g^2)}. \quad (193)$$

On the primitive finite branch $\rho_{\text{RP}} = 1$ this gives

$$r_{\text{src}} := \frac{|\lambda_2|}{|\lambda_1|^2} = \sqrt{\frac{8}{3(4 + \kappa_g^2)}} \leq \sqrt{\frac{2}{3}} = c_2, \quad (194)$$

with equality exactly on the geodesic source branch. Amplitude-preserving multiplicativity is therefore excluded on this primitive provenance branch.

Corollary 5.7 (Conditional source-Hodge jet folding) Assume a marked map is conformal on the three directions in (124) and sends $A = Q'$ to J_{rel} and $-Q^{(3)}$ to J_{com} . Then

$$q_* = 4 + \kappa_g^2, \quad \delta_q := \frac{\Delta_q^2}{\|J_{\text{rel}}\|_{\text{HS}}^2} = 9\kappa_g^2 + \dot{\kappa}_g^2, \quad q_* r_{\text{src}}^2 = \frac{8}{3}. \quad (195)$$

Equivalently, scalar folding is feasible exactly when $0 < r_{\text{src}}^2 \leq 2/3$ and $\delta_q \geq 24/r_{\text{src}}^2 - 36$, with $\kappa_g^2 = 8/(3r_{\text{src}}^2) - 4$ and $\dot{\kappa}_g^2 = \delta_q - 24/r_{\text{src}}^2 + 36$. The zero-defect corner forces $r_{\text{src}}^2 = 2/3$ and $q_* = 4$. This is a marked folding test with source Taylor and Hodge gradings kept distinct.

Physical relative mixing further requires $[H_u, H_d] \neq 0$, and quark CP requires $\det[H_u, H_d] \neq 0$.

Definition 5.8 (Hodge-Eisenstein matrix-completion branch) Let $L_{ij} = E_{ij} - E_{ji}$ and let c_2, c_3 be given by (188). Before assigning a common-to-relative amplitude, put

$$\begin{aligned} A_u(\epsilon) &= (\mathfrak{q} - \frac{1}{2})\epsilon L_{12} + (\mathfrak{t} - \frac{c_2}{2})\epsilon^2 L_{23} - \frac{\mathfrak{h}}{2}\epsilon^3 L_{13}, \\ A_d(\epsilon) &= (\mathfrak{q} + \frac{1}{2})\epsilon L_{12} + (\mathfrak{t} + \frac{c_2}{2})\epsilon^2 L_{23} + \frac{\mathfrak{h}}{2}\epsilon^3 L_{13}. \end{aligned} \quad (196)$$

Set $V_{\text{HE}} = e^{-A_u} e^{A_d}$. On the small positive branch, ϵ is fixed by $s_{12}(V_{\text{HE}}(\epsilon)) = \lambda_{\text{cen}}$; the unit linear 12 coefficient makes this reparametrization locally unique. The physical complex CKM class is attached only after (182).

Proposition 5.9 (Hodge-Eisenstein common-jet closure) *The leading 13 retention and vanishing connected order-four 23 correction give*

$$\mathfrak{t} = c_2(\mathfrak{q} - 3 + \mathfrak{q}^{-1}) + 6c_3(1 - \mathfrak{q}^{-1}), \quad \mathfrak{h} = c_2(1 - \frac{1}{2\mathfrak{q}}) - c_3(2 - \frac{3}{\mathfrak{q}}). \quad (197)$$

Put $D = c_2 - 6c_3$ and $H = \mathfrak{h} - c_2 + 2c_3$. For $D \neq 0$,

$$\mathcal{R}_H := 2H(\mathfrak{t} + 3c_2 - 6c_3 + 2H) + c_2 D = 0, \quad \mathfrak{q}_H = -\frac{D}{2H}. \quad (198)$$

If $H \neq 0$, this is the unique finite higher-jet amplitude; at $D = 0$, closure requires $H = 0$ and gives $\mathfrak{q}_H = 2 + \mathfrak{t}/c_2$. The microscopic gate is

$$\mathcal{R}_H = 0, \quad \Delta_{\mathfrak{q}} = 0, \quad \mathfrak{q}_* = \mathfrak{q}_H, \quad (199)$$

before comparison with $\mathfrak{q}_H/p_H = 4$. On the primitive integral branch, if the blind value is 4, (197) gives the completed coefficients and

$$\begin{aligned} s_{23} &= c_2 \lambda_{\text{cen}}^2 (1 + \lambda_{\text{cen}}^2/3 + O(\lambda_{\text{cen}}^4)), \\ s_{13} &= c_3 \lambda_{\text{cen}}^3 (1 + \kappa_{13}^{\text{HE}} \lambda_{\text{cen}}^2 + O(\lambda_{\text{cen}}^4)), \end{aligned} \quad (200)$$

where $\kappa_{13}^{\text{HE}} = 185/192 - 203\sqrt{6}/384$.

Corollary 5.10 (Relative-logarithm no-go and cubic split) Baker-Campbell-Hausdorff and (197) give

$$\log V_{\text{HE}} = \epsilon L_{12} + c_2 \epsilon^2 L_{23} + (c_3 - c_2/2) \epsilon^3 L_{13} + O(\epsilon^4). \quad (201)$$

The common amplitude cancels through cubic order. Hence relative CKM data cannot determine $\mathfrak{q}$ at this order, and the cubic coefficient is the exact retained/generated sum.

If the blind value in (199) is 4 and the marked incidence is a rank-one C_3 -equivariant $\mathbb{Z}[\omega]$ -linear map whose complexification is multiplication by $4u$ with $u \in \mu_6$, then

$$\text{SNF}(H_{\text{inc}}) = \text{diag}(4, 4), \quad \text{coker } H_{\text{inc}} \simeq (\mathbb{Z}/4\mathbb{Z})^2. \quad (202)$$

A different Smith type excludes this literal rank-one Eisenstein-scalar mechanism.

A common Rees rephasing leaves the Picard projectors invariant and sends $[X, Y] \mapsto e^{3i\phi}[X, Y]$. Thus ρ_{RP} , p , Δ_{pol} , and the normalized commutator cubic are independent of the common determinant phase.

Proposition 5.11 (Primitive cubic and integral CP separation) *The primitive CP-odd cubic is*

$$\mathcal{J}_{2j} = \frac{3}{2} \text{Re} \left(\frac{\det[X, Y]}{(\det X)^3} \right) = \frac{1}{2} \text{Re} \left(\frac{\text{Tr}([X, Y]^3)}{(\det X)^3} \right). \quad (203)$$

On the strict integral branch of Corollary 5.6, (192) is purely imaginary, so cycle-trivial transport gives $\mathcal{J}_{2j} = 0$. A nonzero physical CP phase therefore requires residual/torsor holonomy or another primitive response.

The direct calculations below use the primitive polarized finite branch $\rho_{\text{RP}} = 1$ and $\Delta_{\text{pol}} = 0$; Corollary 5.6 is needed only for its discrete integral orientation. Along the neutral affine tangent put

$$\kappa_{\text{SB}}(u) = 1 + \frac{20}{3}u, \quad \lambda_{\text{cen}}(u) = \frac{\kappa_{\text{SB}}(u)B}{A + \kappa_{\text{SB}}(u)B}. \quad (204)$$

Using (172) and $r_{WZ} = (m_W^{\text{prim}}/m_Z^{\text{prim}})^2$ gives the explicit one-parameter curve

$$\lambda_{\text{cen}}(u) = \frac{4(20u + 3)}{54 - 67u}, \quad r_{WZ}(u) = \frac{\frac{3}{7} - \frac{3}{2}u}{\frac{3}{7} - \frac{3}{2}u + \frac{6}{49}}. \quad (205)$$

At $u = 0$, (173) gives $s_{\text{det}}(0) = \lambda_{\text{cen}}(0) = 2/9$. Eliminating u gives

$$\lambda_{\text{cen}} = \frac{507s_{\text{det}} - 80}{67 + 360s_{\text{det}}}. \quad (206)$$

and, using (175),

$$\lambda_{\text{cen}} = \frac{427 - 507r_{WZ}}{427 - 360r_{WZ}}, \quad \frac{(m_h^{\text{prim}})^2}{(m_Z^{\text{prim}})^2} = \frac{14(831\lambda_{\text{cen}} + 100)}{15(169 - 120\lambda_{\text{cen}})}. \quad (207)$$

The physical small- u domain used below is $0 \leq u < 10/49$, equivalently $B_0 < A(u)$, so the ordered first two neutral amplitudes are preserved.

Proposition 5.12 (One-calibration identifiability) *On every connected branch in this domain,*

$$\lambda'_{\text{cen}}(u) = \frac{5124}{(54 - 67u)^2} > 0, \quad r'_{WZ}(u) = -\frac{196}{(18 - 49u)^2} < 0. \quad (208)$$

Hence either scalar fixes a unique u ; equivalently $u = 6(9\lambda_{\text{cen}} - 2)/(67\lambda_{\text{cen}} + 80)$. Once one observable is declared as the calibration, the other and the Higgs-to- Z ratio in (207) are fixed without retuning.

With $\rho_{\text{RP}} = 1$, exact exponentiation after (199) gives the CKM magnitudes in Table 5; (189) remains independent of this normalization.

The selector and torsor spectra are compatible readings of the same oriented norm-seven μ_6 orbit. The physical holonomy still requires the marked choice that identifies the family transport with the least scalar twist of this orbit. On that branch the one-phase lift gives

$$\begin{aligned} e^{i\delta_{\text{CKM}}^{\min}} &= \frac{143 + 180\sqrt{3}i}{343}, & \delta_{\text{CKM}}^{\min} &= \pi - 6 \arccos \frac{c + w}{2\sqrt{c + 2w}}, \\ \cos \delta_{\text{CKM}}^{\min} &= \frac{143}{343}, & \sin \delta_{\text{CKM}}^{\min} &= \frac{180\sqrt{3}}{343}, & \vartheta_{\Gamma}^{\min} &= \frac{1}{3}\delta_{\text{CKM}}^{\min} = 21.787^\circ. \end{aligned} \quad (209)$$

Galois conjugation preserves $N_E(\xi)$ and the unoriented torsor spectrum while interchanging the two noncentral traces; it sends $\delta_{\text{CKM}}^{\min} \mapsto -\delta_{\text{CKM}}^{\min}$ and hence $J_q \mapsto -J_q$ in (182). The same branch gives the J_q value recorded in Table 5. The branch inherits the statuses in Table 4; Corollary 3.36 identifies its holonomy as the archimedean reading of the same ξ whose ramified Weyl reading supplies the discrete neutral factor.

For invertible direct and neutral susceptibilities, exact triality is tested basis-freely by the scalarity residual of $\mathcal{S}_N P \mathcal{S}_D^{-1}$. Its vanishing is equivalent to $\mathcal{S}_N P = \kappa \mathcal{S}_D$ and implies

$$\mathcal{S}_N^\dagger \mathcal{S}_N = |\kappa|^2 P(\mathcal{S}_D^\dagger \mathcal{S}_D) P^{-1}. \quad (210)$$

Hence all singular values scale by $|\kappa|$ and $\kappa^3 = \det \mathcal{S}_N / \det \mathcal{S}_D$; determinant-only agreement is insufficient.

5.3.4 Charged-lepton vertex polarization and Hodge response

The charged-lepton block is the color-singlet direct channel $L_L \leftrightarrow e_L^c$ in Table 3. At the direct spectral depth of (168), the minimal sourced branch retains the vertex-resolved spectrum and no independent charged edge-transport coefficient. Its spectral representative is selected by

$$R_{\text{vert}}^\ell = H_\ell - \mathbb{E}_{\text{vert}}(H_\ell), \quad R_{\text{vert}}^\ell = 0, \quad (211)$$

so that $H_{\ell,0} \in \mathbb{C}[Z]$ after an auxiliary torsor origin has been chosen. This is the vertex polarization fixed above; a mass prediction additionally requires the completed direct block.

Let x_ℓ be the positive charged-lepton amplitude vector, $x_{\ell,r} = \sqrt{m_{\ell,r}}$ for $r \in \mathbb{Z}_3$. After removal of the overall scale, its real vertex coordinates may be written in the standard $\mathbb{Z}_3$ -symmetric Koide parametrization as

$$x_{\ell,r} = C_\ell \left(a_\ell + \cos \left(\phi_\ell + \frac{2\pi r}{3} \right) \right), \quad r \in \mathbb{Z}_3. \quad (212)$$

Put $\mathfrak{X}_0 = \sum_r x_{\ell,r}$ and $\mathfrak{X}_1 = \sum_r x_{\ell,r} \omega^{-r}$. Then $\mathfrak{X}_0 = 3C_\ell a_\ell$ and $\mathfrak{X}_1 = (3/2)C_\ell e^{i\phi_\ell}$, so $(C_\ell, a_\ell, \phi_\ell)$ is the real polar parametrization of the two independent Fourier coefficients. The balance and clock are the two components of

$$\mathcal{Z}_\ell = \log \frac{\mathfrak{X}_0}{\sqrt{2}\mathfrak{X}_1}, \quad \mathfrak{B}_\ell = 2 \operatorname{Re} \mathcal{Z}_\ell, \quad \phi_\ell = -\operatorname{Im} \mathcal{Z}_\ell. \quad (213)$$

The factor $\sqrt{2}$ is the real multiplicity of the nontrivial character pair in the finite-Heisenberg Parseval identity above. The scale-free residual is

$$R_{\text{bal}}^\ell = \mathfrak{B}_\ell - \Sigma_\ell, \quad (214)$$

where Σ_ℓ is the completed Schur correction to the invariant/complement norm balance. Proposition 3.37 gives directly

$$2a_\ell^2 = e^{\Sigma_\ell}. \quad (215)$$

Hence $\Sigma_\ell = 0$ gives $a_\ell = 1/\sqrt{2}$. Equivalently, $Q_\ell := \sum_r m_{\ell,r}/(\sum_r \sqrt{m_{\ell,r}})^2$ equals $2/3$ exactly when $|\mathfrak{X}_0|^2 = 2|\mathfrak{X}_1|^2$. The Koide value $Q_\ell = 2/3$ [76] and its geometric interpretation [77] are therefore the standard zero-correction balance reproduced by the Hodge-Parseval decomposition. The companion $\mathbb{Z}_3$ phase value $2/9$, attributed to the Brannen-Rosen parametrization and summarized in [78], is used below as a separate clock-lift reference. Near the balance branch, $a_\ell = 1/\sqrt{2} + \delta a_\ell$ gives $\Sigma_\ell = 2\sqrt{2}\delta a_\ell + O(\delta a_\ell^2)$.

For a positive $H_\ell = Y_\ell Y_\ell^\dagger$ with simple eigenvalues h_r and normalized eigenvectors u_r , put $x_\ell = \mathfrak{s}(H_\ell)$ with $\mathfrak{s}(H)_r = \lambda_r(H)^{1/4}$. Standard eigenvalue perturbation gives

$$(D\mathfrak{s}_{H_\ell}[\delta H_\ell])_r = \frac{\langle u_r, \delta H_\ell u_r \rangle}{4h_r^{3/4}}. \quad (216)$$

Writing $\delta\mathfrak{X}_0 = \sum_r (D\mathfrak{s}_{H_\ell}[\delta H_\ell])_r$ and $\delta\mathfrak{X}_1 = \sum_r \omega^{-r} (D\mathfrak{s}_{H_\ell}[\delta H_\ell])_r$, differentiation of (213) gives $D\mathcal{Z}_\ell = \delta\mathfrak{X}_0/\mathfrak{X}_0 - \delta\mathfrak{X}_1/\mathfrak{X}_1$. Define

$$\mathrm{Tr}_{\mathrm{bal}}^{(H_\ell)}(\delta H_\ell) = 2 \operatorname{Re} D\mathcal{Z}_\ell[\delta H_\ell], \quad \mathrm{Tr}_{\mathrm{clk}}^{(H_\ell)}(\delta H_\ell) = -\operatorname{Im} D\mathcal{Z}_\ell[\delta H_\ell] = D\phi_\ell[\delta H_\ell]. \quad (217)$$

These are the paired linear responses of the same Fourier coordinate.

Let a Hermitian perturbation T_3 of type V_3 satisfy Proposition 4.34, with its Schur compression represented in the charged-lepton Hermitian block. At $\Sigma_\ell(0) = 0$ and fixed $\phi_{\ell,0}$,

$$\begin{aligned} \Sigma_\ell(\varepsilon) &= -\varepsilon^2 \operatorname{Tr}_{\mathrm{bal}}^{(H_{\ell,0})} \left(PT_3 Q(K_{HH}^{(0)})^{-1} QT_3 P \right) + O(\varepsilon^3), \\ \phi_\ell(\varepsilon) - \phi_{\ell,0} &= -\varepsilon^2 \operatorname{Tr}_{\mathrm{clk}}^{(H_{\ell,0})} \left(PT_3 Q(K_{HH}^{(0)})^{-1} QT_3 P \right) + O(\varepsilon^3). \end{aligned} \quad (218)$$

Writing $\mathcal{S}_\ell^H$ for the Hermitian completed Schur datum gives $\Sigma_\ell = \operatorname{Tr}_{\mathrm{bal}}^{(H_{\ell,0})}(\mathcal{S}_\ell^H) + O(\|\mathcal{S}_\ell^H\|^2)$ and $\phi_\ell - \phi_{\ell,0} = \operatorname{Tr}_{\mathrm{clk}}^{(H_{\ell,0})}(\mathcal{S}_\ell^H) + O(\|\mathcal{S}_\ell^H\|^2)$. The Brannen-Rosen value $2/9$ is retained below only as an external clock-lift reference. No identification of $\phi_{\ell,0}$ with the neutral determinant coordinate in (173) is imposed. Since (171) has zero torsor-clock degree, the neutral determinant coordinates respond at first order in the scalar tangent, while the minimal charged clock responds only through the noncentral Schur datum in (218). The overall scale C_ℓ and the charged clock origin remain independent.

5.4 Neutral determinant, Schur response, and PMNS

The neutral sector is the $|\Delta(B-L)| = 2$ Schur problem of Table 3. Let D_ν be the neutral Dirac bridge, R_N the heavy singlet family denominator, M_R its scale, and A_L the active Majorana correction.

$$K_N = \begin{pmatrix} A_L & D_\nu^T \\ D_\nu & M_R R_N \end{pmatrix}. \quad (219)$$

The Majorana entries are taken complex symmetric, $A_L^T = A_L$ and $R_N^T = R_N$, so that $K_N^T = K_N$. Their physical neutral bases are defined by Takagi factorization. The heavy scale is separated from its family shape by the convention $|\det R_N| = 1$, so that $M_R = |\det(M_R R_N)|^{1/3}$. This convention removes the rescaling redundancy between M_R and R_N and carries no additional spectral prediction.

The quadratic half-flux normalization used in the minimal neutral branch is

$$S_M = \frac{1}{4} S_{\mathrm{prim}}, \quad M_R = M_{\mathrm{Pl}} e^{-S_M}. \quad (220)$$

The unreduced Planck mass convention $M_{\mathrm{Pl}} = 1.22089 \times 10^{19} \text{ GeV}$ is used for the completed numerical reading. The fraction in (220) is a completed geometric scale assumption beyond the carrier theorem. With $S_{\mathrm{prim}} = 4\pi^2$, this scale is $M_R = M_{\mathrm{Pl}} e^{-\pi^2}$. It does not define a holomorphic square root of L_Γ : a line M with $M^{\otimes 2} \simeq L_\Gamma$ would imply $2c_1(M) = 1$ in $H^2(\mathbb{CP}^1, \mathbb{Z})$, which has no integral solution.

The Pfaffian, seesaw, and Takagi readings are collected by the same complex-symmetric Schur datum. Put $H := M_R R_N$ and $E_{\mathrm{sp}} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$.

Proposition 5.13 (Symmetric Schur-Pfaffian package) *For invertible H , let $\mathbb{A}_N = E_{\text{sp}} \otimes K_N$ and $\mathbb{A}_H = E_{\text{sp}} \otimes H$. Then $\mathbb{A}_N$ and $\mathbb{A}_H$ are skew, $\text{Pf}(\mathbb{A}_N) = -\det K_N$, $\text{Pf}(\mathbb{A}_H) = -\det H$, and the protected reduction is*

$$K_\nu^{\text{eff}} = A_L - D_\nu^T H^{-1} D_\nu, \quad \det K_\nu^{\text{eff}} = \frac{\text{Pf}(\mathbb{A}_N)}{\text{Pf}(\mathbb{A}_H)}, \quad m_1 m_2 m_3 = \left| \frac{\text{Pf}(\mathbb{A}_N)}{\text{Pf}(\mathbb{A}_H)} \right|. \quad (221)$$

Along a branch parameter t , write $H(t) = \sum_{n \geq 0} t^n H_n$, $D_\nu(t) = \sum_{n \geq 0} t^n D_n$, $A_L(t) = \sum_{n \geq 0} t^n A_n$, $H(t)^{-1} = \sum_{n \geq 0} t^n Q_n$, and $K_\nu^{\text{eff}}(t) = \sum_{n \geq 0} t^n K_{\nu,n}$. Then

$$Q_0 = H_0^{-1}, \quad Q_n = -Q_0 \sum_{r=1}^n H_r Q_{n-r}, \quad K_{\nu,n} = A_n - \sum_{i+j+k=n} D_i^T Q_j D_k. \quad (222)$$

If $U_0^T K_{\nu,0} U_0 = \text{diag}(\sigma_i)$ has simple positive Takagi values and $E := U_0^T K_{\nu,1} U_0$, the off-diagonal generator satisfies

$$\text{Re } \Omega_{ij} = -\frac{\text{Re } E_{ij}}{\sigma_i - \sigma_j}, \quad \text{Im } \Omega_{ij} = -\frac{\text{Im } E_{ij}}{\sigma_i + \sigma_j}, \quad i \neq j. \quad (223)$$

Degenerate Takagi subspaces are resolved before this formula is applied.

Proof The Pfaffian identities are the product-orientation identities for $E_{\text{sp}} \otimes K$; block elimination gives the Schur and determinant formulas, coefficient comparison gives the inverse recursion, and differentiation of the Takagi relation gives the final response. $\square$

In the pure type-I subbranch $A_L = 0$, (221) reduces to

$$m_1 m_2 m_3 = \frac{|\det D_\nu|^2}{M_R^3} \quad (|\det R_N| = 1). \quad (224)$$

The half-flux scale fixes the denominator scale, while an absolute light-neutrino reading additionally requires the determinant shape of the neutral Dirac bridge. The skew doubling is auxiliary: its paired singular values come from the two-dimensional E_{sp} factor and impose no additional Takagi degeneracy.

The normalized determinant-suppressed type-I branch used below takes $R_N = \mathbf{1}$ and $A_L = 0$. Before imposing (177), put $\Sigma_\nu = \text{diag}(B_0, A, 1 + S_{\text{prim}}^{-1})$. In the charged-vertex/heavy-singlet basis its Takagi-compatible Dirac orientation is fixed by

$$D_\nu = m_D \Sigma_\nu U_\nu^\dagger, \quad r_\nu := \frac{|\det D_\nu|}{m_D^3} = B_0 A(u) (1 + S_{\text{prim}}^{-1}). \quad (225)$$

On the common lock, the fixed benchmark value $u = 1/(64\pi^2)$ gives $r_\nu = 0.05351$. Substitution in (221) gives

$$K_\nu^{\text{eff}} = -\frac{m_D^2}{M_R} U_\nu^* \Sigma_\nu^2 U_\nu^\dagger, \quad m_i = \frac{m_D^2}{M_R} (\Sigma_\nu)_{ii}^2. \quad (226)$$

The common sign in (226) is absorbed by a common Takagi column phase and does not affect the oscillation quotient. Equation (225) is a normalized neutral-shape and orientation completion; no rank reduction is imposed.

Corollary 5.14 (Scale-free neutral oscillation shape) On the determinant-suppressed type-I branch,

$$\mathcal{R}_\nu := \frac{\Delta m_{21}^2}{\Delta m_{31}^2} = \frac{A^4 - B_0^4}{(1 + S_{\text{prim}}^{-1})^4 - B_0^4}. \quad (227)$$

Under the common lock (177),

$$\mathcal{R}_\nu(u) = \frac{\left(\frac{3}{7} - \frac{3}{2}u\right)^4 - \left(\frac{6}{49}\right)^4}{(1 + 16u)^4 - \left(\frac{6}{49}\right)^4}, \quad (228)$$

which is strictly decreasing for $0 \leq u < 10/49$. Eliminating u with $r := m_W^2/m_Z^2$ gives the calibration-free cross-sector test

$$\mathcal{R}_\nu = \frac{6^4[r^4 - (1-r)^4]}{(273 - 337r)^4 - 6^4(1-r)^4}. \quad (229)$$

Thus any one scalar anchor predicts both the remaining neutral-cell ratios and $\mathcal{R}_\nu$ on the additional lock; without (177), the neutrino shape remains an independent completed parameter.

The charged-lepton Hermitian block determines a unitary basis U_e , while the complex symmetric effective neutral block (221) determines its Takagi basis U_ν . The lepton mixing matrix is the relative basis

$$U_{\text{PMNS}} = U_e^\dagger U_\nu. \quad (230)$$

A common Riesz frame cancels from (230): if $U_e = RT_e$ and $U_\nu = RT_\nu$, then $U_{\text{PMNS}} = T_e^\dagger T_\nu$. Thus only the relative sectoral Hermitian/Takagi responses enter the mixing matrix. PMNS compares the $B - L$ preserving charged vertex polarization with the $B - L$ breaking neutral Schur basis. The charged baseline (211) and the circulant neutral denominator select the complementary polarizations of the A_2 phase plane in (49); their leading relative frame is the real Fourier frame, with susceptibility controlled by the resolved neutral gaps.

At oscillation depth, right multiplication $U_\nu \mapsto U_\nu D_\eta$ by a diagonal phase matrix leaves the oscillation probabilities invariant. The neutrinoless-double-beta effective mass,

$$m_{\beta\beta} = \left| \sum_i m_i (U_{\text{PMNS}})_{ei}^2 \right|, \quad (231)$$

depends on the relative column phases. These phases are therefore not basic on the oscillation-only mixing quotient and must be retained when (231) is protected downstream.

A useful diagnostic denominator on the central carrier is the circulant Majorana shape

$$R_N^{\text{circ}} = m\mathbf{1} - r(S + S^\dagger). \quad (232)$$

Its eigenvalues are $m - 2r, m + r, m + r$. If $m = 2r + \delta$ with $0 < \delta \ll r$, then

$$(R_N^{\text{circ}})^{-1} = \frac{1}{3r + \delta} \mathbf{1} + \frac{1}{3} \left(\frac{1}{\delta} - \frac{1}{3r + \delta} \right) J, \quad J_{ab} = 1. \quad (233)$$

The scale-shape convention gives $\delta(3r + \delta)^2 = 1$ for this positive circulant representative. With $x = \delta/r$, the two parameters reduce to the single dimensionless softness x , with $r = [x(3 + x)^2]^{-1/3}$. The denominator is diagonal in the transport frame and its exact double eigenspace is the $A_{2,\mathbb{R}}$ plane of (49). A small clock-even perturbation $Z + Z^\dagger$ resolves the canonical real Fourier frame of Proposition 3.37. For the physical column ordering, U_{TBM} places the democratic vector in the second column, and $L_{ij} = E_{ij} - E_{ji}$ is taken in this ordered basis.

Lemma 5.15 (TBM torsor Lie closure) Put $H_0 = H_S(0)$. The operators in (73) satisfy

$$U_{\text{TBM}}^\top G U_{\text{TBM}} = -L_{13}, \quad U_{\text{TBM}}^\top \left(-\frac{\sqrt{2}}{3} [K_Z, H_0] \right) U_{\text{TBM}} = L_{23}, \quad U_{\text{TBM}}^\top \left(\frac{\sqrt{2}}{3} [[K_Z, H_0], G] \right) U_{\text{TBM}} = L_{12}. \quad (234)$$

Hence G and $[K_Z, H_0]$ generate $\mathfrak{so}(3)$ on the real Fourier frame.

Proof The first two identities are the real-Fourier readings of (73); the third is their commutator. $\square$

Since G fixes the invariant line $P_1 \mathcal{H}_{\text{cen}}$, a neutral correction contained in the circulant algebra preserves the democratic PMNS column in the minimal charged vertex branch. With the democratic vector assigned to the second column, this gives

$$|(U_{\text{PMNS}})_{e2}|^2 = \frac{1}{3}, \quad \sin^2 \theta_{12} = \frac{1}{3 \cos^2 \theta_{13}}. \quad (235)$$

The restricted pure-circulant repair therefore does not reproduce the central solar and reactor values of [79]. The required non-circulant real rotation channels are fixed by Lemma 5.15 without adjoining a new family generator: G supplies the 13 direction, the normalized detector commutator supplies the 23 direction, and their commutator supplies the 12 direction.

The finite Lie closure fixes the available real 13, 23, and 12 directions. In the same ordered real-Fourier frame, direct evaluation gives $(U_{\text{TBM}}^\top K_Z U_{\text{TBM}})_{23} = 1/\sqrt{2}$, equal to the zero-balance factor a_ℓ from (215). This $1/\sqrt{2}$ coefficient is the finite-module content of the 23 normalization; it is written as $\Xi_{23} = \sqrt{3/8}$ only in the convention of (238). Its physical use still requires unit microscopic descent. The other channel amplitudes are not fixed by Lie closure alone.

Transposition preserves each low-symbol space $\mathcal{A}_r$ by Proposition 3.37. Hence, for a heavy inverse in $\mathcal{A}_0$ and degree-pure $D_1 \in \mathcal{A}_1$, $D_2 \in \mathcal{A}_2$, the mixed symmetric Schur contraction $D_1^\top H^{-1} D_2 + D_2^\top H^{-1} D_1$ lies in $\mathcal{A}_0$. A non-circulant cubic 12 response is therefore not fixed by low-symbol degree alone.

Proposition 5.16 (Canonical Eisenstein neutral relay) *Let $K \in M_2(\mathbb{Z})$ have odd trace t and discriminant -3 . Then $J = 2K - t\mathbf{1}$ obeys $J^2 = -3\mathbf{1}$ and $W = (J - \mathbf{1})/2$ obeys $W^2 + W + \mathbf{1} = 0$. Since $\mathbb{Z}[\omega]$ is a PID, one integral marked basis puts $W = W_0 := \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}$ up to orientation, so every cell on the same marked relay is $K = a\mathbf{1} + W_0$ with $a = (t + 1)/2$ and $\det K = a^2 - a + 1$. The central affine map*

$$\Phi(K) = K + \frac{\det K - \text{Tr } K}{2} \mathbf{1} \quad (236)$$

preserves the same W_0 and sends $(\text{Tr } K, \det K) = (5, 7) \mapsto (7, 13) \mapsto (13, 43)$. Thus one passed integral seed fixes the noncentral matrix form of the full relay; later cells carry only the central trace update.

The ordered real-Fourier convention is

$$U_{\text{TBM}} = \begin{pmatrix} \sqrt{2/3} & 1/\sqrt{3} & 0 \\ -1/\sqrt{6} & 1/\sqrt{3} & 1/\sqrt{2} \\ -1/\sqrt{6} & 1/\sqrt{3} & -1/\sqrt{2} \end{pmatrix}, \quad R_{ij}(\theta) = \exp(\theta L_{ij}), \quad (237)$$

with L_{ij} as in Definition 5.8; the phased column rotation has $(R_{23})_{23} = \sin \alpha e^{-i\phi}$ and $(R_{23})_{32} = -\sin \alpha e^{i\phi}$, with diagonal 23 entries $\cos \alpha$.

Definition 5.17 (Normalized neutral three-channel branch) For physical same-marking descent, the relay normalizations are the consecutive marked trace ratios $\Xi_{12} := \text{Tr } K_1 / \text{Tr } K_0$ and $\Xi_{13} := \text{Tr } K_2 / \text{Tr } K_1$; Proposition 5.16 then gives 7/5 and 13/7. The retained neutral basis, expressed in the charged-vertex eigenframe, uses (234),

$$U_\nu = U_{\text{TBM}} R_{12}(\beta_\nu) R_{13}(\gamma_\nu) R_{23}(\alpha_\nu, \phi_\nu), \quad \alpha_\nu = \frac{2}{\sqrt{3}} \Xi_{23} \lambda_{\text{cen}}, \quad \gamma_\nu = -\Xi_{13} \lambda_{\text{cen}}^2, \quad \beta_\nu = -\Xi_{12} \lambda_{\text{cen}}^3, \quad \phi_\nu = \frac{4\pi}{3} - \frac{2}{5} r_\vartheta \vartheta_\Gamma. \quad (238)$$

The target values are

$$(\Xi_{23}, \Xi_{13}, \Xi_{12})_{\text{target}} = \left(\sqrt{\frac{3}{8}}, \frac{13}{7}, \frac{7}{5} \right). \quad (239)$$

The 23 target has finite-module normalization. The 12 and 13 targets are simultaneous conditional outputs of the trace-ratio descent when the physical marked cells pass the relay operator test; no separate continuous relay coefficients are then available.

Proposition 5.18 (Pre-PMNS neutral channel tomography) *If λ_{cen} is fixed independently by (207), the neutral Lie jets give*

$$\hat{\Xi}_{23} = \frac{\sqrt{3} \alpha_\nu}{2 \lambda_{\text{cen}}}, \quad \hat{\Xi}_{13} = -\frac{\gamma_\nu}{\lambda_{\text{cen}}^2}, \quad \hat{\Xi}_{12} = -\frac{\beta_\nu}{\lambda_{\text{cen}}^3}. \quad (240)$$

Thus the three targets are tested before Takagi exponentiation. On the affine-relay branch they additionally obey the necessary scalar gate

$$\hat{\Xi}_{12} \hat{\Xi}_{13} = \frac{13}{5}. \quad (241)$$

On the unit-23 branch,

$$3 \sin^2 \theta_{12}^{\text{PMNS}} + 2 \sin^2 \theta_{13}^{\text{PMNS}} = 1 - 2\sqrt{2} \Xi_{12} \lambda_{\text{cen}}^3 + O(\lambda_{\text{cen}}^4), \quad J_{\text{PMNS}} = \frac{\lambda_{\text{cen}}}{6\sqrt{3}} \sin \phi_\nu + O(\lambda_{\text{cen}}^3). \quad (242)$$

The first relation tests the cubic 12 channel independently of Ξ_{13} and ϕ_ν at the displayed order.

The $4\pi/3$ offset is the ramified Weyl-cocycle phase fixed by (70); in the unsymmetrized clock-shift frame it is the ω^2 coefficient of the oriented ZS^2 term in (74). The second term in ϕ_ν is the integrated outer- C attachment (151). The directions in (234) and this determinant-centered phase weight are fixed before the microscopic descent factors implicit in (239) are specified. The transpose-grading observation in the neutral branch keeps the cubic response at completed Jacobian depth. On the physical common-lift branch, $r_\vartheta = 1$ follows from (148); then (238) and (209) give $\phi_\nu = 4\pi/3 - (2/15) \delta_{\text{CKM}}^{\text{min}}$, with no independent continuous leptonic CP seed. The CP invariant is read from the standard rephasing-invariant product $J_{\text{PMNS}} = \Im(U_{e1} U_{\mu 2} U_{e2}^* U_{\mu 1}^*)$.

In the finite-Heisenberg reading, the quark blocks perturb the same transport polarization, while the selected lepton branch compares the vertex and Fourier-dual polarizations. The inverse Sylvester response (183) supplies the remaining spectral susceptibility: direct gaps suppress CKM motion, while the neutral character doublet is soft before clock-even resolution. This remains conditional on the stated polarization and neutral

completions.

A symmetric perturbation δK_N induces the skew perturbation $E_{\text{sp}} \otimes \delta K_N$ of Proposition 5.13. Preservation of the neutral quadratic-form class therefore requires no additional family-space skew constraint; the remaining restrictions are the $B - L$ filtration, the retained gaps, and the chosen determinant orientation.

The projective-color expectation (71) may also be applied to a Hermitian representative of the effective neutral block, for example $K_\nu^{\text{eff}}(K_\nu^{\text{eff}})^\dagger$. The resulting entropy measures the non-circulant component of this Hermitian representative, while the basis response is controlled by the resolved neutral gaps. Hence the direct and neutral sectors use the same sectoral entropy with different spectral susceptibilities: large direct gaps suppress CKM motion through (184), whereas the soft denominator in (233) can amplify PMNS motion. Outside the normalized type-I shape (225), the shapes of D_ν , R_N , and A_L remain completed data. The relative-basis reading of the normalized three-channel branch is fixed by (238).

5.5 Radial, contact, and common scale completion

5.5.1 Radial and decay diagnostics

The neutral bosonic cell gives a separate radial test. Let $\delta A := \partial_{\log v} A$, $\delta B := \partial_{\log v} B$, and $\delta C := \partial_{\log v} C$ at the reference point. If the finite neutral cell is radially locked, these derivatives vanish and the standard hWW and hZZ normalizations are recovered. A trace-neutral leakage direction satisfies $\delta A + \delta B + \delta C = 0$. The corresponding first radial coupling modifiers are

$$\kappa_W - 1 = \frac{1}{2} \frac{\delta A}{A}, \quad \kappa_Z - 1 = \frac{1}{2} \frac{\delta A + \delta B}{A + B}. \quad (243)$$

If radial motion is tangent to the same affine u line, then $\delta A = -3\dot{u}/2$, $\delta B = 0$ and the leakage amplitude cancels: $\kappa_Z - 1 = r_{WZ}(\kappa_W - 1)$. This parameter-free relation is a conditional completed-cell test. Additional scalar directions invisible to P_{obs} and the filtered labels are removed by Proposition 5.2; projected-curvature stress remains a separate completed comparison.

Fermion masses and Yukawa hierarchies remain singular values of the compressed weak-bridge blocks (166). A vortex-Yukawa relation, if imposed, is therefore a sectoral branch condition on those matrix elements. In a locked scalar island one may record the branch condition as $y_f = \cosh \varphi_f - 1$ and $g_{hff} = m_f/v$ at zero radial leakage. A nonzero $\partial_{\log v} \varphi_f$ is a sectoral leakage diagnostic; the exterior channel remains fixed.

Decay channels are read as allowed punctures of the Riesz gap. For a parent island i and a final channel F , the kinematic excess is

$$\Delta_{i \rightarrow F} = m_i - \sum_{a \in F} m_a. \quad (244)$$

A channel is open only when (244) is non-negative and the corresponding exterior, weak-bridge, or contact matrix element is nonzero. Standard Higgs decay formulae and loop form factors are used only as comparison readings of the bridge-resolvent calculation [80].

5.5.2 Doubled top-form contact and scale conversion

The contact sector is the doubled top-form class in Table 3. Its baryon-violating component obeys (165); the remaining contractions preserve B and L separately. The contact coefficient is a completed Schur-Kuranishi datum separated from the one-Higgs bridge and from the Majorana denominator. The separation is preserved by Proposition 5.2.

A direct tunnelling assignment $M_X = M_{\text{Pl}} \exp(-S_{\text{prim}})$ would put the baryon-violating contact near the electroweak scale and is therefore excluded as a proton-decay reading. The proton-decay comparison convention is the standard GUT effective-operator one [81]. The doubled top-form assignment places the contact at the tower scale. For an order-one finite coefficient in its baryon-violating component and $M_X \simeq M_{\text{Pl}}$, the standard dimension-six estimate gives $\tau_p \sim 10^{48}$ yr. Under the alternative completed-branch assignment $M_X = M_{\text{Pl}}/(2\pi)^2 \simeq 3 \times 10^{17}$ GeV, the same estimate gives $\tau_p \sim 10^{41}$ yr. Here $(2\pi)^2$ is used as the tower-period normalization; no identification with the exponential action reading of S_{prim} is used beyond their equality under (177). An observed $p \rightarrow e^+ \pi^0$ signal near 10^{35} yr would rule out the minimal doubled-top-form contact assignment.

The trace normalizations (34) and (35) fix the relative finite normalization of the three gauge factors. A common coupling prediction additionally requires a matching scale, the kinetic normalization of the completed gauge block, and the heavy threshold spectrum generated by the same boundary-cycle completion. The $SU(3)$ coupling therefore belongs to the completed scale test together with the electroweak couplings, and its low-energy value is left to this common matching problem.

Threshold conversion and running are applied after the finite branch and its heavy denominators have been fixed. The matching map is evaluated on the same Schur-compressed operator, with heavy-field elimination organized by the covariant derivative expansion of [82]. One-loop matching across separated mass scales is represented by [83]. Entries generated by one heavy boundary-cycle completion inherit one matching prescription within the same reconstruction class; their masses and threshold positions need not coincide. Independent sector-by-sector threshold functions constitute an additional completed input.

A threshold-free common-kinetic diagnostic may be evaluated before the heavy spectrum is specified. With one common gauge-block kinetic coefficient, (34) identifies the $SU(2)$ and $SU(3)$ matching conditions, while α_1 is taken in the canonical $k_Y = 5/3$ normalization of (35). Using the one-loop Standard-Model coefficients $b_i = (41/10, -19/6, -7)$ and the reference electroweak and strong-coupling inputs of [69], the central crossings are $\mu_{23} = 9.7 \times 10^{16}$ GeV from $\alpha_2 = \alpha_3$ and $\mu_{12} = 1.0 \times 10^{13}$ GeV from $\alpha_1 = \alpha_2$. At μ_{23} one finds $\alpha_1^{-1} - \alpha_2^{-1} = -10.6$, about 23% of α_2^{-1} . The threshold-free one-scale subbranch is therefore excluded at this order.

In the fermionic kinetic subbranch of Definition 4.27, Proposition 4.33 supplies $f = 3$ Dirac-form adjoint pairs. The structural dictionary (45) gives $\Delta b_i = 4T(\text{adj}_i)$, hence $(\Delta b_1, \Delta b_2, \Delta b_3) = (0, 8, 12)$, $M_2 = 3M_H/80$, and $M_3 = 3M_H/5$. At the Standard-Model crossing μ_{23} , preservation of $\alpha_2 = \alpha_3$ gives

$$\frac{\mu_{23}}{M_3} = d_F^{w/(c-w)} = 16^2, \quad \frac{\mu_{23}}{M_2} = d_F^{c/(c-w)} = 16^3, \quad \Delta_{12}^{\text{th}} = \frac{48 \log 2}{\pi} = 10.5905. \quad (245)$$

The exponents are the solution of the one-loop matching equation; for the selected split they become integers because $c - w = 1$. Once the normalized high-gap and kinetic completion is imposed, the gap coefficient and $153.6 = d_F^2/k_Y$ are dependent readings of the same structural integers. The central inputs give $M_H^{\text{RG}} = \mu_{23}/153.6 = 6.3151 \times 10^{14}$ GeV and the independent half-flux completion gives $M_R = 6.3148 \times 10^{14}$ GeV. Propagating the current $\alpha_s(M_Z)$ and $\hat{s}_Z^2$ input uncertainties summarized in [69] gives an input-only one-loop logarithmic width $\sigma_{\log M_H} \simeq 0.107$, equivalently $M_H^{\text{RG}} \simeq 6.32_{-0.64}^{+0.71} \times 10^{14}$ GeV at one standard deviation. The central offset from M_R is about 0.007 of this input-only width, so the proximity is retained as a common-scale comparison; two-loop running, finite matching, and threshold uncertainties remain outside (245).

For a local no-refit audit, let $\mathcal{C}$ collect observables used in calibration and $\mathcal{T}$ those reserved for testing. At a regular point p_0 , first-order constancy of $\mathcal{T}$ along the calibration fiber is equivalent to $\ker DC(p_0) \subseteq \ker DT(p_0)$, or equivalently $\text{row } DT(p_0) \subseteq \text{row } DC(p_0)$. The coordinate-invariant defect $r_{\text{def}} := \text{rank} \left(\begin{smallmatrix} DC \\ DT \end{smallmatrix} \right) - \text{rank } DC$ counts the remaining first-order test sensitivities; $r_{\text{def}} = 0$ is local no-refit identifiability provided the test observables were not themselves used in calibration. At the effective level an unrestricted $M_3(\mathbb{C})$ family coefficient space is flavor-surjective, so quantitative flavor comparisons acquire content only after the microscopic or completed coefficient restrictions recorded below have been frozen.

5.6 Quantitative status and numerical readings

The quantitative relations are evaluated after spectral-before-mixing descent and reduced Schur closure. Reference conventions follow [69] and [79]; the final frozen-branch comparison uses the 2026 Particle Data Group review [84], with [85] retained for oscillation-fit cross-checks. The ledger separates structural data, marked descent, normalized completion, and absolute scale. One value of u is used throughout every u -dependent numerical row, and downstream readings inherit the status of every required upstream gate.

Table 4: Consolidated provenance and falsification ledger.

Datum or output	Status	Decisive requirement or falsifier
primitive defect, A_2 , minimal $3 + 2$ carrier, global $\mathbb{Z}_6$	structural	primitive source grades, faithful marked support, determinant and exterior trace identities
three-point family torsor and integral marking	structural/projective	Proposition 3.29; non-torsion determinant phase is central, not a literal lattice automorphism
constitutive projector and $(1, 4, 5; 7)$ packet	exact certificates with marked provenance	(96), source Hodge sequence, and norm-seven arithmetic must agree
first family response	microscopic gate	(138); central-normalized projective part must match the marked regular C_3 cycle
primitive second response/Rees polarization	microscopic/projective gate	require (191), $Y_{\perp} \neq 0$, $\rho_{\text{RP}} = 1$, and $\Delta_{\text{pol}} = 0$
source-Hodge attachment	conditional blind gate	(195) plus the feasibility cone before comparison with four
physical common connection	2:3:5 conditional curvature gate	(150); failure cannot be repaired by u

Datum or output	Status	Decisive requirement or falsifier
neutral Eisenstein seed and 23, 13, 12 descent	conditional relay	marked Proposition 5.16 and (240); one seed fixes the noncentral relay form
one- u neutral/direct branch	one calibration plus no-refit tests	Proposition 5.12 and (229); the u - S_{prim} lock remains conditional
Majorana and absolute scales	completed subbranches	unresolved Majorana phases, κ_Ω , kinetic matching, and dimensionful normalization remain separate
equivariant Callias/global attachment	global conditional	target index character $1 \oplus \chi \oplus \chi^2$ and a common-base gapped Riesz-Callias deformation

The arithmetic packet refines only the corresponding finite rows. Absolute normalization and global Callias attachment remain separate; the projected rate and family determinant phase are linked only after the physical common 2:3:5 lift is realized.

For the direct Hodge completion, a reference common-scale conversion of the PDG quark-mass inputs [69] gives $Q_d^{\text{obs}} = 0.74803 \pm 0.00280$ and $Q_u^{\text{obs}} = 0.88763 \pm 0.00232$, to be compared with 3/4 and 8/9 without assuming exact renormalization-group invariance. Using the observed CKM angles as leading Rees proxies, the left-hand side of (189) is 0.36065 ± 0.02858 , while the Hodge-Eisenstein unitary completion gives 0.36683 against the structural value 1/3. The corresponding leading inversions are $\rho_{\text{RP}}^{(23)} = 1.024 \pm 0.037$ and $\rho_{\text{RP}}^{(13)} = 0.983 \pm 0.025$; they share s_{12} and remain diagnostics rather than independent reconstructions once Proposition 5.9 is imposed.

5.6.1 Common- u numerical ledger

The numerical illustration freezes the reference value $u = 1/(64\pi^2)$ as a benchmark before any comparison is made. In the one-calibration reading, this value is replaced by the unique u obtained from the predeclared calibration observable, and every u -dependent row is recomputed together. Table 5 therefore records one frozen branch rather than a collection of independently adjusted sectoral values. Absolute quantities whose normalization requires additional input remain marked separately.

Table 5: Frozen-branch numerical ledger. The same u is used in every u -dependent row. Gaussian pulls are shown only when the one-dimensional uncertainty used in the comparison is displayed; otherwise a central-value or fit diagnostic is reported.

Quantity	Frozen branch	Reference	Comparison / status
m_W/m_Z	0.881372	PDG 2026: 0.88136 ± 0.00015	tree-level matching: $+0.08\sigma$
m_h/m_Z	1.373422	PDG 2026: 1.37222 ± 0.00121	tree-level matching: $+0.99\sigma$
$\mathcal{R}_\nu$	0.0296554	NuFIT 6.1: 0.029558	central offset $+9.74 \times 10^{-5}$ fit diagnostic central offsets
CKM angles	$s_{12} = 0.2250096$	PDG 2026: $s_{12} = 0.22517$	-1.60×10^{-4}
	$s_{23} = 0.0420361$	$s_{23} = 0.04189$	$+1.46 \times 10^{-4}$
	$s_{13} = 0.0037353$	$s_{13} = 0.003763$ PDG 2026	-2.77×10^{-5}
CKM CP	$J_q = 3.1262 \times 10^{-5}$	$J_q = (3.16^{+0.13}_{-0.11}) \times 10^{-5}$	-0.31σ
	$\delta_{\text{CKM}} = 65.360^\circ$	$\delta_{\text{CKM}} = 1.154 \pm 0.025$ rad PDG 2026	-0.53σ
CKM triangle	$\bar{\rho} = 0.16074$	$\bar{\rho} = 0.1576$	central offsets $+0.00314$
	$\bar{\eta} = 0.34986$	$\bar{\eta} = 0.3556$	-0.00574
	$\sin 2\beta = 0.71030$	$\sin 2\beta = 0.710$ PDG 2026	$+0.00030$
PMNS angles	$\sin^2 \theta_{12} = 0.30890$	$\sin^2 \theta_{12} = 0.307 \pm 0.012$	$+0.16\sigma$
	$\sin^2 \theta_{13} = 0.02241$	$\sin^2 \theta_{13} = 0.0217 \pm 0.0007$	$+1.01\sigma$
	$\sin^2 \theta_{23} = 0.47033$	$\sin^2 \theta_{23} = 0.533 \pm 0.018$	$\Delta_{23} = -0.06267$; octant diagnostic central displacement -10.7°
PMNS δ_{PMNS}	207.13°	PDG 2026: $1.21^{+0.19}_{-0.22}\pi$	likelihood diagnostic
CP tower $n = 1, 7$	$180.000^\circ, 65.360^\circ$ $n = 13, 19: 96.613^\circ, 39.521^\circ$ $n = 31, 37: 126.310^\circ, 28.290^\circ$	primitive commensurate scan	$n = 1$ nonsimple $n = 7$ selected
CP tower $n \geq 13$	$n = 43, 49: 134.465^\circ, 49.279^\circ$	higher simple classes	arithmetic diagnostic

Quantity	Frozen branch	Reference	Comparison / status
charged shape	$a_\ell = 0.70710912, Q_\ell = 0.66666446$ $\phi_\ell = 0.22222476$	balance and clock closures	u -independent diagnostic
charged m_τ	$m_\tau^{\text{bal}} = 1776.9690 \text{ MeV}$		
PMNS J_{PMNS}	$m_\tau^{\text{clk}} = 1776.9665 \text{ MeV}$ -0.01539	separation 2.5 keV $ J /J_{\text{max}} = 0.456$	inherited closures dependent reading phase envelope
PMNS absolute masses	$1.016 \leq m_{\beta\beta} \leq 4.250 \text{ meV}$ $m_\beta = 8.97 \text{ meV}$	unresolved Majorana phases	3.89 meV point subbranch

5.6.2 Common- u central-family and CKM readings

The central cell is the one-dimensional curve (205). For the frozen reference value $u = 1/(64\pi^2)$, (204) gives

$$\lambda_{\text{cen}} = 0.2250096. \quad (246)$$

In the one-calibration reading, a predeclared scalar observable instead fixes u uniquely by Proposition 5.12; no second value of u is introduced in the CKM or neutral sectors.

The CKM hierarchy is then assembled from independent structural layers. The valuations (185) and Lemma 5.5 fix the tangent hierarchy, $\rho_{\text{RP}} = 1$ fixes the primitive norm ratio, and (67) supplies $q_H/p_H = 4$ before Proposition 5.9 gives the unitary magnitude representative. The commensurate $n = 7$ class is preserved through Proposition 3.31 and (182); Proposition 5.11 isolates the remaining primitive coefficient phase. For (246), $J_q/\lambda_{\text{cen}}^6 = 0.240882$, while the entropic susceptibility remains CP-even.

The low-denominator commensurate scan in Table 5 is exhaustive for $n \leq 49$; $n = 1$ is the nonsimple endpoint and $n = 7$ is the least-denominator simple branch. The same matrix completion gives $A = 0.830271$, $R_b = 0.385015$, and $\sin 2\beta = 0.71030$, whereas direct identification of the Rees path coefficients would give 0.73012 at the same phase. The difference isolates the magnitude-completion step. The branch value $|V_{cb}| = 0.0420361$ may also be compared with the inclusive and exclusive determinations summarized in [69]; the angle comparison of [86] is retained as a secondary flavor diagnostic. Neither comparison enters the determination of u . Proposition 5.2 excludes an observable-invisible lower-valuation $1 \leftrightarrow 3$ repair.

5.6.3 Charged-lepton balance benchmark

The charged-lepton diagnostic evaluates (214) on $L_L \leftrightarrow e_L^c$, independently of the quark relative-basis problem and neutral Majorana denominator. Using the PDG masses [69] with torsor ordering (τ, e, μ) gives $\Sigma_\ell^{\text{obs}} = (0.66 \pm 1.52) \times 10^{-5}$ and $\phi_\ell - 2/9 = (2.54 \pm 6.26) \times 10^{-6}$. These probe (215) and (217), while the response-order separation follows from the scalar tangent (172) and the noncentral Schur response (218).

The zero-correction balance $\Sigma_\ell = 0$ and the clock-lift reference $\phi_\ell = 2/9$, using the same inputs m_e, m_μ , give $m_\tau^{\text{bal}} = 1776.9690 \text{ MeV}$ and $m_\tau^{\text{clk}} = 1776.9665 \text{ MeV}$, separated by 2.5 keV. Simultaneous exact closure is not imposed. A proposed super tau-charm threshold study reaches projected 1 keV precision [87]; the complementary residuals are $\Sigma_\ell(m_\tau^{\text{clk}}) = 4.26 \times 10^{-7}$ and $\phi_\ell(m_\tau^{\text{bal}}) - 2/9 = -1.75 \times 10^{-7}$.

5.6.4 Neutral determinant-scale and PMNS diagnostics

The half-flux scale completion (220) gives $M_R = 6.31 \times 10^{14} \text{ GeV}$. In the unit-determinant pure type-I subbranch of (224), $|\det D_\nu|^{1/3} = v_F/\sqrt{2} = 174.10 \text{ GeV}$ gives $(m_1 m_2 m_3)^{1/3} = 0.0480 \text{ eV}$. Combining this prefactor with the normal-ordering splittings of [85] gives $m_1 \simeq 0.0409 \text{ eV}$ and $\sum_i m_i \simeq 0.1474 \text{ eV}$.

For flat ΛCDM , the DESI+CMB analysis [88] reports the 95% limits $\sum m_\nu < 0.072 \text{ eV}$ for the prior $\sum m_\nu > 0$ and $\sum m_\nu < 0.113 \text{ eV}$ for the hierarchy-motivated prior $\sum m_\nu > 0.059 \text{ eV}$. The unit-determinant subbranch lies above both limits. With the same oscillation splittings, the two limits translate through (224) to $|\det D_\nu|/m_D^3 \lesssim 0.22$ and $\lesssim 0.64$, respectively, where $m_D = v_F/\sqrt{2}$.

The normalized branch (225) gives $r_\nu = 0.05351$. With $m_D = 174.10 \text{ GeV}$ and the half-flux scale, the Takagi factorization (226) gives

$$m_1 = 0.720 \text{ meV}, \quad m_2 = 8.72 \text{ meV}, \quad m_3 = 50.46 \text{ meV}, \quad \sum_i m_i = 0.05990 \text{ eV}. \quad (247)$$

The corresponding squared differences are $7.55 \times 10^{-5} \text{ eV}^2$ and $2.55 \times 10^{-3} \text{ eV}^2$. Independently of their common scale, (227) gives $\mathcal{R}_\nu = 0.0296554$, while the zero-correction limit $u = 0$, $S_{\text{prim}}^{-1} = 0$ gives 0.0335187; the NuFIT 6.1 normal-ordering central ratio is 0.029558 [85]. The adjoint threshold identity (245) independently fixes M_H^{RG} in the same scale region, while (220) fixes M_R . The normalized Dirac orientation remains completed neutral data; the pure type-I determinant identity is (224). The seesaw comparison is represented by [89].

For the relative basis, (238) is evaluated with the same frozen u through (246), the phase (209), and the common-lift rate $r_\vartheta = 1$. The resulting PMNS reading is listed in Table 5; [85] is retained as the dedicated oscillation-fit comparison.

Proposition 5.19 (Lower-octant rigidity of the normalized neutral branch) *On the domain $0 \leq u < 10/49$, (205) gives $2/9 \leq \lambda_{\text{cen}} < 347/494$. With (237), (238), and the fixed phase from (209), let $v := R_{12}(\beta_\nu)R_{13}(\gamma_\nu)R_{23}(\alpha_\nu, \phi_\nu)e_3$, $A := -v_1/\sqrt{6} + v_2/\sqrt{3}$, and $B := v_3/\sqrt{2}$. Then $|U_{\tau 3}|^2 - |U_{\mu 3}|^2 = -4B \text{Re } A$. Outward-rounded interval evaluation on 50 equal subintervals of the stated λ_{cen} range gives $|U_{\tau 3}|^2 - |U_{\mu 3}|^2 > 3.4 \times 10^{-3}$, hence $\sin^2 \theta_{23} < 1/2$ throughout the branch.*

The reference point has $\sin^2 \theta_{23} = 0.47033$; an upper-octant determination would therefore falsify this fixed neutral realization rather than select another calibration. The joint T2K+NOvA likelihood [90] is retained as the current octant comparison.

The oscillation branch leaves two Majorana column phases unresolved. For the masses in (247) and the fixed PMNS moduli, the exact polygon bound gives $1.016 \text{ meV} \leq m_{\beta\beta} \leq 4.250 \text{ meV}$, while the phase-independent beta-decay mass is $m_\beta = 8.97 \text{ meV}$. The no-extra-Takagi-phase choice gives the interior point $m_{\beta\beta} = 3.89 \text{ meV}$.

5.7 Output status and falsification

Table 4 records the decisive chain: projector and R_1 , the primitive second-response fingerprint and Rees polarization, source-Hodge feasibility, the curvature-level 2:3:5 gate, and one canonical Eisenstein neutral seed with unit marked descent. After these gates are frozen, Proposition 5.12 permits one scalar calibration of u and (229) gives a direct cross-sector no-refit test. The numerical ledger is Table 5; absolute normalization and equivariant Riesz-Callias attachment remain separate.

6 Discussion and Conclusions

6.1 Results and reconstruction mechanism

The mechanism can be read from the geometric defect outward. A codimension-three timelike defect supplies an oriented two-sphere link, the primitive positive line, and the two non-scalar source grades $V_1[1] \oplus V_2[2]$. Atomicity fixes the primitive line degree, while Proposition 3.6 identifies the active normal module with the principal $V_1 \oplus V_2$ decomposition of compact A_2 . The Coxeter plane then supplies the equianharmonic lattice used by the finite arithmetic construction.

Protected reconstruction asks for the least finite carrier that represents both marked grades faithfully. Under the intersection-closure and faithful-central-marking assumptions, the Borel-Weil cutoff gives $E_3 \oplus E_2 \simeq N_\Gamma^{\mathbb{C}} \oplus S(N_\Gamma)$, while Theorem 3.10 supplies the independent theta-rank check. Determinant and Clifford closure give the $S(U(3) \times U(2))$ Levi form, its global $\mathbb{Z}_6$ quotient, the even exterior package, and $k_Y = 5/3$. The quotient $[E_\omega/\mu_6]$ and Proposition 3.25 supply the three-point family torsor; Corollary 3.36 selects the first simple norm-seven commensurate class.

The realization layer is operator-first. The isolated projector and R_1 are tested before the primitive second-response fingerprint and Rees polarization; source-Hodge feasibility and the curvature residual (150) then test whether the same microscopic branch carries the finite arithmetic. The regular integral family cycle and norm-seven determinant phase are algebraically compatible as the projective and central parts of one lift; their physical marked identification remains a first-response test. Proposition 5.16 reduces the neutral 12/13 descent to one marked Eisenstein seed and central updates.

Once these gates pass, one universal scalar u is fixed once and all remaining local dimensionless outputs are evaluated without sectoral retuning. The relation (229) makes part of this content calibration-free, while Proposition 5.19 gives a branch-level atmospheric falsifier. Absolute action normalization and the equivariant Riesz-Callias identification remain separate. Thus the mechanism runs from Lorentzian defect geometry to a protected graded source, the minimal $3 + 2$ carrier, finite response gates, one marked family/neutral arithmetic, one frozen u , and downstream CKM, neutral, and PMNS tests; failure of an upstream residual terminates the branch.

6.2 One-calibration quantitative test

The numerical test is summarized by the single frozen- u ledger in Table 5. If u is fixed experimentally, the designated calibration observable is removed from the prediction scorecard and every remaining u -dependent row is evaluated without retuning. The displayed reference uses the fixed benchmark value $u = 1/(64\pi^2)$ and compares current dimensionless outputs with the 2026 PDG values; the dedicated oscillation-fit comparison remains [85]. Reactor and solar discrimination is represented by [91] and [92], with prospective octant and CP sensitivity represented by [93] and [94].

On the determinant-suppressed mass branch the unresolved Majorana phases give $1.016\text{ meV} \leq m_{\beta\beta} \leq 4.250\text{ meV}$ and $m_\beta = 8.97\text{ meV}$; 3.89 meV is the no-extra-Takagi-phase point. The common adjoint threshold identity (245) and the neutral half-flux scale remain absolute completed readings outside the one- u dimensionless scorecard.

6.3 Relation to geometric and spectral constructions

The curvature reconstruction used here is closest in spirit to the algebraic program initiated by Rainich [95], while the Codazzi input is compared with the structural analysis of [96]. The trace-adjusted Codazzi coefficient adds a graded boundary source and finite reconstruction step to that geometric setting.

A complementary geometric comparison is supplied by the first-order separation of [97] and the projective-link twistor setting of [98]. These comparisons concern the realization layer; the finite internal package is selected by the boundary grading and protected reconstruction.

In almost-commutative geometry, the internal algebra is introduced as spectral data in [99], with dynamics organized by the spectral action of [68]. Dynamical principal-bundle models provide a geometric comparison [100]. Here the finite algebra and its integral grading are outputs of the primitive boundary reconstruction.

Modular-invariant flavor models provide a closer comparison for the family sector. In their minimal formulation, Yukawa couplings are modular forms of a modulus τ , with finite modular groups constraining the flavor representations [101]. Generalized CP can restrict CP violation to the modulus sector [102]. The equianharmonic fixed point $\tau = \omega$ carries a residual $\mathbb{Z}_3^{ST}$ symmetry; stabilization near this point can generate fermion hierarchies [103], and modular A_4 models provide explicit quark realizations in its neighbourhood [104]. The present reconstruction shares the equianharmonic CM point and its order-three action, while E_ω , the rank-three torsor, and the finite carrier are obtained from the defect boundary data rather than from a modulus-dependent Yukawa sector. At the orbifold level the comparison remains structural: the modular curve has signature $(2, 3, \infty)$, whereas $[E_\omega/\mu_6]$ is tubular of type $(6, 3, 2)$.

6.4 Further directions

The remaining local programme is now narrow: compute an actual R_1 , verify its central-normalized regular C_3 projective class, test (191) before full Rees polarization, evaluate (150) and the source-Hodge feasibility cone, and extract one physical neutral seed in the fixed Eisenstein class of Proposition 5.16. The optional zero-input programme is separate: rank-one Clifford scalarization must derive the factor $1/d_F$, while a microscopic action theorem must fix $\kappa_\Omega = 1$.

The global target is the regular equivariant Riesz-Callias class rather than index three alone. A common-base gapped comparison of the determinant line, Riesz bundle and Callias kernel is required; boundary and APS tools are represented by [105], [106], [107], and [108], while line operators are compared with [45]. Einstein-Weyl and minitwistor extensions [109], [110] remain optional. The filtered-functor problem is still open: [46], [42], [111], and [112] provide comparison frameworks for relating tubular degrees to the retained Rees brackets. Experimental tests remain downstream of these gates and of the single frozen calibration.

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